DO316 v4.18 – EX316 Exam Questions

All Tasks Overview

- Deploy OpenShift Virtualization
- Resume a node from maintenance mode
- Configure user access
- Deploy a virtual machine and configure access
- Configure a web server in a virtual machine
- Deploy a multihomed virtual machine
- Manage storage for virtual machines
- Create UDN
- Create VM on UDN
- Create a virtual machine template
- Create a service using VM load balancing
- Create a virtual machine snapshot
- Configure virtual machine migration
- Clone root disk of a VM
- Prepare cluster for OADP backups
- Backup VMs using OADP
- Configure a liveness probe
- Prepare VM for node failure

==========

Login Instructions in exam:

[desktop@desktop]$ ssh opsadm@workbentch.ocp4.example.com

[opsadm@workbentch]$ oc login -u kubeadmin -p <password given in info page> –<server detail> from info page like https://api.ocp4.example.com:6443>

[opsadm@workbentch]$ oc whoami

==========

Q-1 - Deploy OpenShift Virtualization

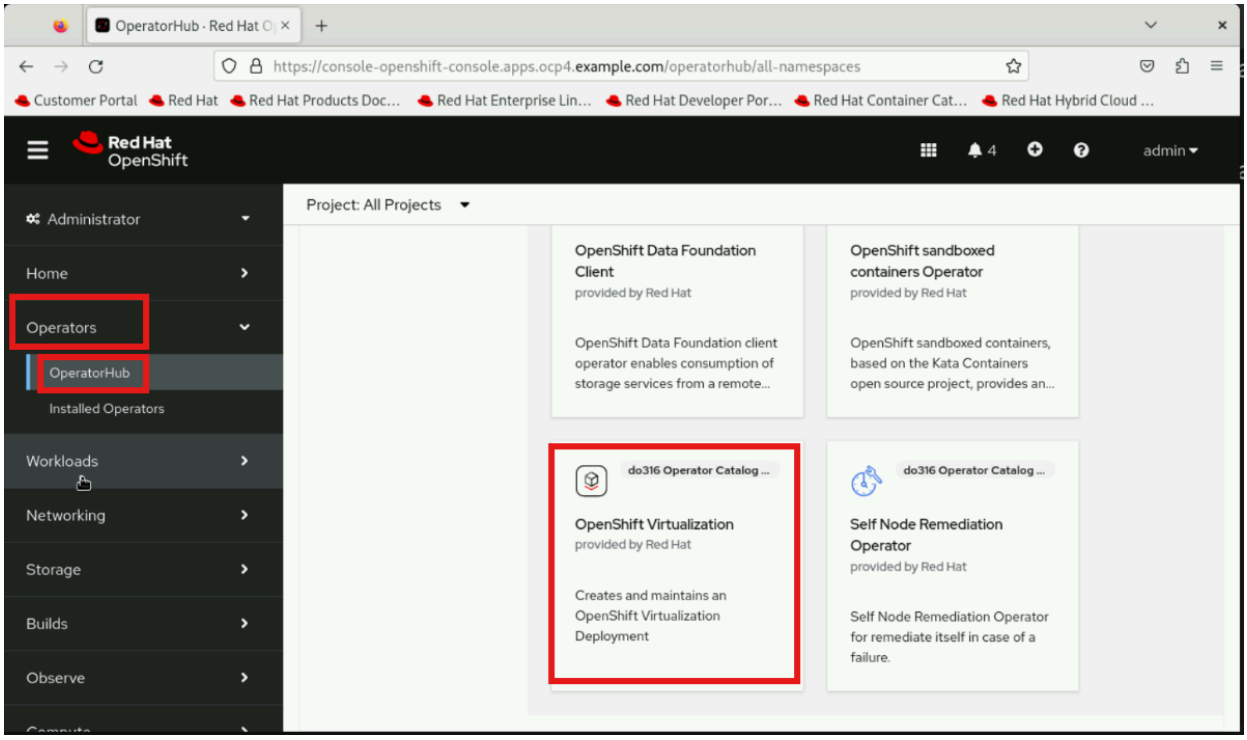
- Deploy OpenShift Virtualization in the **openshift-cnv** project
- Install and configure an **operator** with the same **kubevirt-hyperconverged** in the same projece

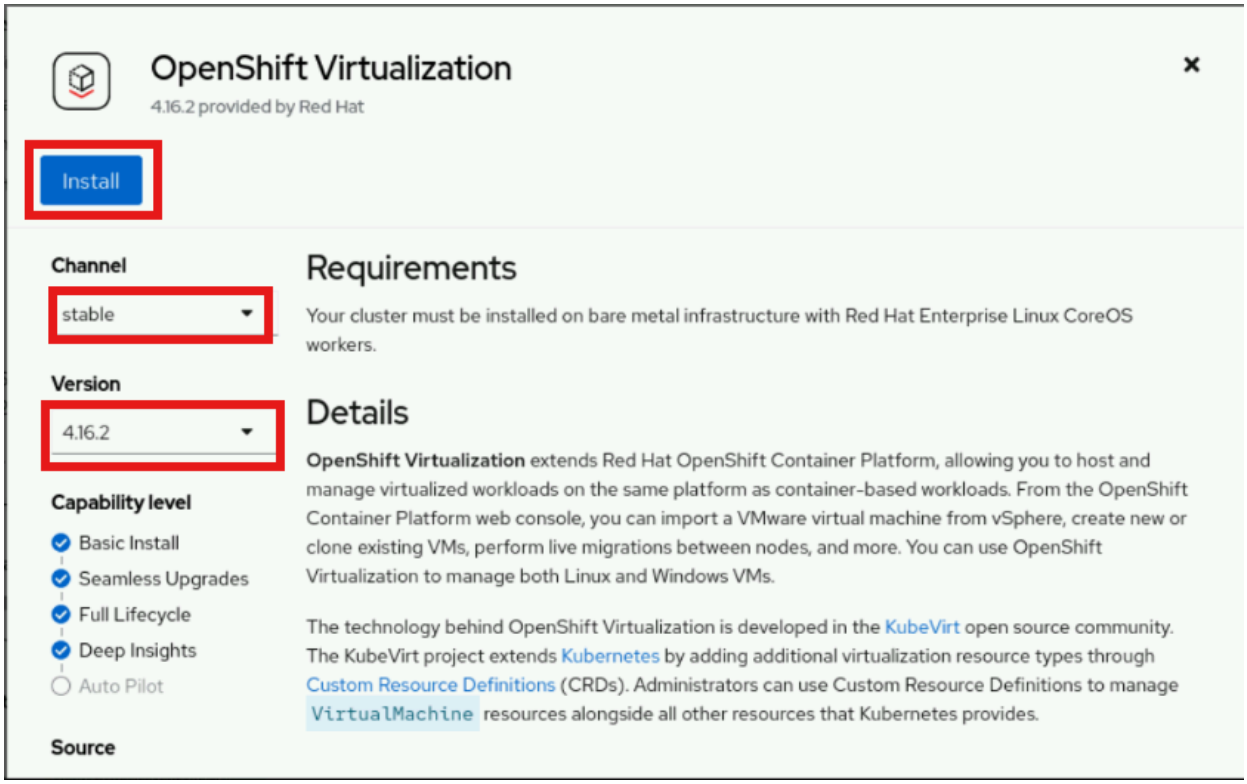
ANS:

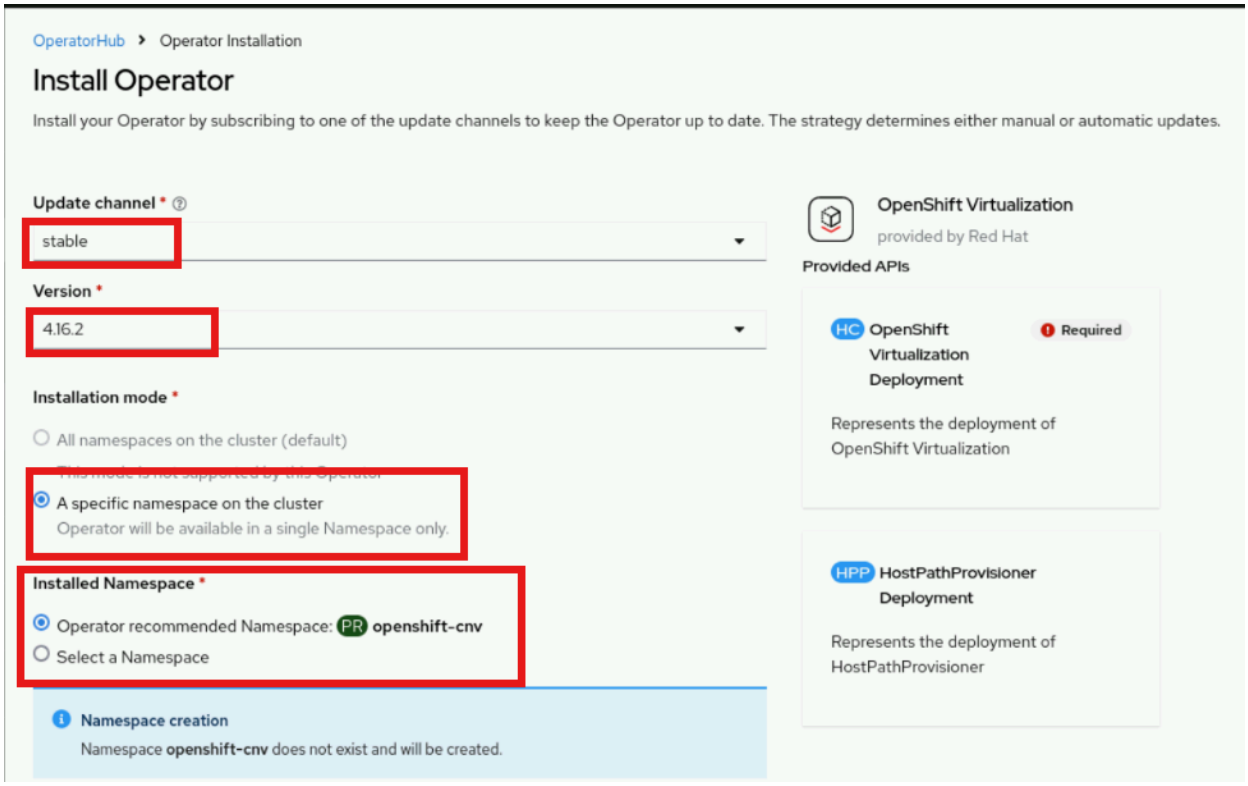
```
[student@workstation ~]$ oc login -u admin -p redhatocp https://api.ocp4.example.com:6443
Login successful.

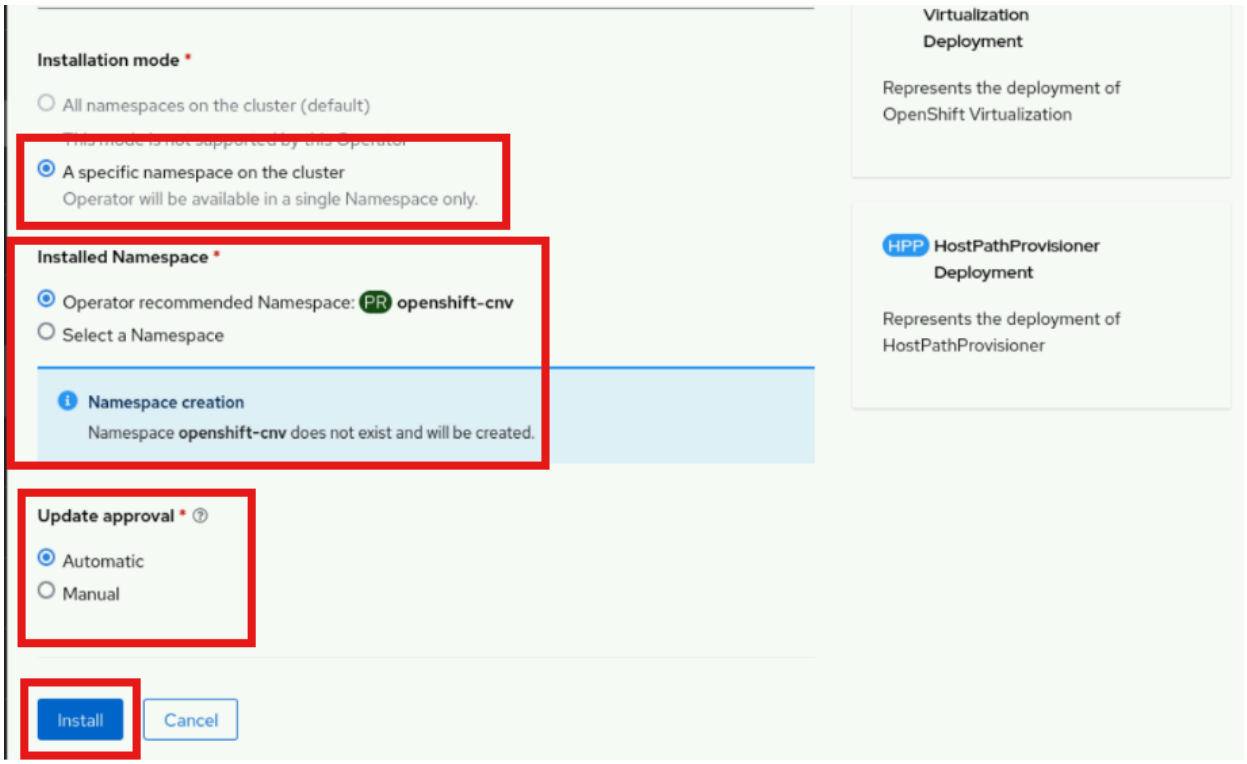
You have access to 74 projects, the list has been suppressed. You can list all projects with
'oc projects'

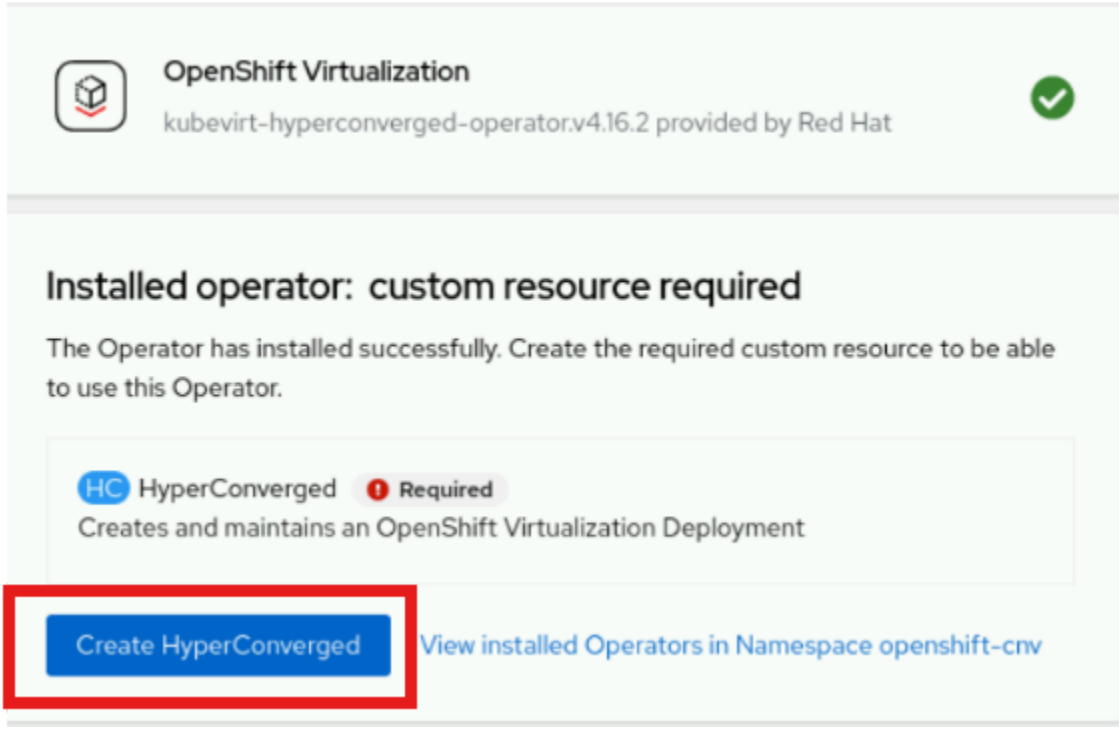
Using project "default".
Welcome! See 'oc help' to get started.
[student@workstation ~]$ oc whoami --show-console
https://console-openshift-console.apps.ocp4.example.com
[student@workstation ~]$
```

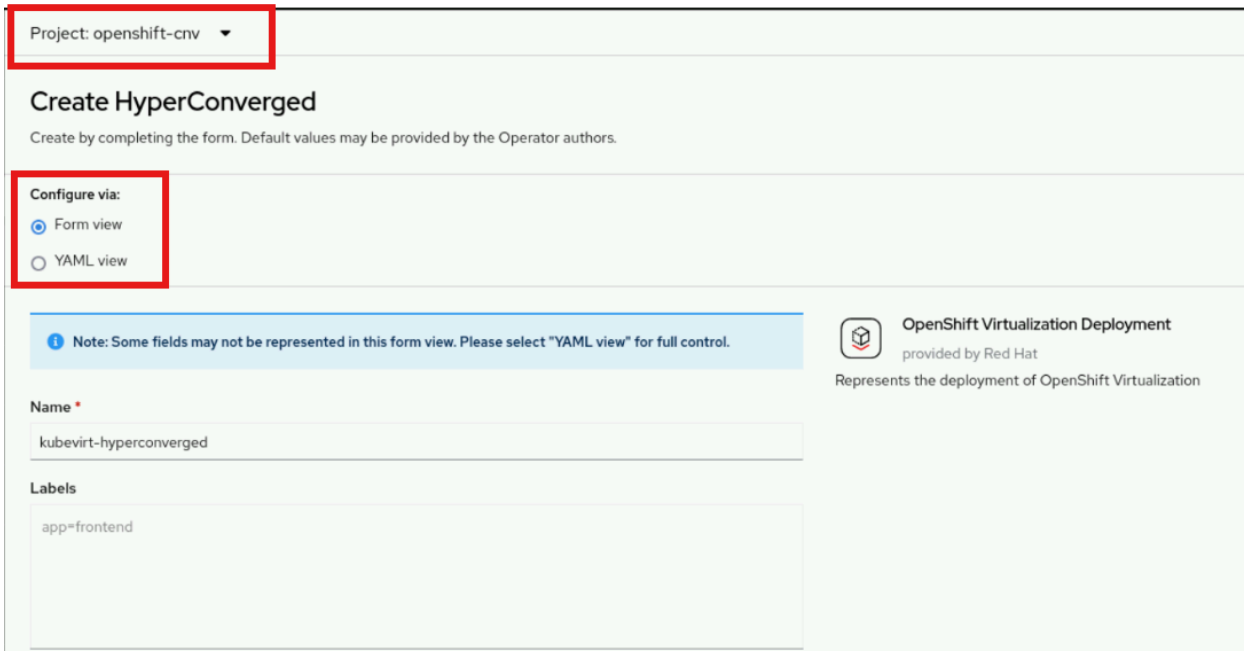


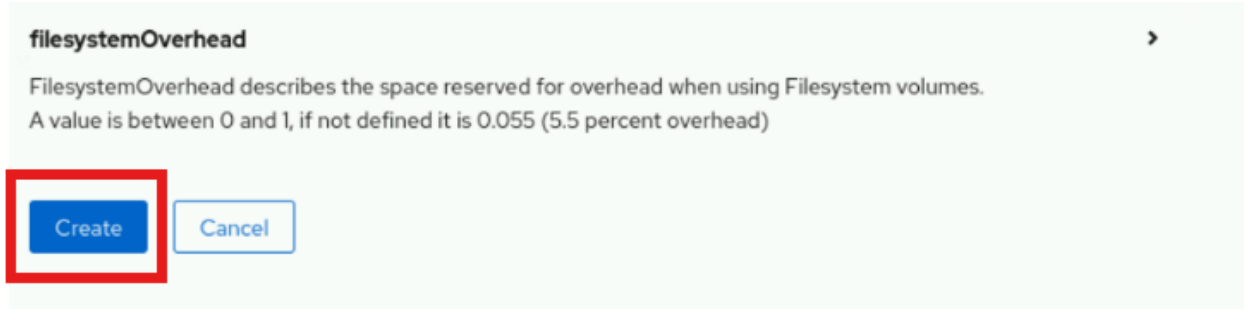

OpenShift Virtualization
4.16.2 provided by Red Hat
Install
Channel
stable
Version
4.16.2
Capability level
Basic Install
Seamless Upgrades
Full Lifecycle
Deep Insights
Auto Pilot
Source
Requirements
Your cluster must be installed on bare metal infrastructure with Red Hat Enterprise Linux CoreOS workers.
Details
OpenShift Virtualization extends Red Hat OpenShift Container Platform, allowing you to host and manage virtualized workloads on the same platform as container-based workloads. From the OpenShift Container Platform web console, you can import a VMware virtual machine from vSphere, create new or clone existing VMs, perform live migrations between nodes, and more. You can use OpenShift Virtualization to manage both Linux and Windows VMs.
The technology behind OpenShift Virtualization is developed in the KubeVirt open source community. The KubeVirt project extends Kubernetes by adding additional virtualization resource types through Custom Resource Definitions (CRDs). Administrators can use Custom Resource Definitions to manage VirtualMachine resources alongside all other resources that Kubernetes provides.

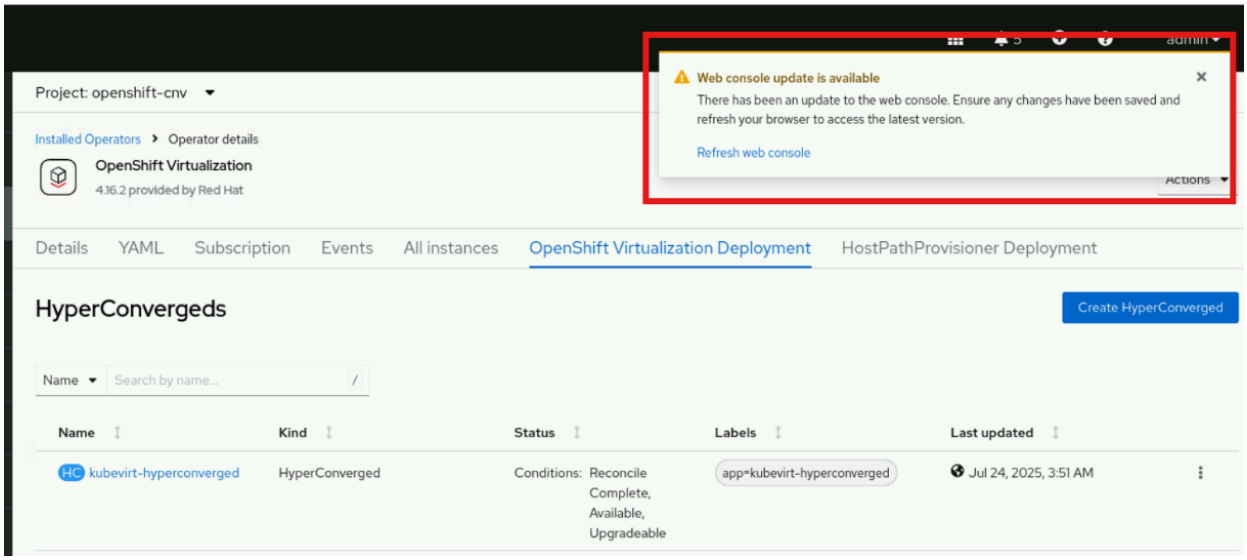

OperatorHub › Operator Installation
Install Operator
Install your Operator by subscribing to one of the update channels to keep the Operator up to date. The strategy determines either manual or automatic updates.
Update channel *
stable
Version *
4.16.2
Installation mode *
All namespaces on the cluster (default)
A specific namespace on the cluster
Operator will be available in a single Namespace only.
Installed Namespace *
Operator recommended Namespace: PR openshift-cnv
Select a Namespace
Namespace creation
Namespace openshift-cnv does not exist and will be created.
OpenShift Virtualization
provided by Red Hat
Provided APIs
HC OpenShift Virtualization Deployment
Required
Represents the deployment of OpenShift Virtualization
HPP HostPathProvisioner Deployment
Represents the deployment of HostPathProvisioner


Virtualization Deployment
Represents the deployment of OpenShift Virtualization
Installation mode *
All namespaces on the cluster (default)
A specific namespace on the cluster
Operator will be available in a single Namespace only.
HPP HostPathProvisioner Deployment
Represents the deployment of HostPathProvisioner
Installed Namespace *
Operator recommended Namespace: PR openshift-cnv
Select a Namespace
Namespace creation
Namespace openshift-cnv does not exist and will be created.
Update approval *
Automatic
Manual
Install
Cancel


OpenShift Virtualization
kubevirt-hyperconverged-operator.v4.16.2 provided by Red Hat
Installed operator: custom resource required
The Operator has installed successfully. Create the required custom resource to be able to use this Operator.
HC HyperConverged
Required
Creates and maintains an OpenShift Virtualization Deployment
Create HyperConverged
View installed Operators in Namespace openshift-cnv


Project: openshift-cnv
Create HyperConverged
Create by completing the form. Default values may be provided by the Operator authors.
Configure via:
Form view
YAML view
Note: Some fields may not be represented in this form view. Please select "YAML view" for full control.
OpenShift Virtualization Deployment
provided by Red Hat
Represents the deployment of OpenShift Virtualization
Name *
kubevirt-hyperconverged
Labels
app=frontend


filesystemOverhead
FilesystemOverhead describes the space reserved for overhead when using Filesystem volumes.
A value is between 0 and 1, if not defined it is 0.055 (5.5 percent overhead)
Create
Cancel



==========

Q-2 - Resume a node from maintenance mode

Resume the node that is currently in maintenance mode so that the following conditions are true:

- The node is able to schedule pods.
- The node shows no status other than Ready.

ANS

```
[student@workstation ~]$ oc get nodes
NAME       STATUS                     ROLES                         AGE    VERSION
master01   Ready                      control-plane,master,worker   104d   v1.29.14+7cf4c05
master02   Ready                      control-plane,master,worker   104d   v1.29.14+7cf4c05
master03   Ready                      control-plane,master,worker   104d   v1.29.14+7cf4c05
worker01   Ready                      worker                        103d   v1.29.14+7cf4c05
worker02   Ready,SchedulingDisabled   worker                        103d   v1.29.14+7cf4c05
[student@workstation ~]$ oc adm uncordon worker02
node/worker02 uncordoned
[student@workstation ~]$ oc get nodes
NAME       STATUS   ROLES                         AGE    VERSION
master01   Ready    control-plane,master,worker   104d   v1.29.14+7cf4c05
master02   Ready    control-plane,master,worker   104d   v1.29.14+7cf4c05
master03   Ready    control-plane,master,worker   104d   v1.29.14+7cf4c05
worker01   Ready    worker                        103d   v1.29.14+7cf4c05
worker02   Ready    worker                        103d   v1.29.14+7cf4c05
[student@workstation ~]$
```

==========

Q-3 - Configure user access

Configure user access so that the following conditions are true:

For **ramtops** project:

- User **verence-ii** has the necessary roles needed to create and manage virtual machines
- The user **moist** can view and access the virtual machine performance metrics
- The user **moist** is allowed to start, stop, pause or restart the virtual machine

For **ubarwald** project:

- User **moist** has the necessary roles needed to create and manage virtual machines

NOTE: The password for user **verence-ii** and for user **moist** is **Atenorth3**

ANS:

➔ oc adm policy add-role-to-user admin verence-ii -n ramtops
➔ oc adm policy add-role-to-user edit moist -n ramtops
➔ oc adm policy add-role-to-user admin moist -n ubarwald
➔ oc adm policy add-role-to-user kubevirt.io:admin verence-ii -n ramtops
➔ oc adm policy add-role-to-user kubevirt.io:edit moist -n ramtops
➔ oc adm policy add-role-to-user kubevirt.io:admin moist -n ubarwald

```
[student@workstation users]$ oc adm policy add-role-to-user kubevirt.io:admin verence-ii -n ramtops
clusterrole.rbac.authorization.k8s.io/kubevirt.io:admin added: "verence-ii"
[student@workstation users]$ oc adm policy add-role-to-user kubevirt.io:edit moist -n ramtops
clusterrole.rbac.authorization.k8s.io/kubevirt.io:edit added: "moist"
[student@workstation users]$ oc adm policy add-role-to-user kubevirt.io:admin moist -n ubarwald
clusterrole.rbac.authorization.k8s.io/kubevirt.io:admin added: "moist"
[student@workstation users]$
```

→ verify via logged in every user and check can they able to create vm or not or using below command

→ oc auth can-i –list –namespace ramtops –as verence-ii

→ oc auth can-i –list –namespace ramtops –as moist

→ oc auth can-i –list –namespace ubarwald –as moist

```
[student@workstation users]$ oc auth can-i create vm -n ramtops --as verence-ii
yes
[student@workstation users]$ oc auth can-i create vm -n ubarwald --as moist
yes
[student@workstation users]$
```

==========

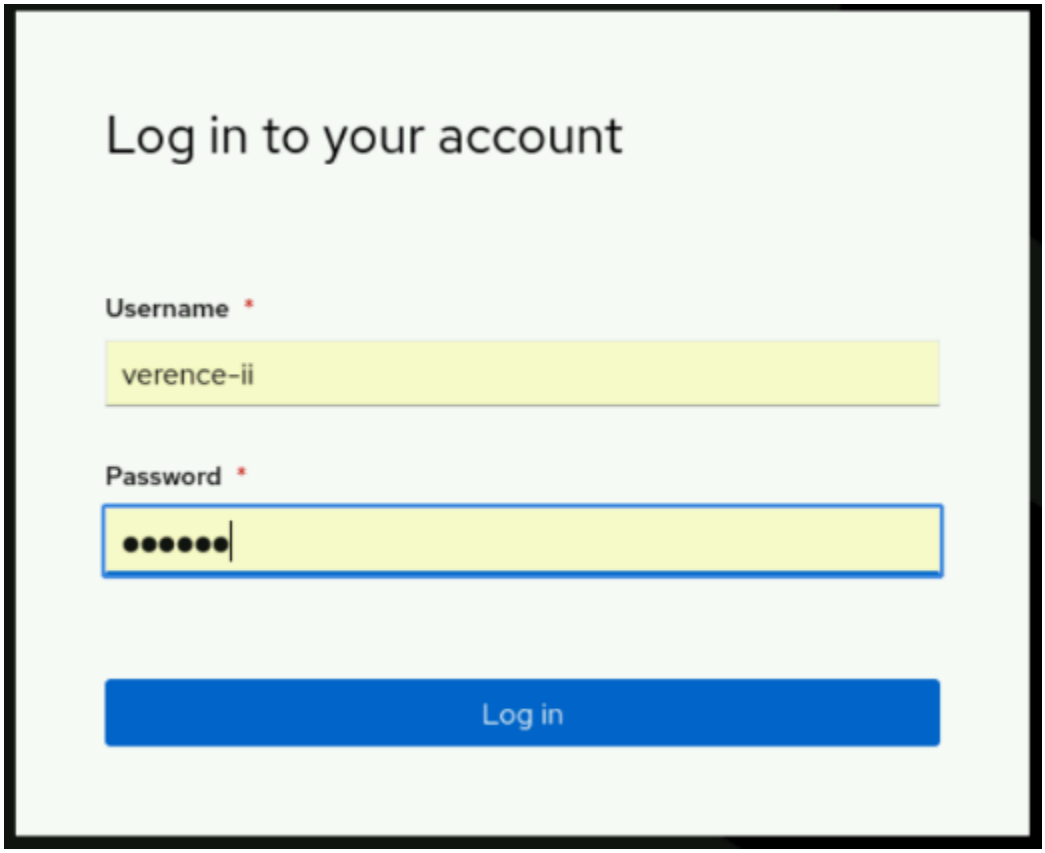
Q-4 - Deploy a virtual machine and configure access to it

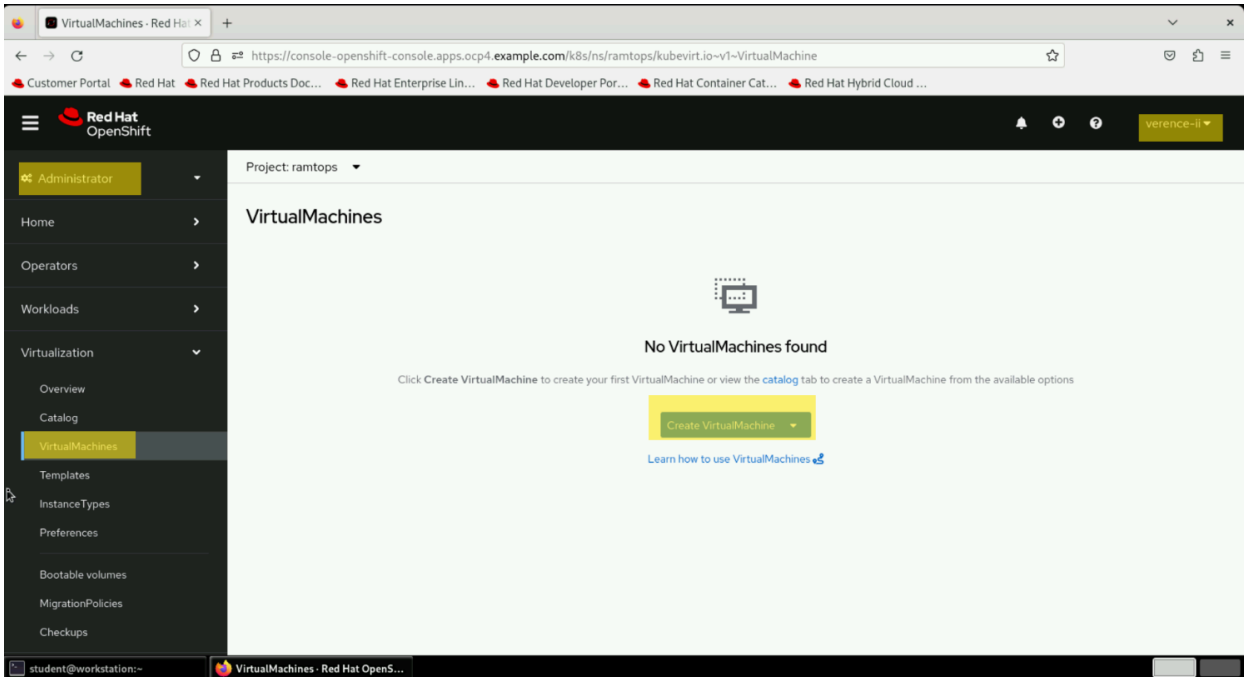
Deploy a **Red Hat Enterprise Linux 9.2** virtual machine so that the following conditions are true:

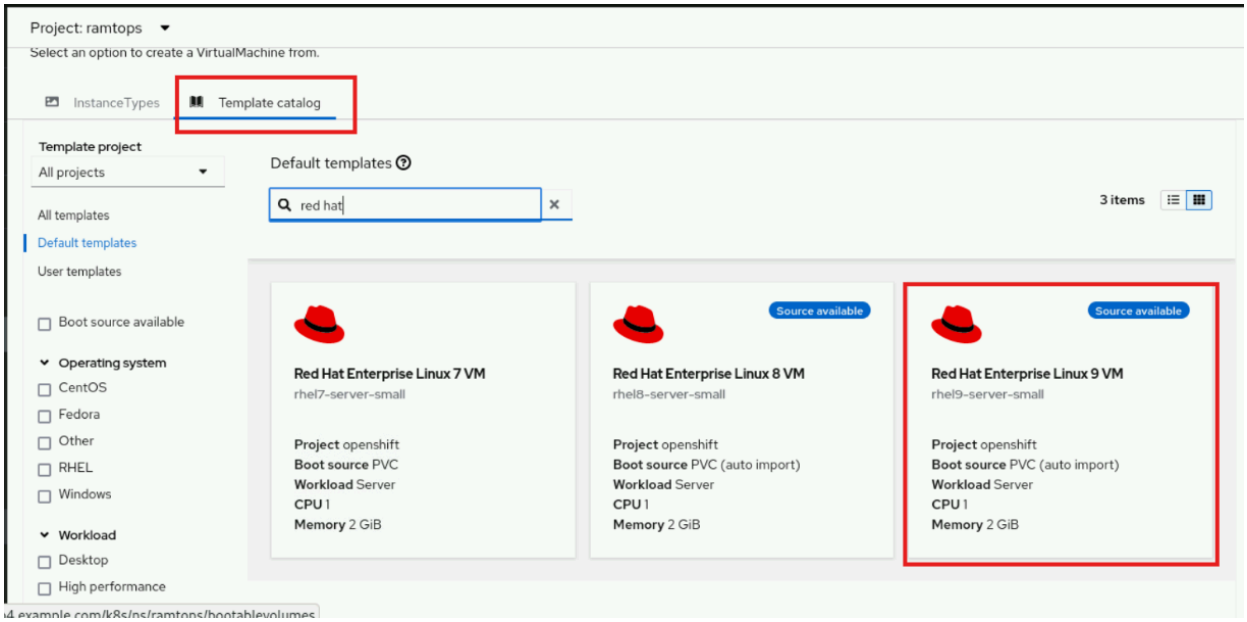
- The project is **ramtops**
- The virtual machine name is **lancre**
- User **verence-ii** creates the virtual machine **lancre**
- The image used to create the persistent volume claim for the virtual machine boot source is http://utility.lab.example.com:8080/openshift4/images/rhel-9-helloworld.qcow2 **#this link only works in lab, in exam they have given**
- The storageClassName is **ocs-external-storagecluster-ceph-rbd**
- The persistent volume claim size is **30Gi**

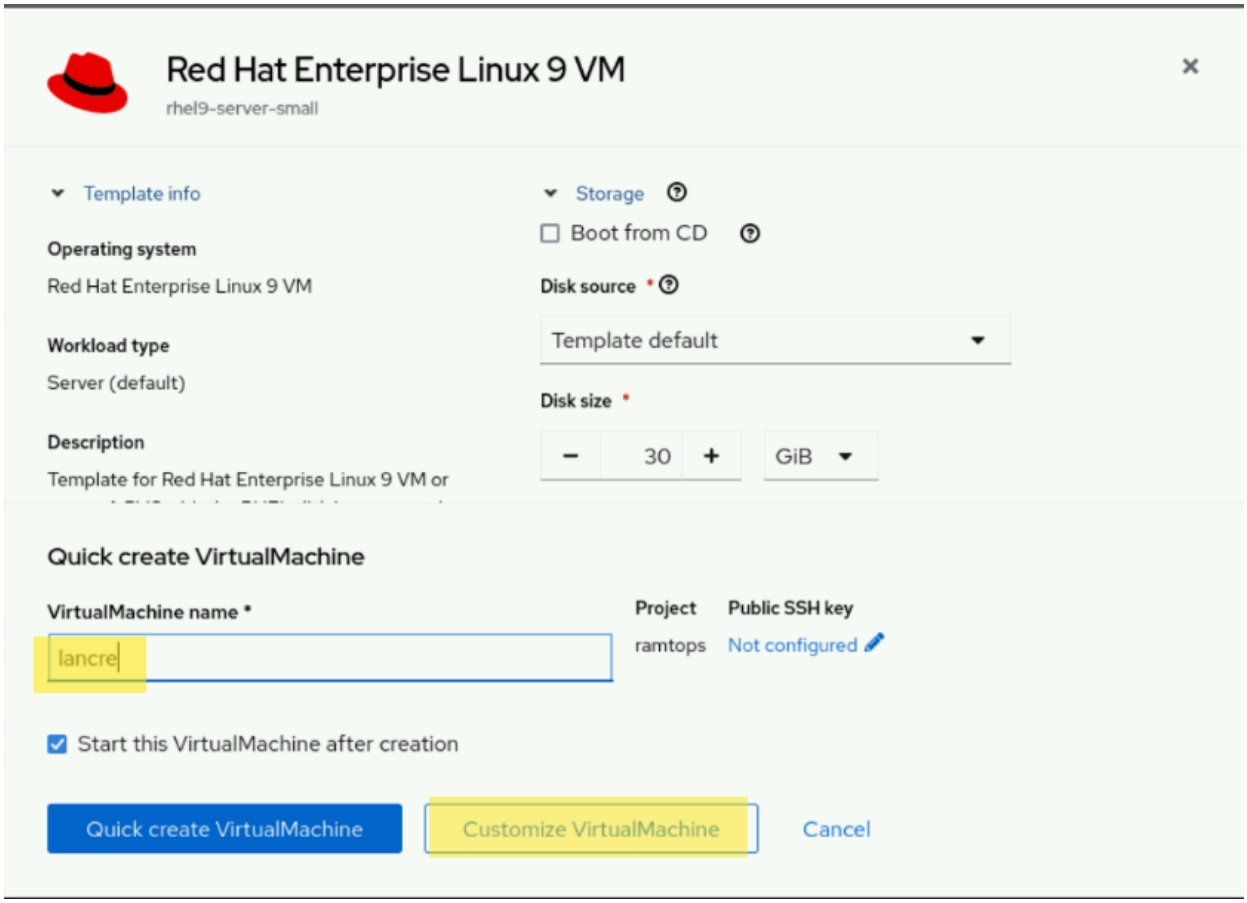
- The workload type of the virtual machine is **server**
- The flavor for the virtual machine is **small**
- The network interface name is **default**
- The user **verence-ii** with password **Atenorth3** exists in the **cloud-init** definition
- The ssh key **/home/opsadms/.ssh/id_rsa_ex316.pub** from user **opsadm** at **workbench.domain3.example.com** has been added as an authorized SSH key via the **cloud-init** definition

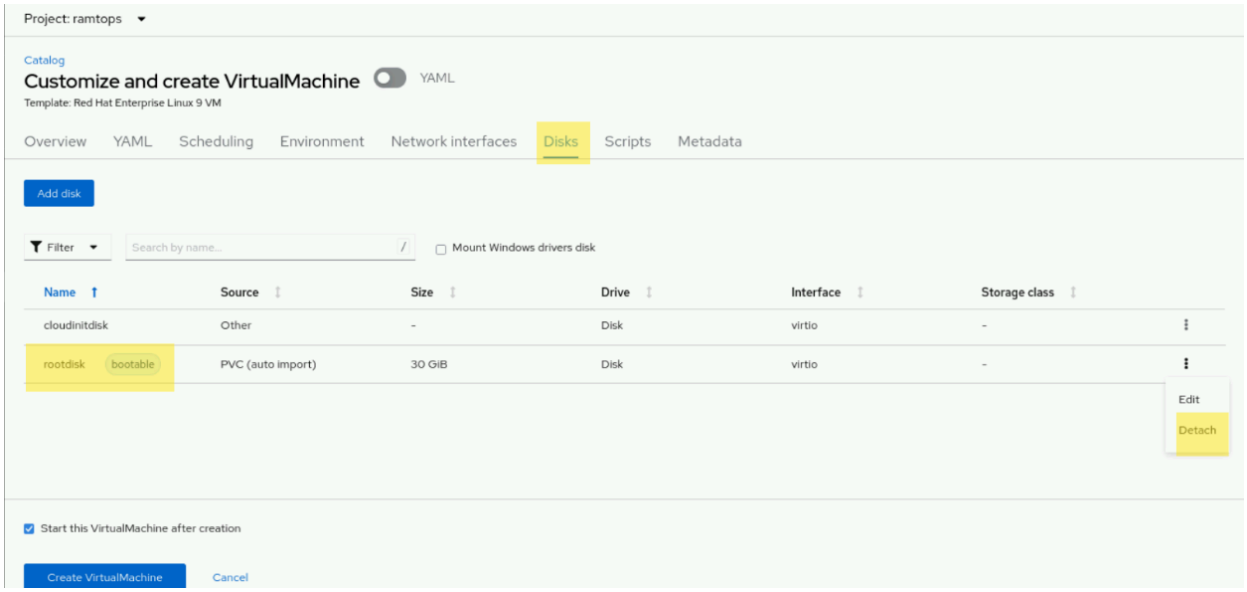
ANS:

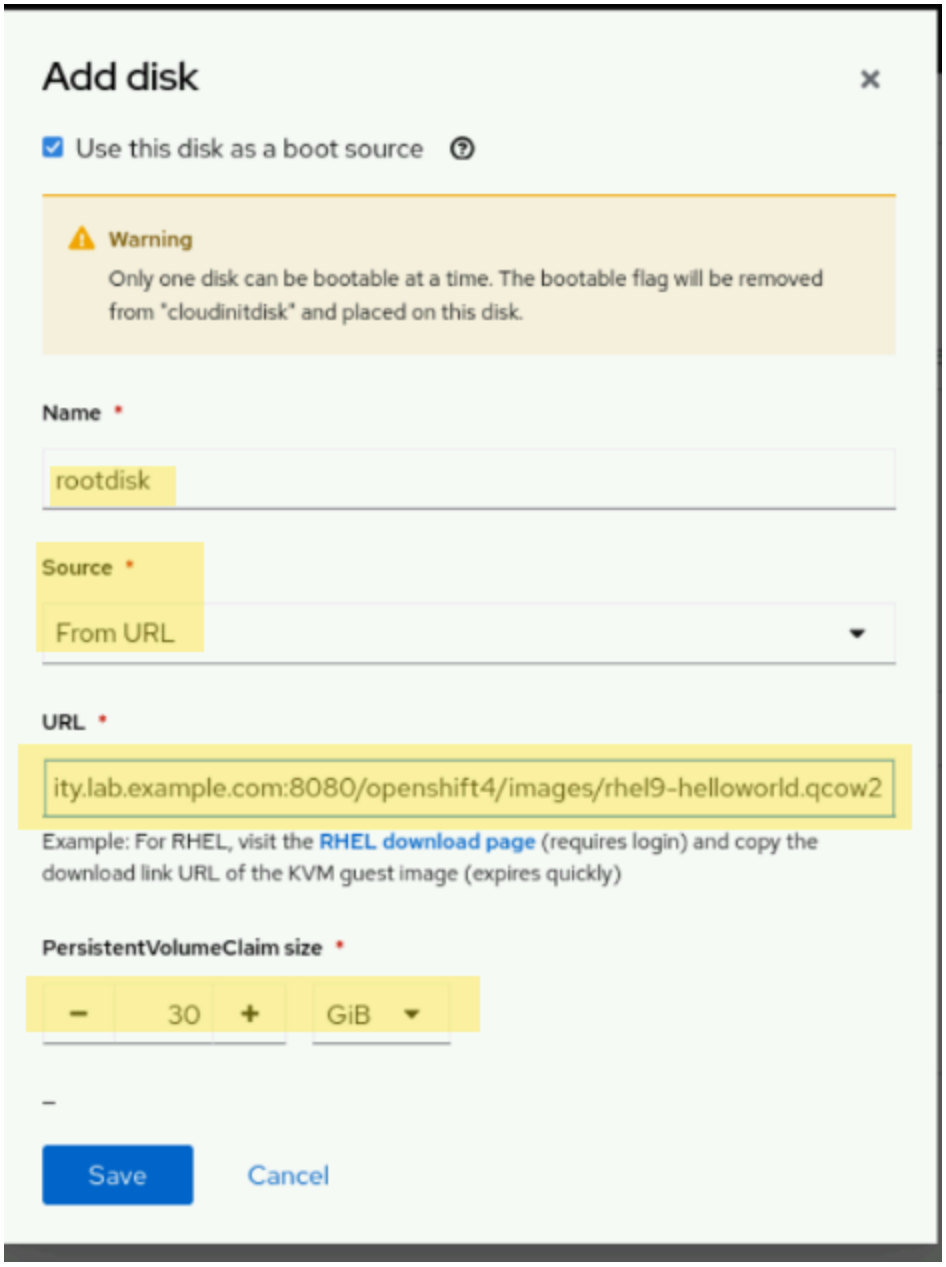


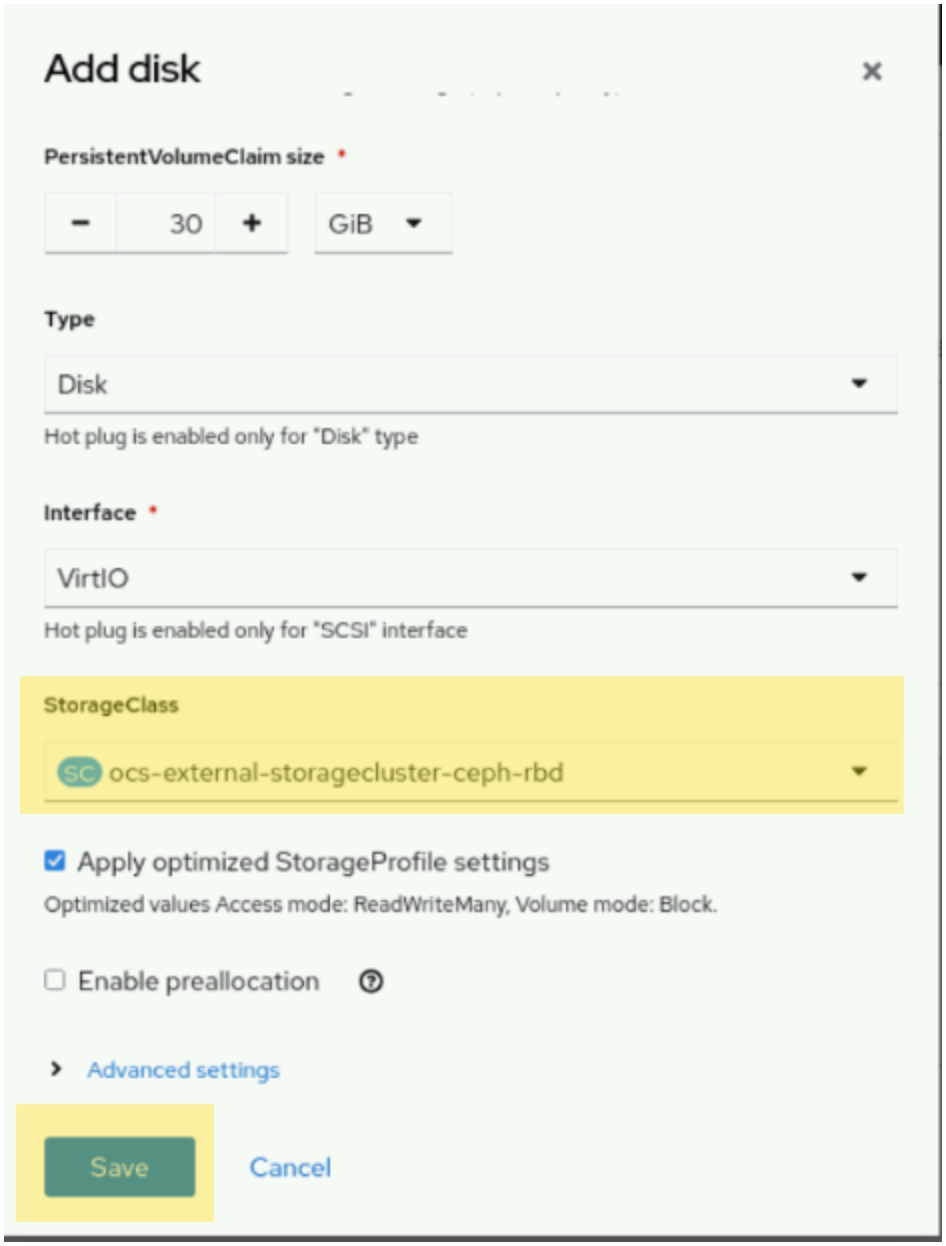

VirtualMachines · Red Ha
https://console-openshift-console.apps.ocp4.example.com/k8s/ns/ramtops/kubevirt.io~v1~VirtualMachine
Customer Portal
Red Hat
Red Hat Products Doc...
Red Hat Enterprise Lin...
Red Hat Developer Por...
Red Hat Container Cat...
Red Hat Hybrid Cloud ...
Red Hat OpenShift
verence-ii
Administrator
Home
Operators
Workloads
Virtualization
Overview
Catalog
VirtualMachines
Templates
InstanceTypes
Preferences
Bootable volumes
MigrationPolicies
Checkups
Project: ramtops
VirtualMachines
No VirtualMachines found
Click Create VirtualMachine to create your first VirtualMachine or view the catalog tab to create a VirtualMachine from the available options
Create VirtualMachine
Learn how to use VirtualMachines
student@workstation:~
VirtualMachines · Red Hat OpenS...


Project: ramtops
Select an option to create a VirtualMachine from.
InstanceTypes
Template catalog
Template project
All projects
All templates
Default templates
User templates
Boot source available
Operating system
CentOS
Fedora
Other
RHEL
Windows
Workload
Desktop
High performance
Default templates
red hat
3 items
Red Hat Enterprise Linux 7 VM
rhel7-server-small
Project openshift
Boot source PVC
Workload Server
CPU 1
Memory 2 GiB
Source available
Red Hat Enterprise Linux 8 VM
rhel8-server-small
Project openshift
Boot source PVC (auto import)
Workload Server
CPU 1
Memory 2 GiB
Source available
Red Hat Enterprise Linux 9 VM
rhel9-server-small
Project openshift
Boot source PVC (auto import)
Workload Server
CPU 1
Memory 2 GiB


Red Hat Enterprise Linux 9 VM
rhel9-server-small
Template info
Operating system
Red Hat Enterprise Linux 9 VM
Workload type
Server (default)
Description
Template for Red Hat Enterprise Linux 9 VM or
Storage
Boot from CD
Disk source *
Template default
Disk size *
30
GiB
Quick create VirtualMachine
VirtualMachine name *
lancre
Project
ramtops
Public SSH key
Not configured
Start this VirtualMachine after creation
Quick create VirtualMachine
Customize VirtualMachine
Cancel

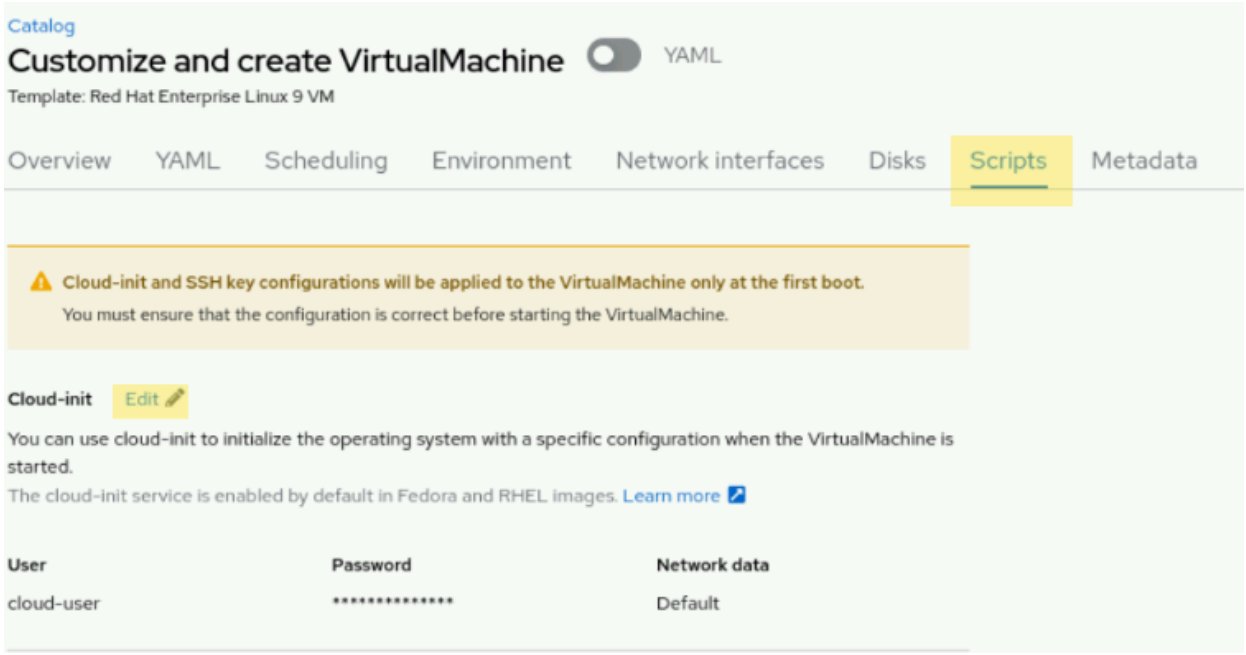

Project: ramtops
Catalog
Customize and create VirtualMachine
YAML
Template: Red Hat Enterprise Linux 9 VM
Overview
YAML
Scheduling
Environment
Network interfaces
Disks
Scripts
Metadata
Add disk
Filter
Search by name...
Mount Windows drivers disk
Name
Source
Size
Drive
Interface
Storage class
cloudinitdisk
Other
-
Disk
virtio
-
rootdisk
bootable
PVC (auto import)
30 GiB
Disk
virtio
-
Edit
Detach
Start this VirtualMachine after creation
Create VirtualMachine
Cancel


Add disk
Use this disk as a boot source
Warning
Only one disk can be bootable at a time. The bootable flag will be removed from "cloudinitdisk" and placed on this disk.
Name *
rootdisk
Source *
From URL
URL *
ity.lab.example.com:8080/openshift4/images/rhel9-helloworld.qcow2
Example: For RHEL, visit the RHEL download page (requires login) and copy the download link URL of the KVM guest image (expires quickly)
PersistentVolumeClaim size *
30
GiB
Save
Cancel

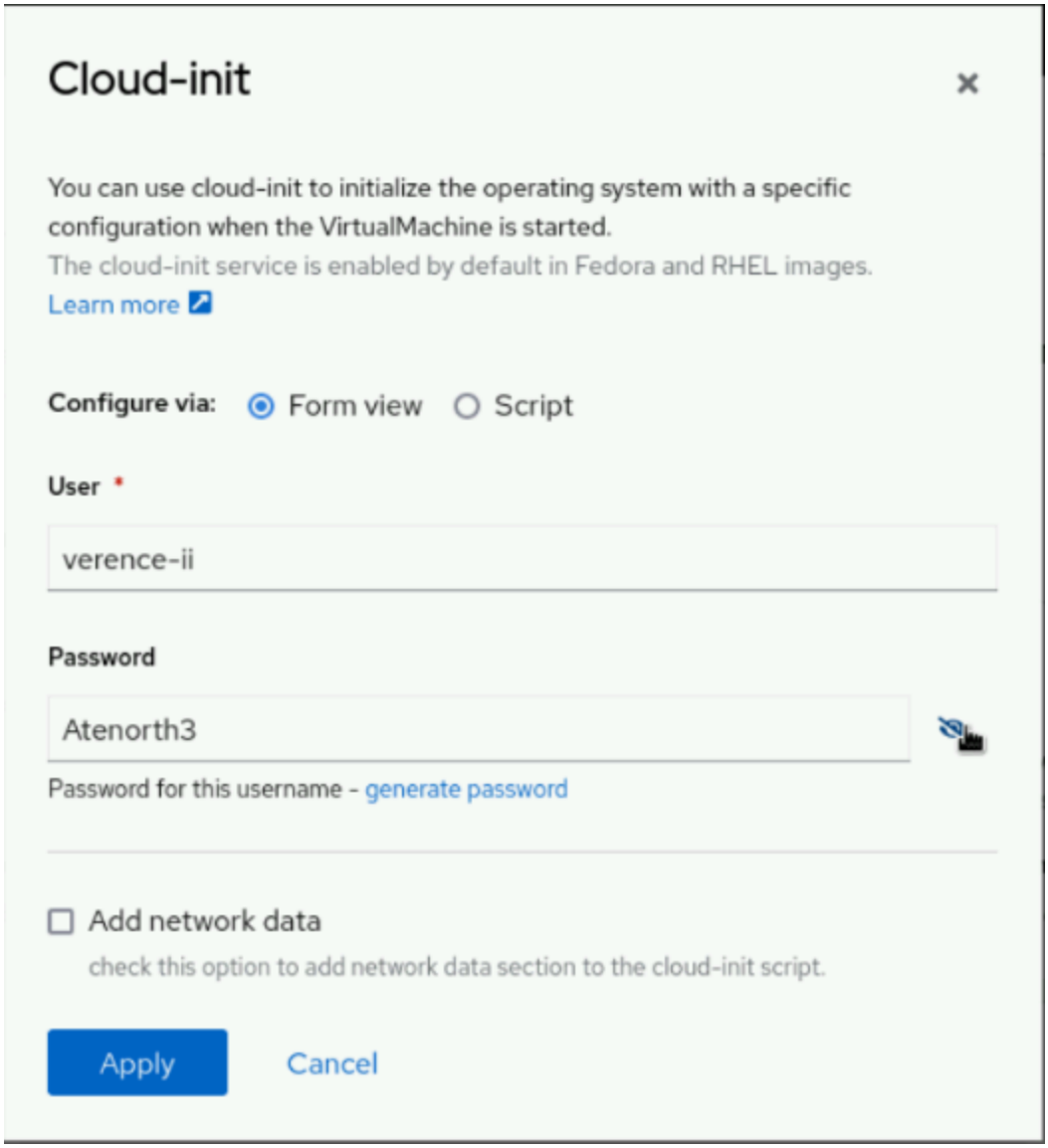

Add disk
PersistentVolumeClaim size *
30
GiB
Type
Disk
Hot plug is enabled only for "Disk" type
Interface *
VirtIO
Hot plug is enabled only for "SCSI" interface
StorageClass
SC ocs-external-storagecluster-ceph-rbd
Apply optimized StorageProfile settings
Optimized values Access mode: ReadWriteMany, Volume mode: Block.
Enable preallocation
Advanced settings
Save
Cancel


Add disk
Filter
Search by name...
Mount Windows drivers disk
Name
Source
Size
Drive
Interface
Storage class
cloudinitdisk
Other
-
Disk
virtio
-
rootdisk
bootable
URL
30 GiB
Disk
virtio
ocs-external-storagecluster-ceph-rbd

Add password and ssh detail from cloud-init script



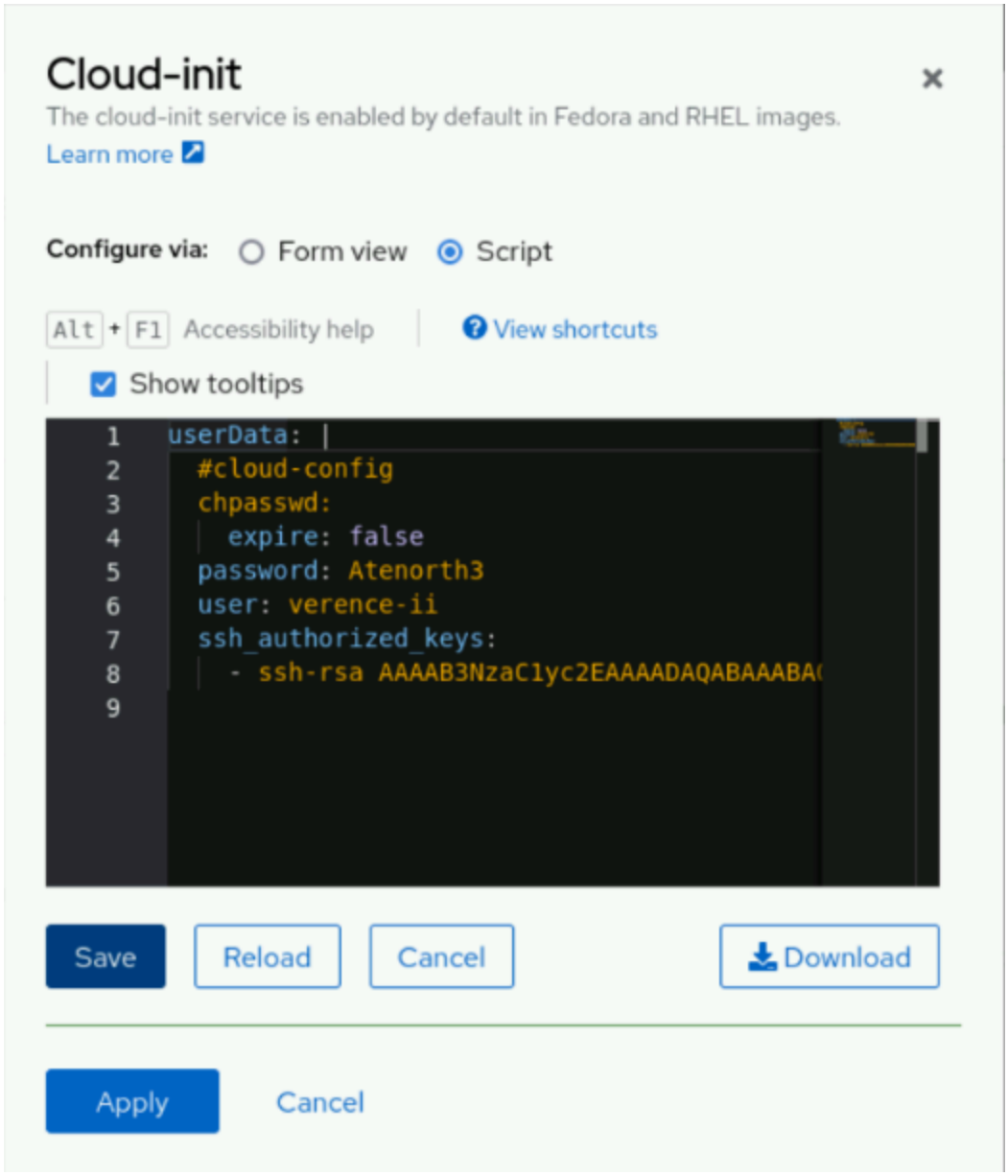
→ Here mention your username and password and click on apply

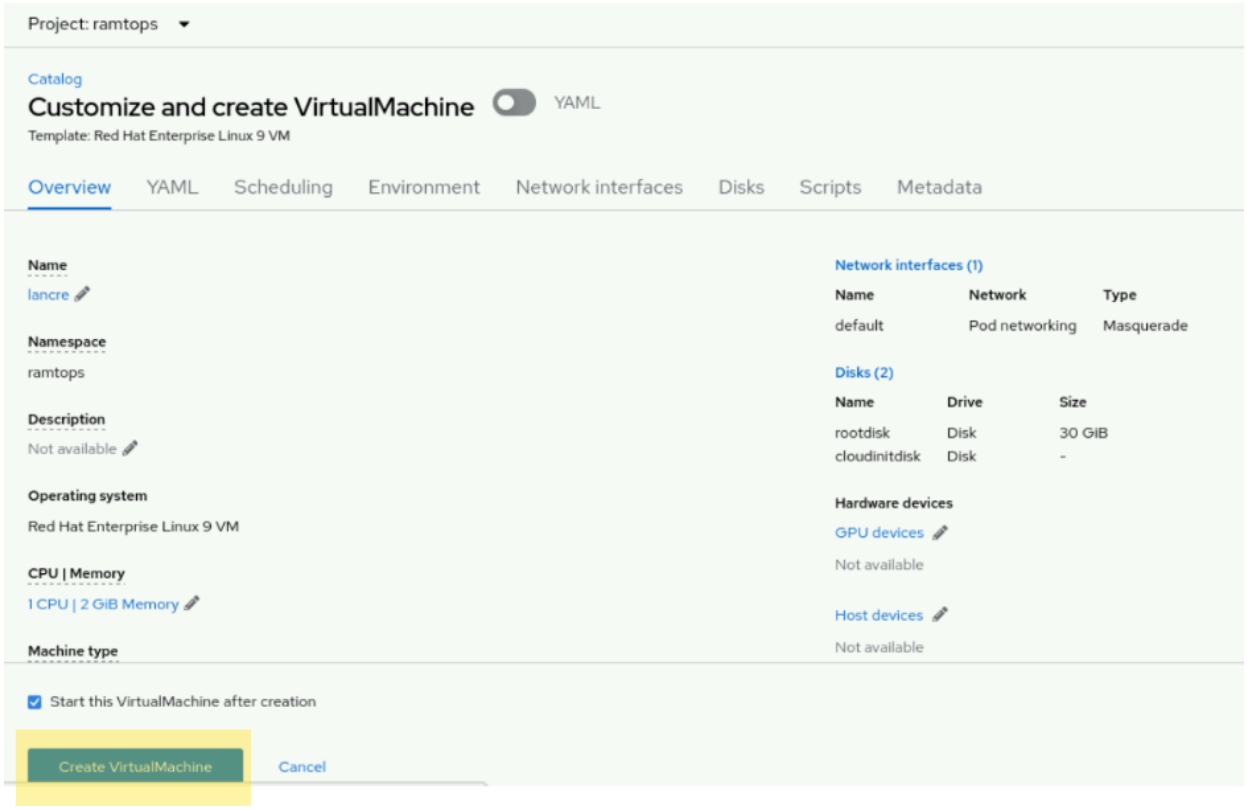


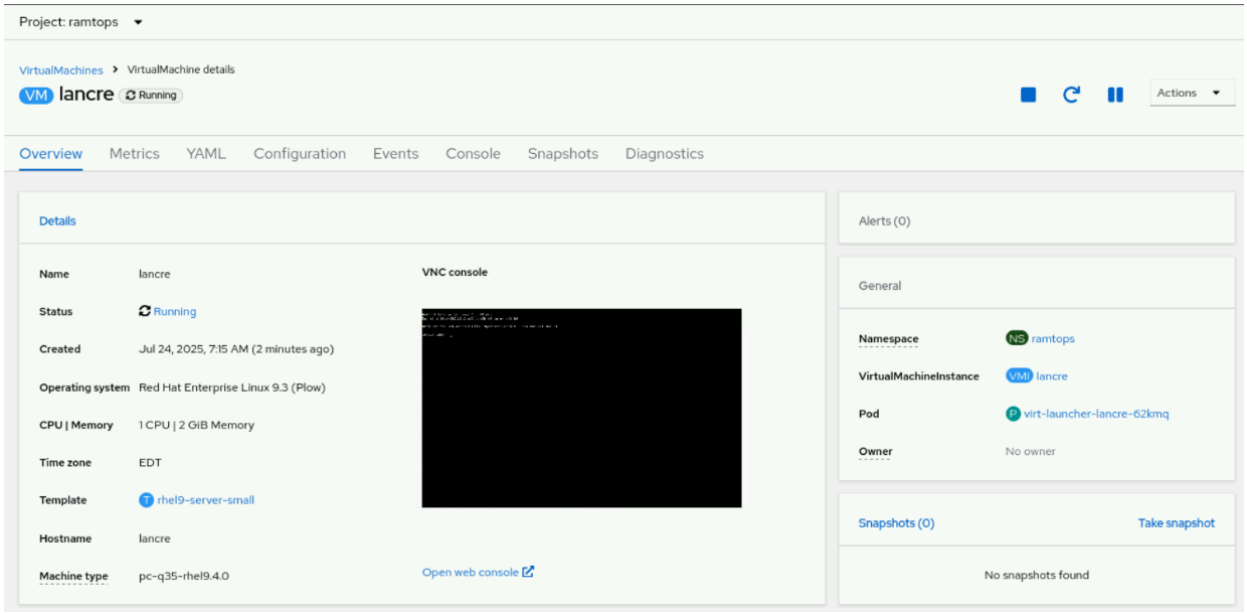
→ now copy the ssh private key

```
[student@workstation ~]$ cat ~/.ssh/lab_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDGtUW3ismHyuCW4CDdTVOOOq6aySdtYenXFWWx7HJa4VTepkG00aaLI
d9ocra10hc+MB0GTJMCyabDv3i8NKdi6GDH/aOLVsp/Ewy8DEzZMBlJDCt4v2i4/wU4liw6KgEFkZs+5hnqU8d4QzldyG
J5onr+AGvFOKG68CS0BBl40Z1twf1HhCyx8k6nzD2ovlkxWRFZKPAFrtPCBVvQDkOfVFZF+lwzaSztgAjbFZ4A9jqQyUY
x4kOJ5DtRef36ucdUdVQale0+8lICl7/gb142SPpYfhxe88/BJScLPRjvVNeu1TxRmoHtVazqnAoRxQYAn2MoI6AG+w6Q
uZf8f7aL LabGradingKey
[student@workstation ~]$
```

→ again click on edit from cloud-init section in webUI and this time click on script, and add details like below image, click on save then click on apply


Cloud-init
The cloud-init service is enabled by default in Fedora and RHEL images.
Learn more
Configure via:
Form view
Script
Alt + F1 Accessibility help
View shortcuts
Show tooltips
userData: |
#cloud-config
chpasswd:
expire: false
password: Atenorth3
user: verence-ii
ssh_authorized_keys:
- ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABA
Save
Reload
Cancel
Download
Apply
Cancel





→ verify from your console, it should logged in

```
[student@workstation ~]$ oc get vmi
NAME          AGE    PHASE     IP           NODENAME   READY
lancre        96s    Running   10.9.0.34    master01   True
vm1-testing   150m   Running   10.8.1.225   master03   True
[student@workstation ~]$ virtctl ssh -i ~/.ssh/lab_rsa verence-ii@lancre
Warning: Permanently added 'vmi/lancre.apple' (ED25519) to the list of known hosts.
Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
[verence-ii@lancre ~]$
```

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Q-5 - Configure a web server in a virtual machine

Configure a web server in the **lancre** virtual machine so that the following conditions are true:

- The **httpd** package that provides the web server is installed in the virtual machine.

  You can download a file <link of repo> into **/etc/yum.repos.d** on the virtual machine to configure repositories

- The httpd service is enabled and started automatically during the boot process of the virtual machine
- The file <link of html file> exits in the path /var/www/html of the virtual machine
- The web server serves the file **service.html** to pods in the **ramtops** project
- A network policy named **allow-webserver** exists in the **ramtops** project

- A ClusterIP service that allows web traffic into the **lancre** virtual machine, and the **ramtops** project must not have any other ClusterIP service either than this one
- The network policy restricts access to the virtual machine **lancre** allowing only the members of project **ramtops** to access TCP port 80
- Other projects cannot reach the virtual machine **lancre** at TCP port 80

NOTES:

- The virtual machine will be rebooted before grading.
- The sample yaml files are available from <link for sample yal file>. You will need to replace any string marked with an @ sign with the correct value for the purpose.
- The container image <link of container image> is available for testing purposes.
- When creating Network Policies through the GUI, you may see the error message "The server does not allow the method on the requested sources" You will still create the network Policy to create the appropriate objects."

ANS:

- Login to virtual machine from webconsole using given user or execute "virtctl console lancre" (make sure you have logged in console or in webconsole using "Verence-ii" user
- Execute sudo -i cmd to become root
- Change into directory /etc/yum.repos.d
- Execute "curl -LO <repo link>
- Verify repo using "yum repolist all" cmd
- Execute below cmds
- yum install httpd -y
- Systemctl enable –now httpd

- Execute curl http://localhost/service.html cmd and verify you can access the page which is same as /var/www/html/service.html page
- Now add a label to the vm using below cmd
- Oc edit vm
- Got to .spec.template.lable and assign lable of your choice example app: lancre
- Save it and restart it using virtctl restart lancre cmd
-

```
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      app: lancre
      kubevirt.io/domain: lencre
      kubevirt.io/size: small
      network.kubevirt.io/headlessService: headless
```

```
[student@workstation ~]$ oc get vmi --show-labels
NAME     AGE     PHASE     IP           NODENAME   READY   LABELS
lancre   4m53s   Running   10.10.0.24   master01   True    app=lancre,kubevirt.io/domain=l
encre,kubevirt.io/nodeName=master01,kubevirt.io/size=small,network.kubevirt.io/headlessSer
vice=headless
```

- Create clusterip service like below

```
[student@workstation ~]$ virtctl expose vmi lancre --name lancre-svc --port 80 --target-po
rt 80 --type ClusterIP
Service lancre-svc successfully created for vmi lancre
[student@workstation ~]$ oc get svc
NAME          TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
headless      ClusterIP   None             <none>        5434/TCP   19m
lancre-svc    ClusterIP   172.30.190.197   <none>        80/TCP     4s
[student@workstation ~]$ oc get endpoints
NAME          ENDPOINTS          AGE
headless      10.10.0.24:5434    19m
lancre-svc    10.10.0.24:80      9s
[student@workstation ~]$
```

- Test the service using creating a temp pod using the link give in question paper

```
[student@workstation ~]$ oc run test-pod --image registry.ocp4.example.com:8443/ubi9/ubi --command sleep infinity
pod/test-pod created
[student@workstation ~]$ oc get pods
NAME                          READY   STATUS    RESTARTS   AGE
test-pod                      1/1     Running   0          14s
virt-launcher-lancre-zxbwr    1/1     Running   0          11m
```

```
[student@workstation ~]$ oc exec -it test-pod -- /bin/bash
bash-5.1$ curl http://lancre-svc/service.html
<!DOCTYPE html>
<html>
    <head>
        <title>Hello, World!</title>
    </head>
    <body>
        <p>Hello, World! Welcome to Red Hat Training.</p>
    </body>
</html>
```

- For network policy, check the labels of project ramtops using “oc get project ramtops –show-lables” cmd and create the network policy like below

```
[student@workstation ~]$ oc get namespaces ramtops --show-labels
NAME      STATUS   AGE   LABELS
ramtops   Active   28h   kubernetes.io/metadata.name=ramtops,pod-security.kubernetes.io/audit-version=v1.24,pod-security.kubernetes.io/audit=restricted,pod-security.kubernetes.io/warn-version=v1.24,pod-security.kubernetes.io/warn=restricted
```

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-webservers
spec:
  podSelector:
    matchLabels:
      app: lancre
  policyTypes:
  - Ingress
  ingress:
    - from:
      - namespaceSelector:
          matchLabels:
            kubernetes.io/metadata.name: ramtops
        podSelector: {}
      ports:
        - protocol: TCP
          port: 80
```

- Apply it using oc apply -f <file name>

```
[student@workstation ~]$ oc get netpol
NAME               POD-SELECTOR   AGE
allow-webservers   app=lancre     4s
[student@workstation ~]$ oc describe netpol allow-webservers
Name:         allow-webservers
Namespace:    ramtops
Created on:   2026-03-26 00:50:19 -0400 EDT
Labels:       <none>
Annotations:  <none>
Spec:
  PodSelector:     app=lancre
  Allowing ingress traffic:
    To Port: 80/TCP
    From:
      NamespaceSelector: kubernetes.io/metadata.name=ramtops
      PodSelector: <none>
  Not affecting egress traffic
  Policy Types: Ingress
```

- After applying policy check from the test pod, you should access it

PLEASE NOTE: make sure to check the network policy name if they asked create using same name and check the labels of namespace, and application what you have created, define proper labels.

Also, in notes sections they have provided image to create pod for testing. Create pod in different namespace and try to access your application from that pod, it should not accessible. You can create pod before creating network policy to check its result.

==========

Q- 6 Deploy a multihomed virtual machine

Deploy a **Red Hat Enterprise Linux 9.2** virtual machine with two network interfaces so that the following conditions are true

- The Project is **uberwald**
- The virtual machine name is **lipwig**
- User **moist** Create the virtual machine **lipwig**
- The workload type of the virtual machine is **server**
- The flavor for the virtual machine is **small**
- The user **moist** with password **Atenorth3** exists in the **cloud-init** definition
- The ssh key **/home/opsadm/id_rsa_ex316.pub** from user **opsadm** at **workbench.domain3.example.com** has been added as an authorized SSH key via the **cloud-init** definition

Storage configuration

- The image used to create the persistent volume claim for the virtual machine boot source is : **http://utility.lab.example.com:8080/openshift4/images/rhel-9-hello world.qcow2**
- The storageClassName is : **ocs-external-storagecluster-ceph-rbd**
- The persistent volume claim size is 30Gi

The first network interface configuration:

- The first network interface name is **default**
- The first network interface is attached to the **pod Networking (default)** network
- The first network interface is **virtio**

The second network interface configuration:

- The second network interface name is **nic-0**
- The second network interface is attached to the **ankh-morpork** network
- The second network interface type is bridge

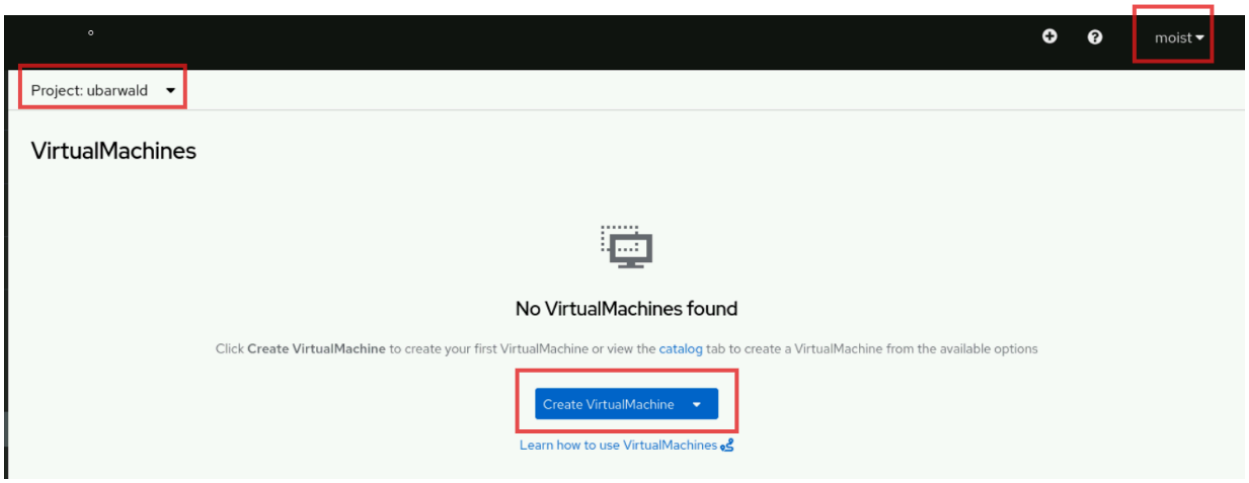
- The IP address of the second network interface is provided by **Openshift**
- The model for the second network interface is **virtio**

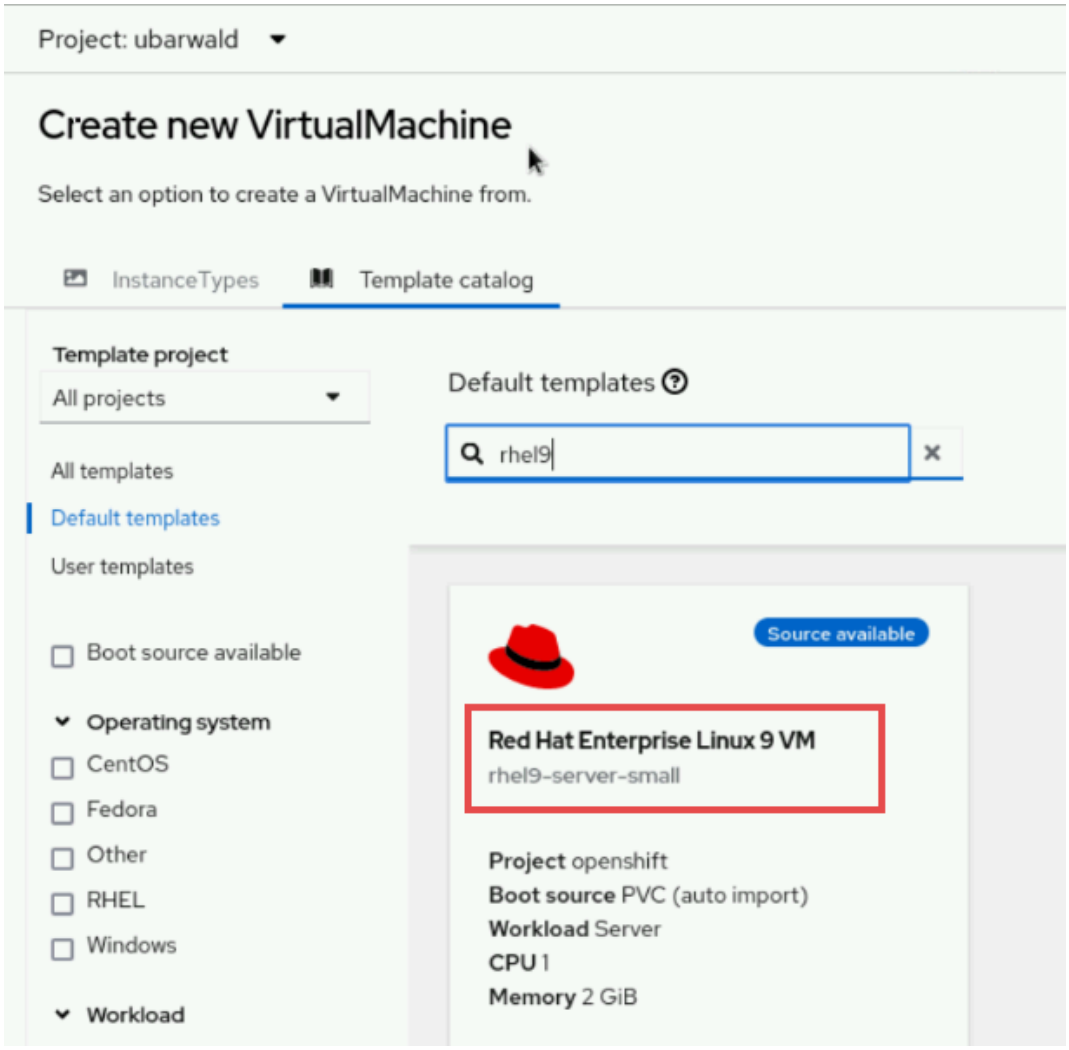
NOTE: The dialog box for the storage sizing may not function correctly, but it is set by default to 30 GiB.

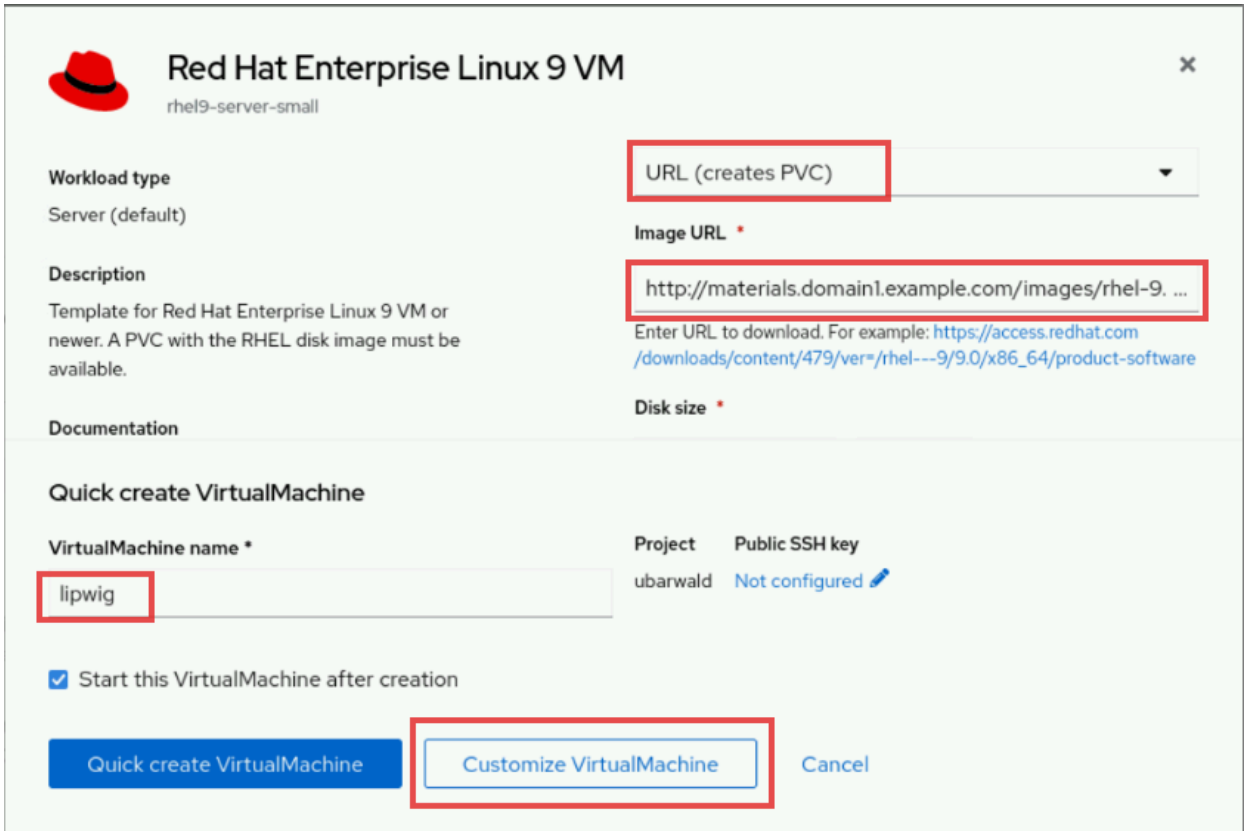
ANS:

NOTE: if you are using 4.16 lab then you may need to execute "lab start multihomed-nmstate", then only you should see ens4 interface in worker01 and worker02

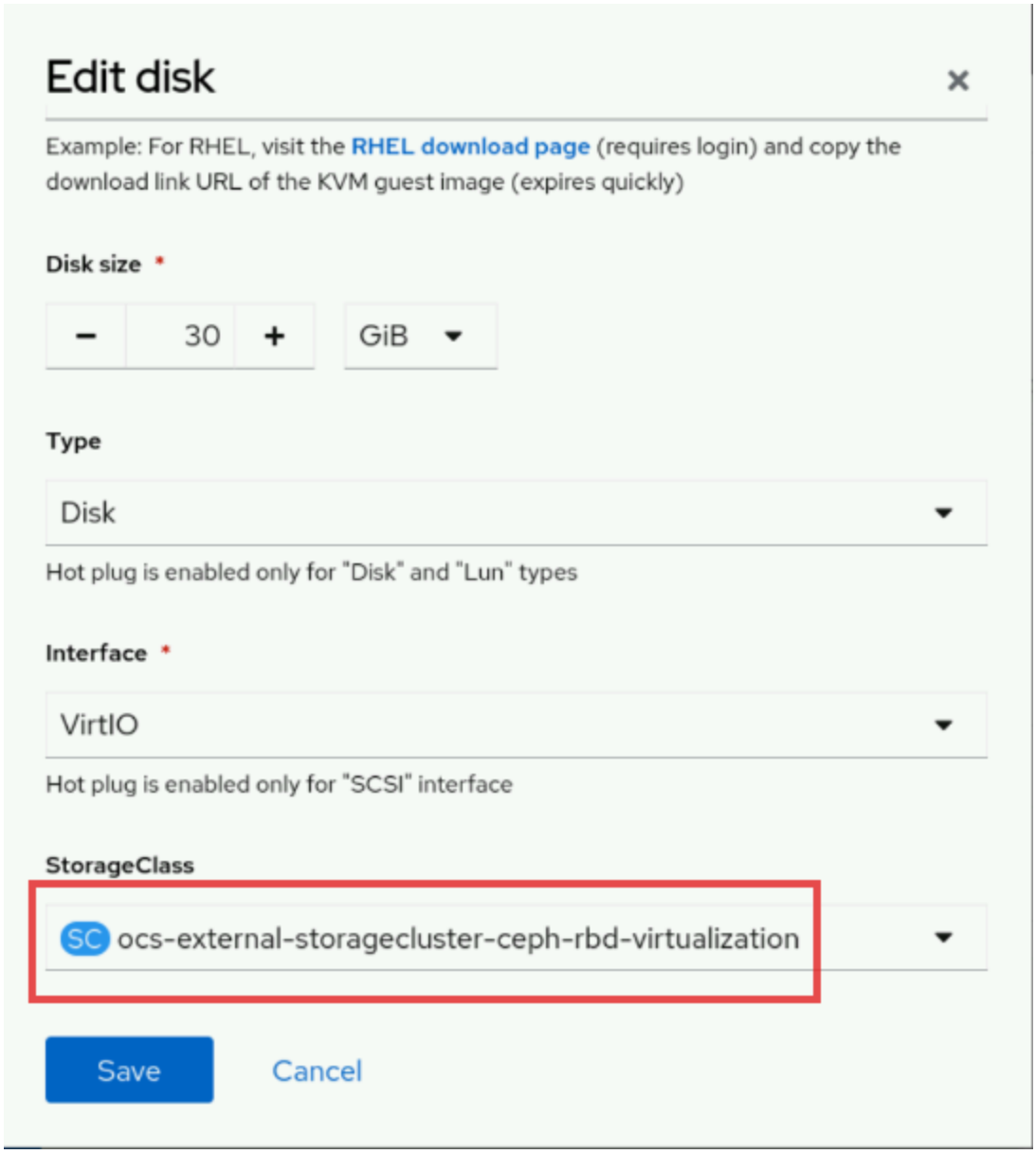
1. Make sure you are in ubarwald project and logged in as moist user and create a vm

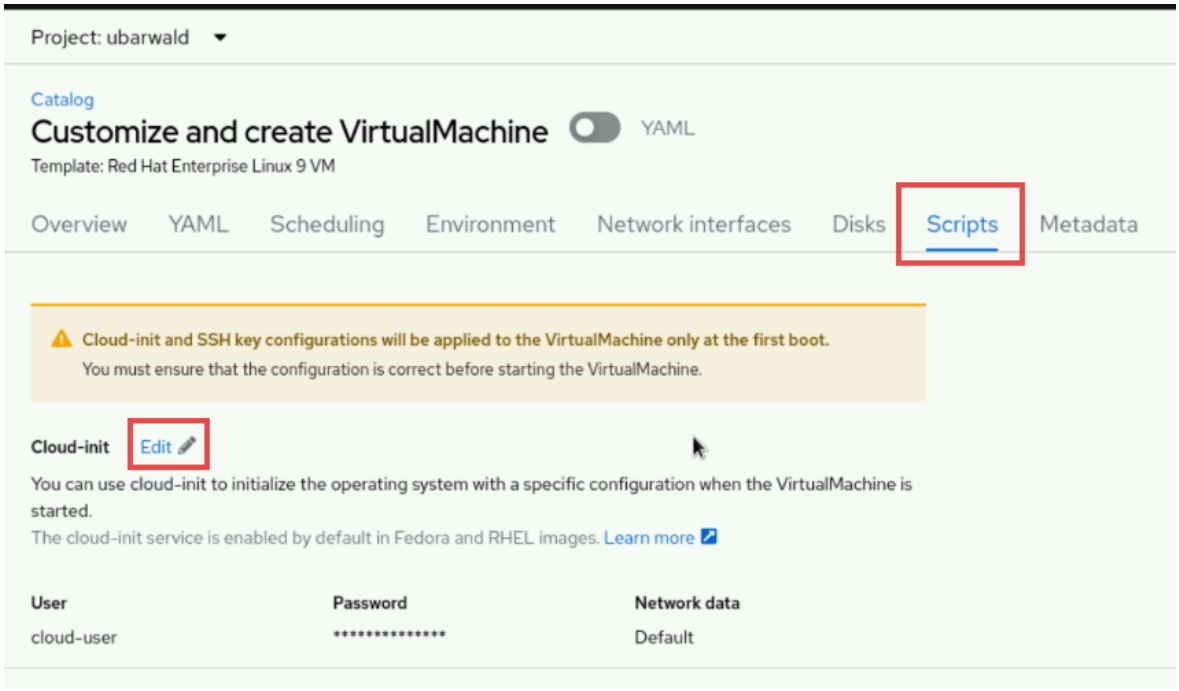



Project: ubarwald
Create new VirtualMachine
Select an option to create a VirtualMachine from.
InstanceTypes
Template catalog
Template project
All projects
All templates
Default templates
User templates
Boot source available
Operating system
CentOS
Fedora
Other
RHEL
Windows
Workload
Default templates
rhel9
Source available
Red Hat Enterprise Linux 9 VM
rhel9-server-small
Project openshift
Boot source PVC (auto import)
Workload Server
CPU 1
Memory 2 GiB

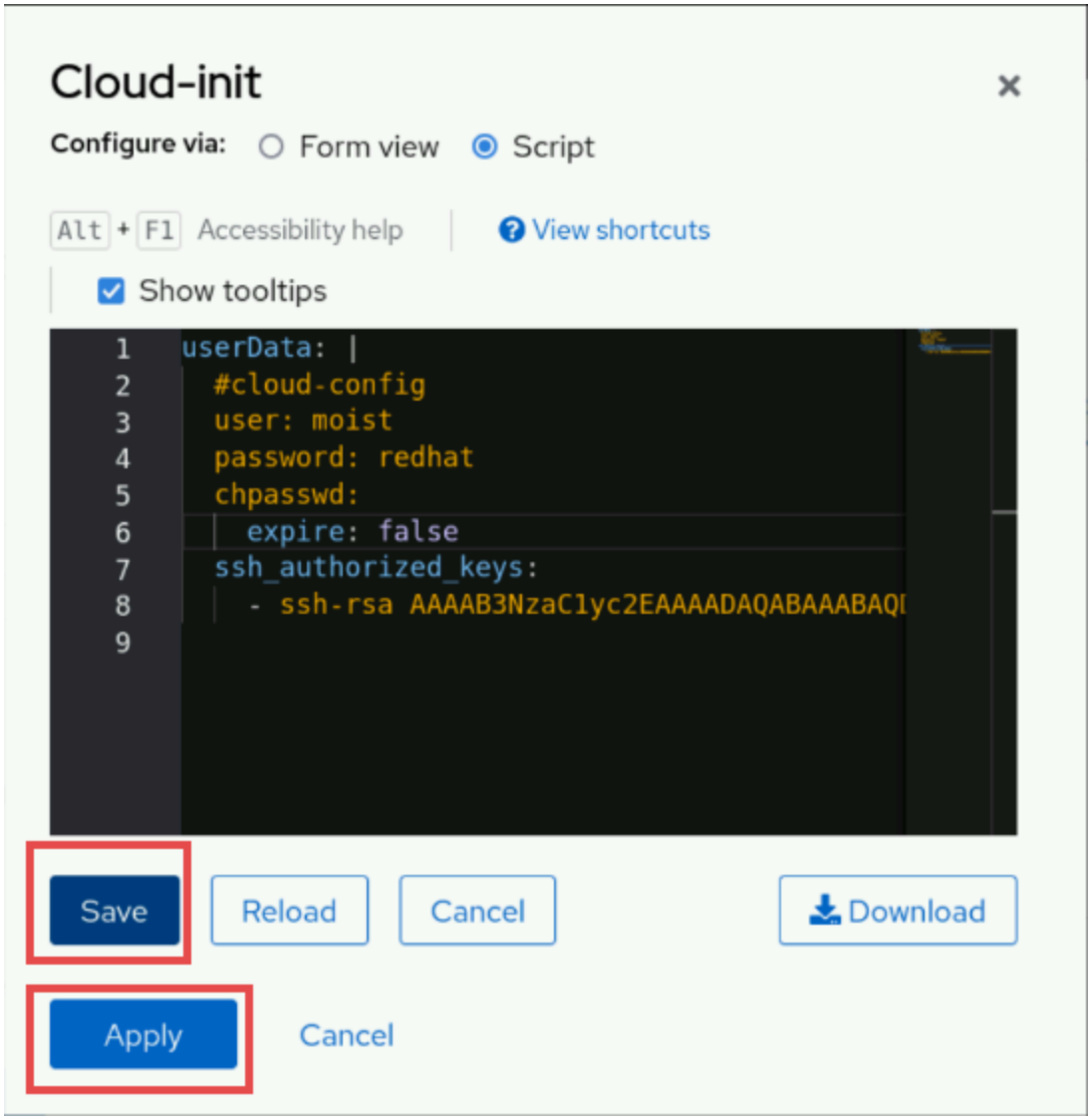


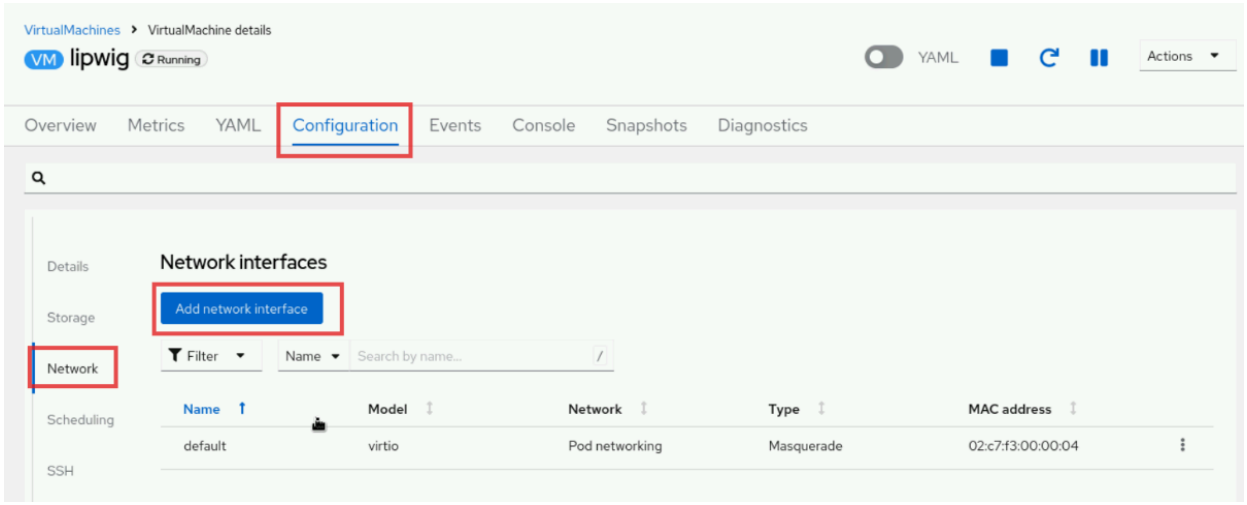
→ make sure your disk has storage class as below, if not click on edit and change the storage class

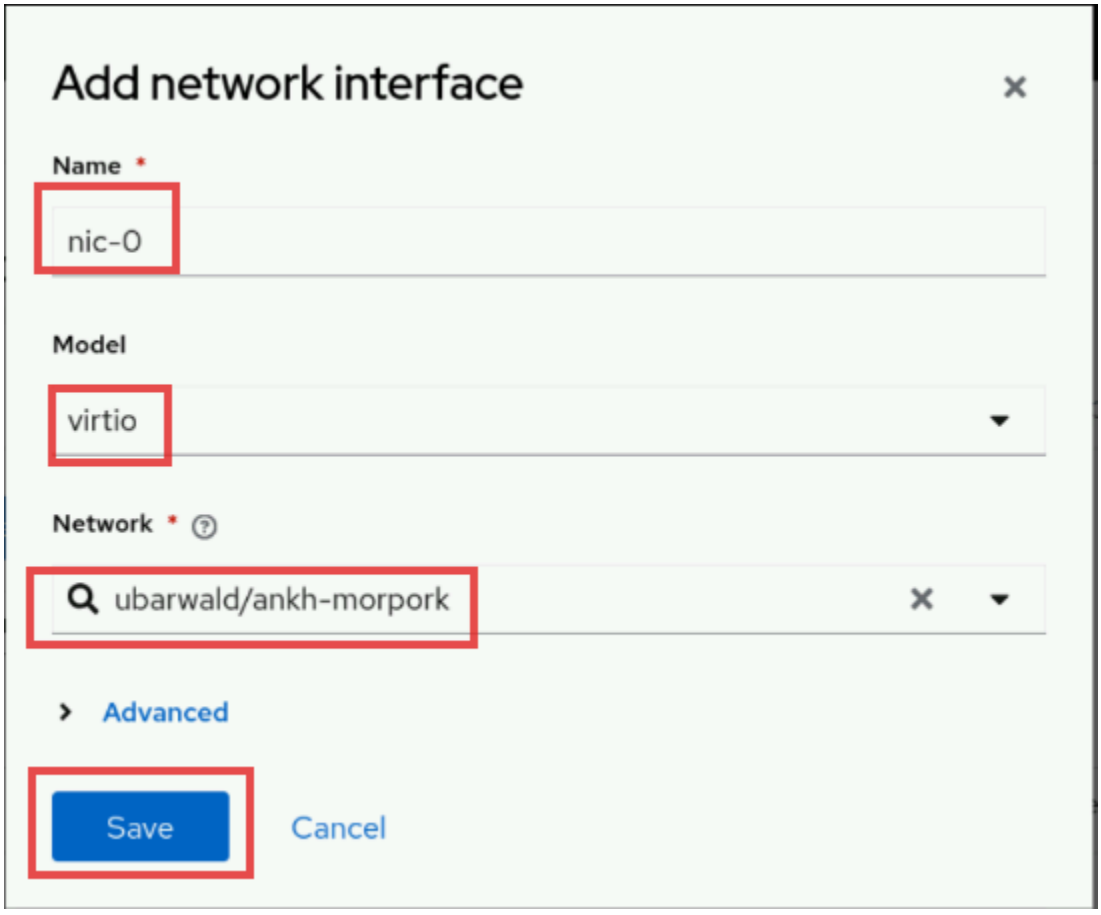

Edit disk
Example: For RHEL, visit the RHEL download page (requires login) and copy the download link URL of the KVM guest image (expires quickly)
Disk size *
30
GiB
Type
Disk
Hot plug is enabled only for "Disk" and "Lun" types
Interface *
VirtIO
Hot plug is enabled only for "SCSI" interface
StorageClass
SC ocs-external-storagecluster-ceph-rbd-virtualization
Save
Cancel

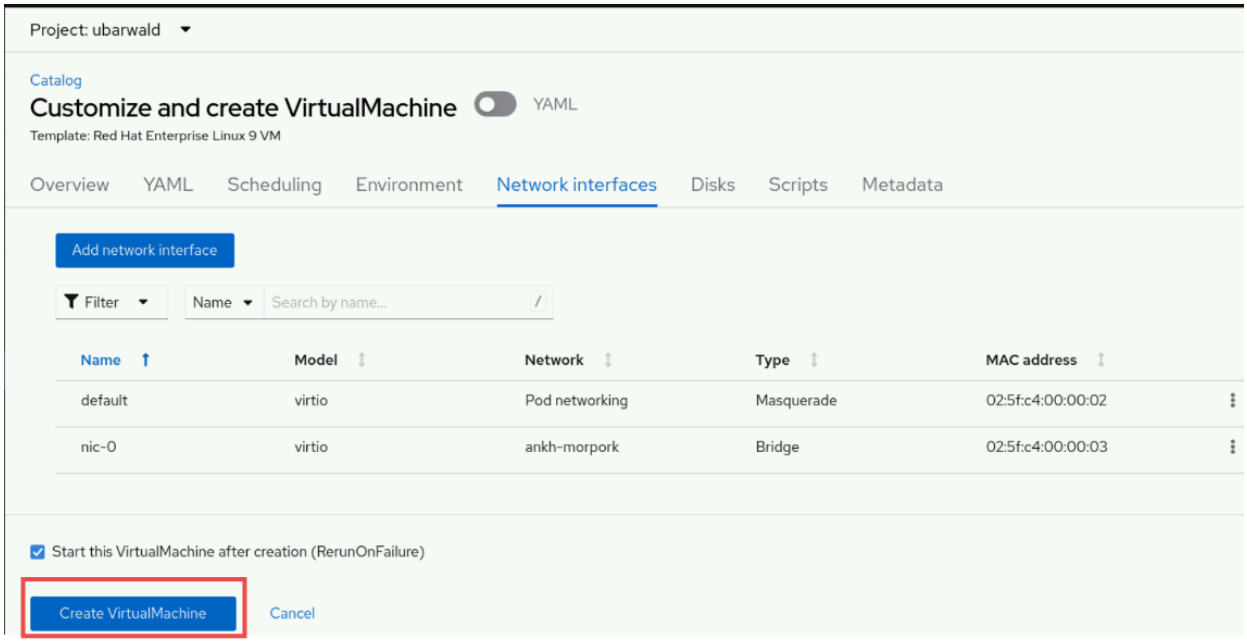


- Copy the ssh key from opsadm machine

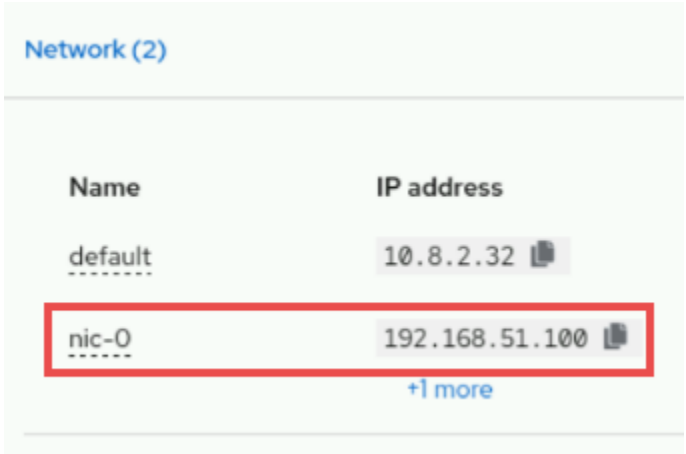

Cloud-init
Configure via:
Form view
Script
Alt + F1 Accessibility help
View shortcuts
Show tooltips
1 userData: |
2 #cloud-config
3 user: moist
4 password: redhat
5 chpasswd:
6 expire: false
7 ssh_authorized_keys:
8 - ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQ
9
Save
Reload
Cancel
Download
Apply
Cancel


VirtualMachines
VirtualMachine details
VM
lipwig
Running
YAML
Actions
Overview
Metrics
YAML
Configuration
Events
Console
Snapshots
Diagnostics
Details
Storage
Network
Scheduling
SSH
Network interfaces
Add network interface
Filter
Name
Search by name...
Name
Model
Network
Type
MAC address
default
virtio
Pod networking
Masquerade
02:c7:f3:00:00:04

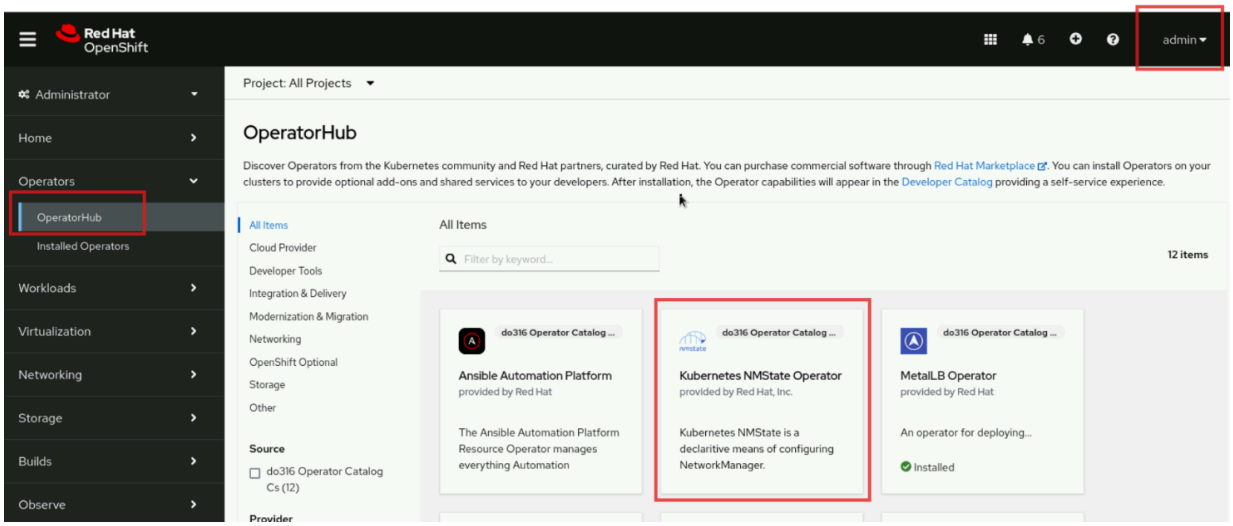

Add network interface
Name *
nic-0
Model
virtio
Network *
ubarwald/ankh-morpork
Advanced
Save
Cancel

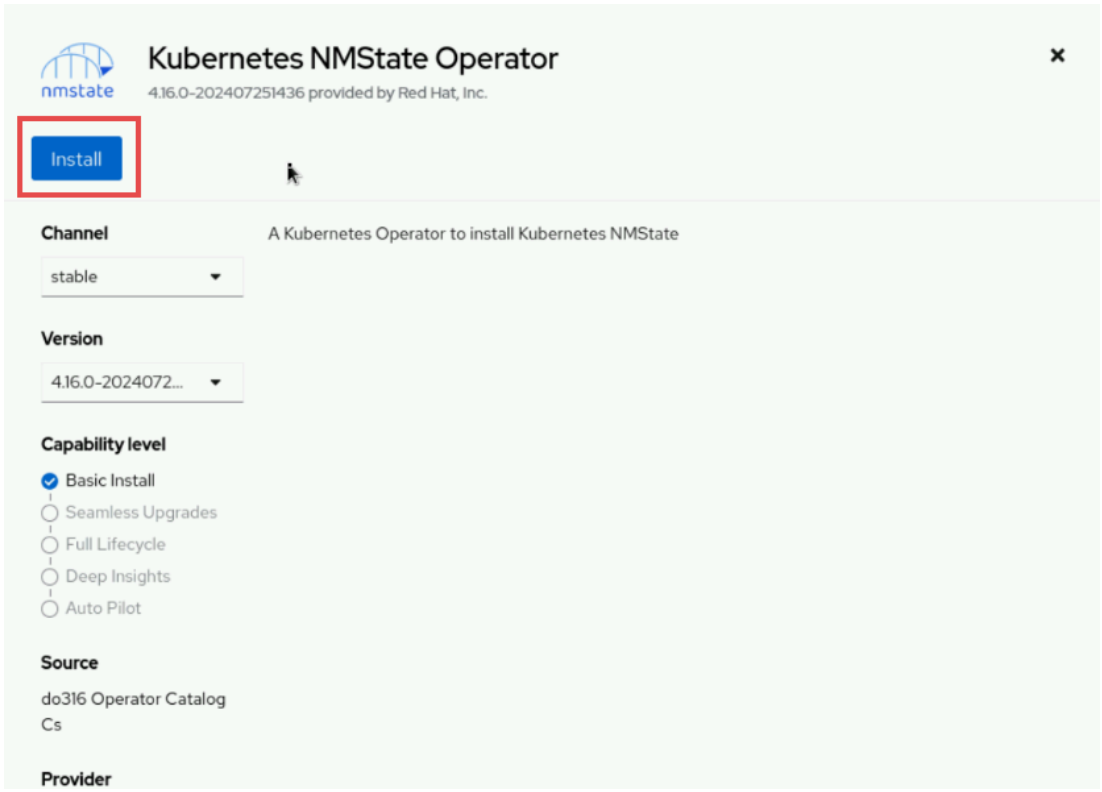


- Once machine created, check into network status, it should shows two interface and an ip add

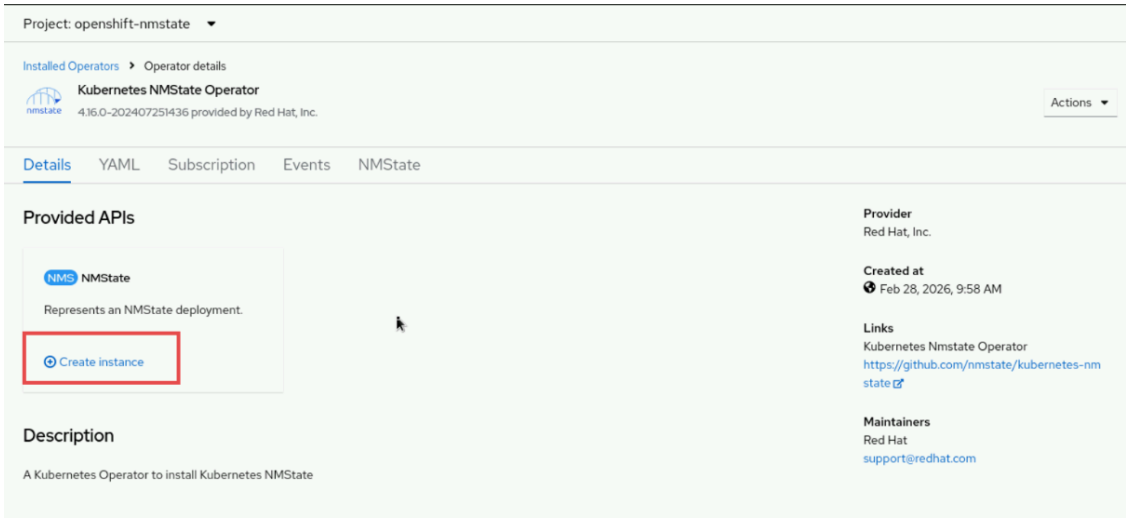


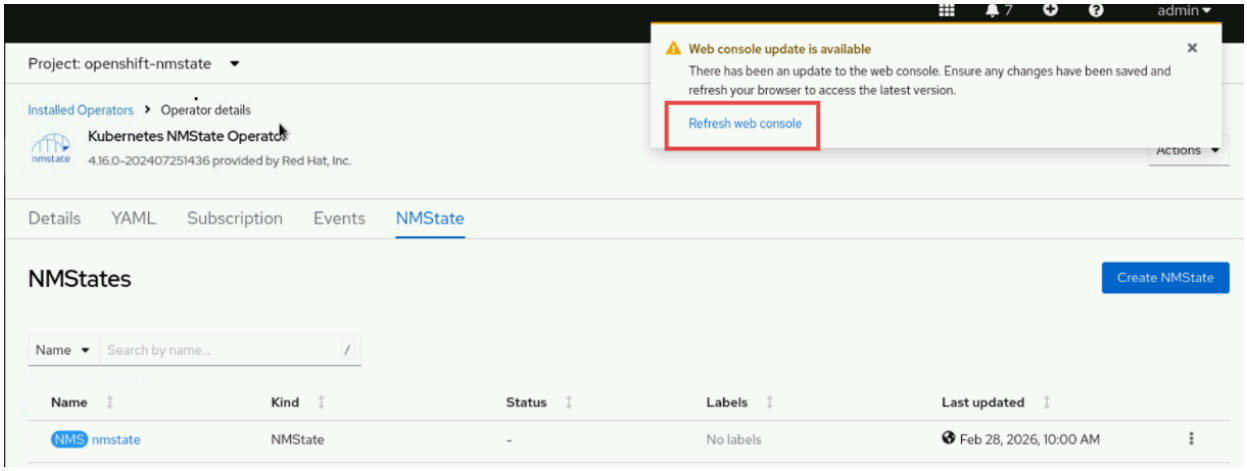
If you not able to see the ankh-morpork network in network setting then do following:



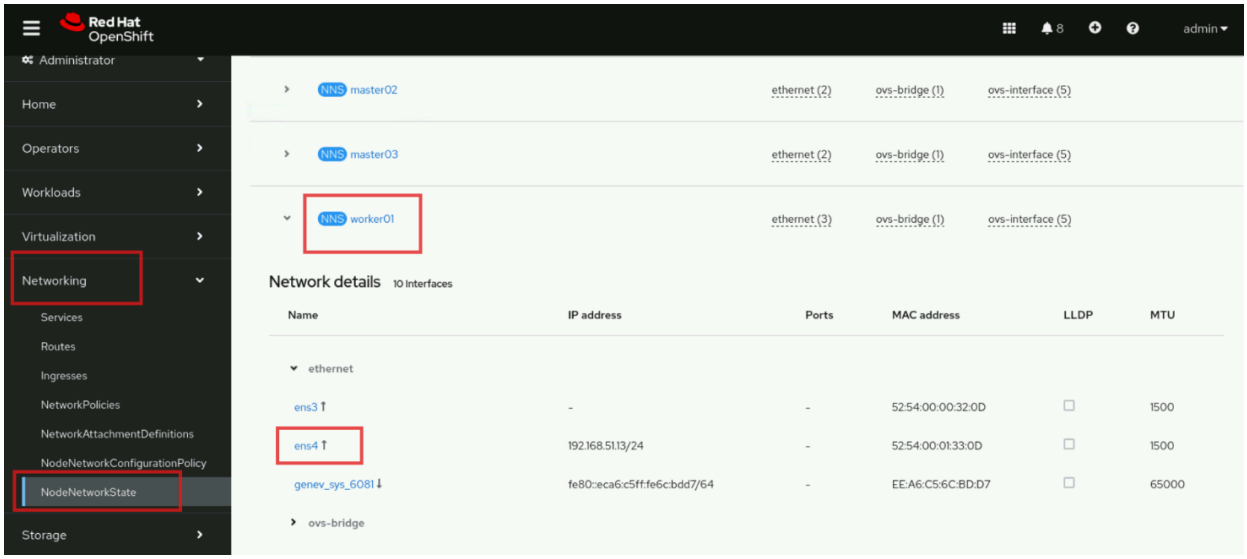


- Once it is installed, or if it says, it's already installed, then go to installed operator page, and open NMState operator, and click on create NMstate instance.

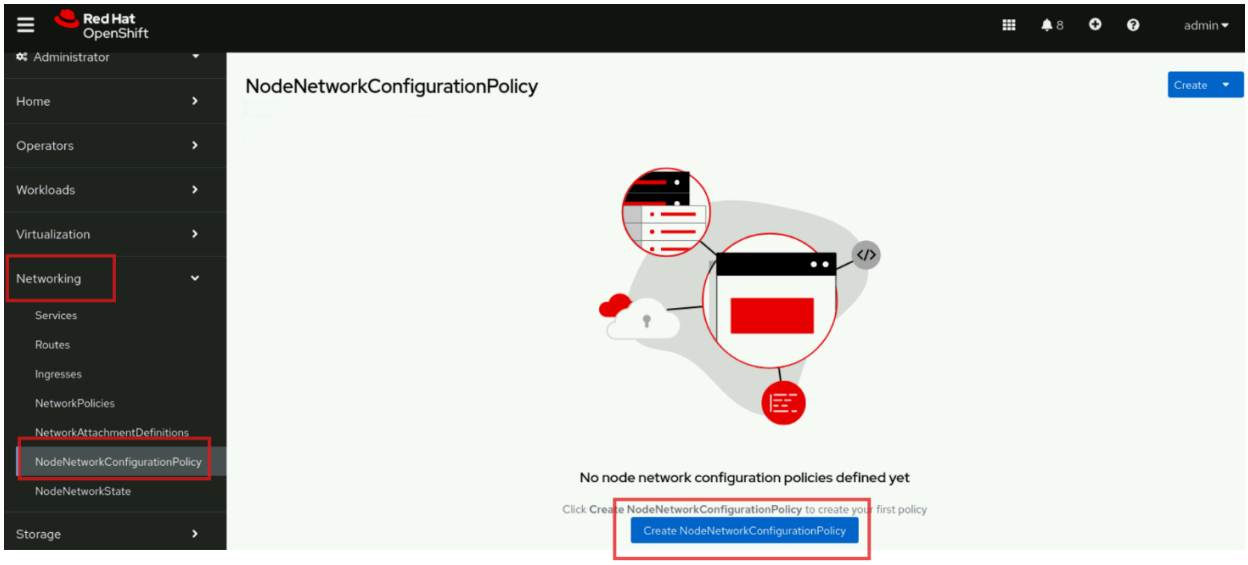


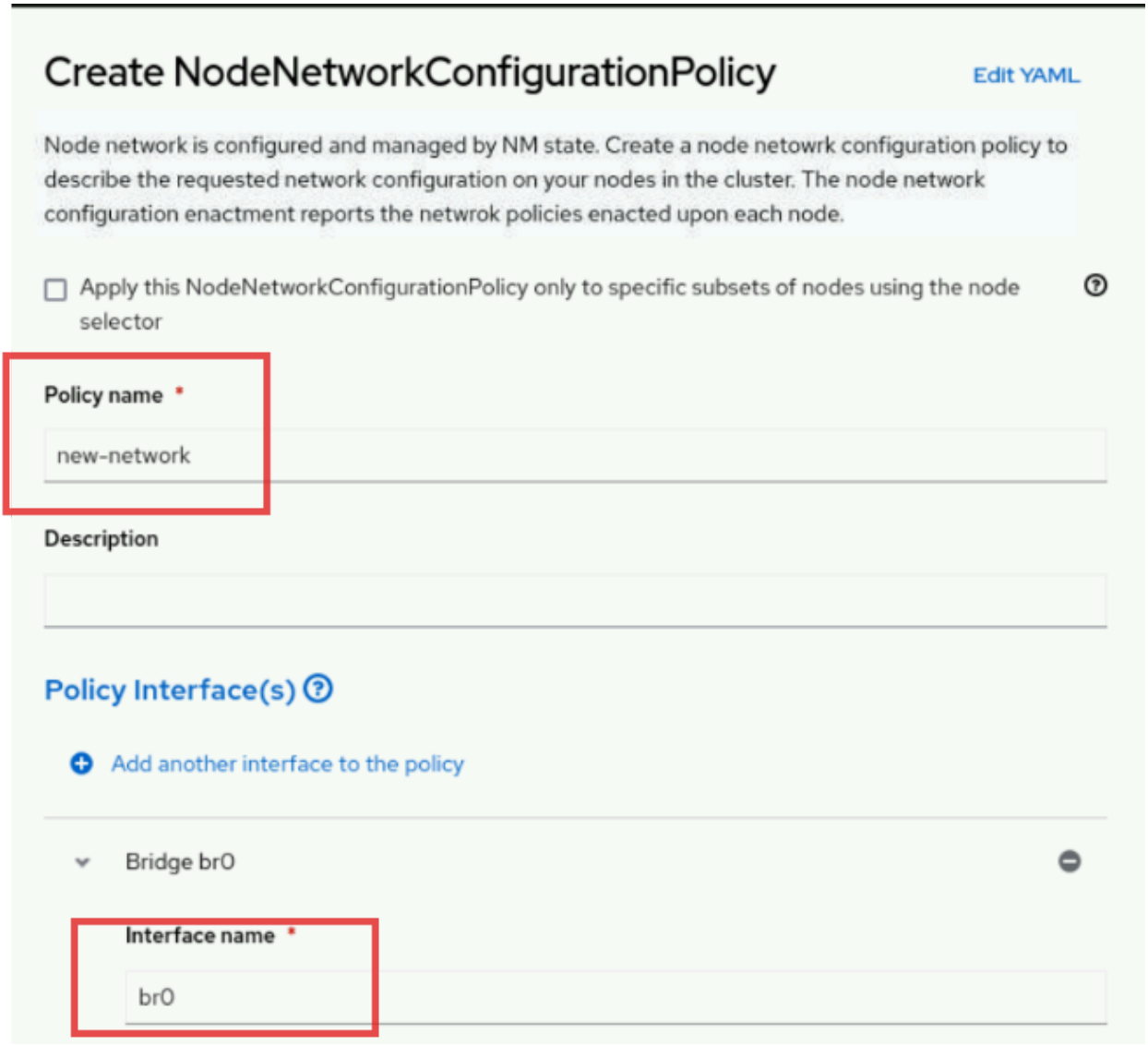


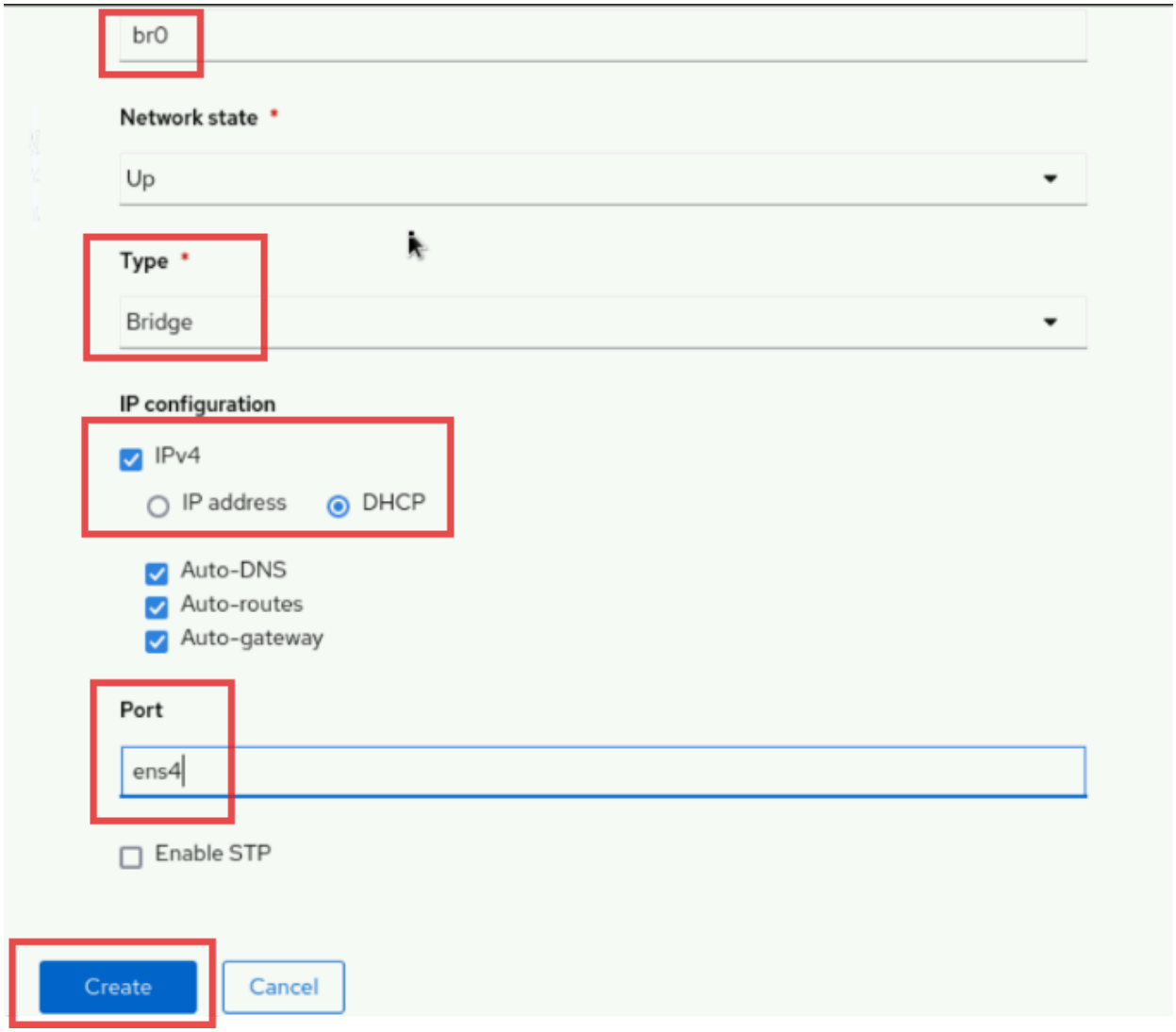
- Now go to network ⟶ NodeNetworkState ⟶ and check additional interface, in below page “ens4” is additional interface available in all machines



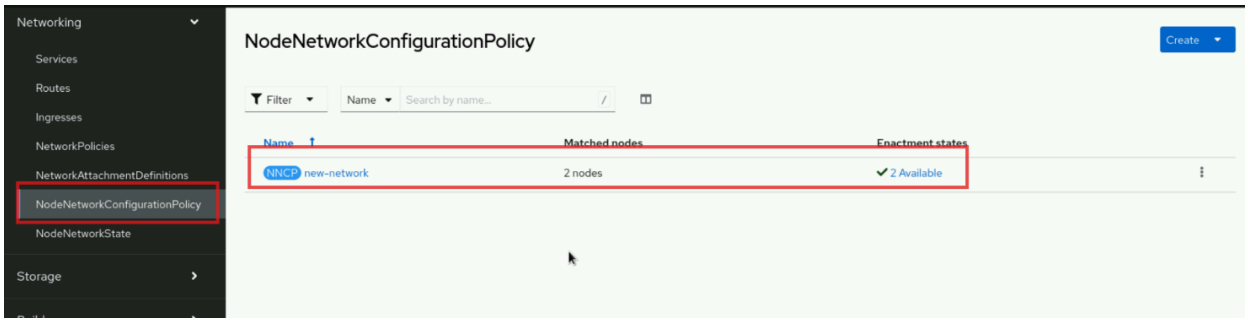
- Once verified, now go to network → NodeNetworkConfigurationPolicy → create

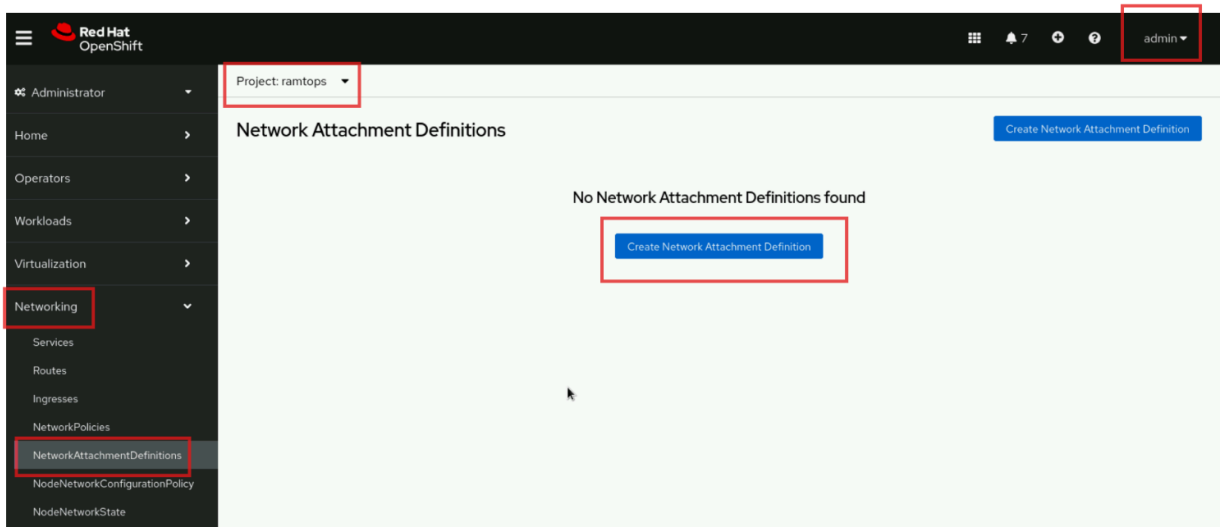


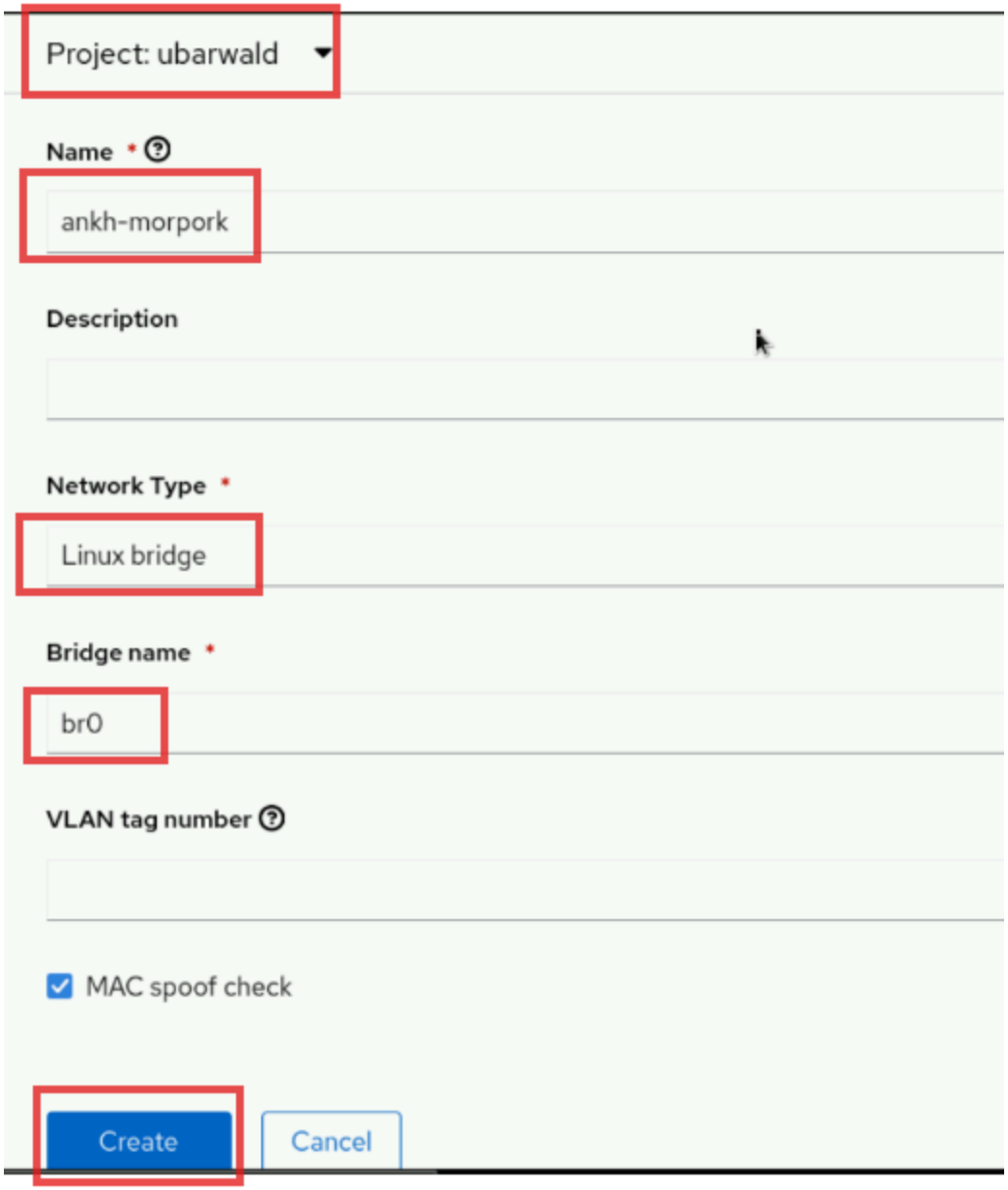

Create NodeNetworkConfigurationPolicy
Edit YAML
Node network is configured and managed by NM state. Create a node netowrk configuration policy to describe the requested network configuration on your nodes in the cluster. The node network configuration enactment reports the netwrok policies enacted upon each node.
Apply this NodeNetworkConfigurationPolicy only to specific subsets of nodes using the node selector
Policy name *
new-network
Description
Policy Interface(s)
Add another interface to the policy
Bridge br0
Interface name *
br0

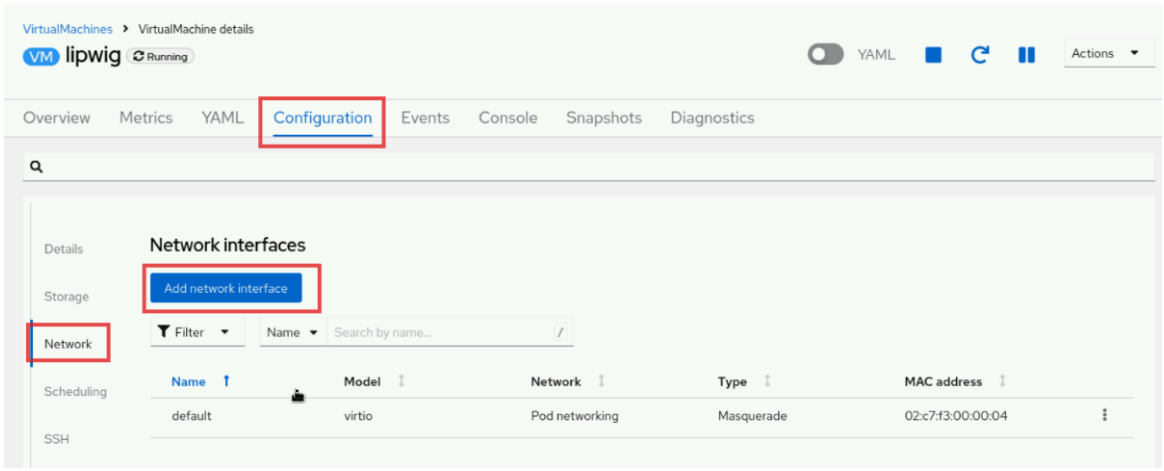


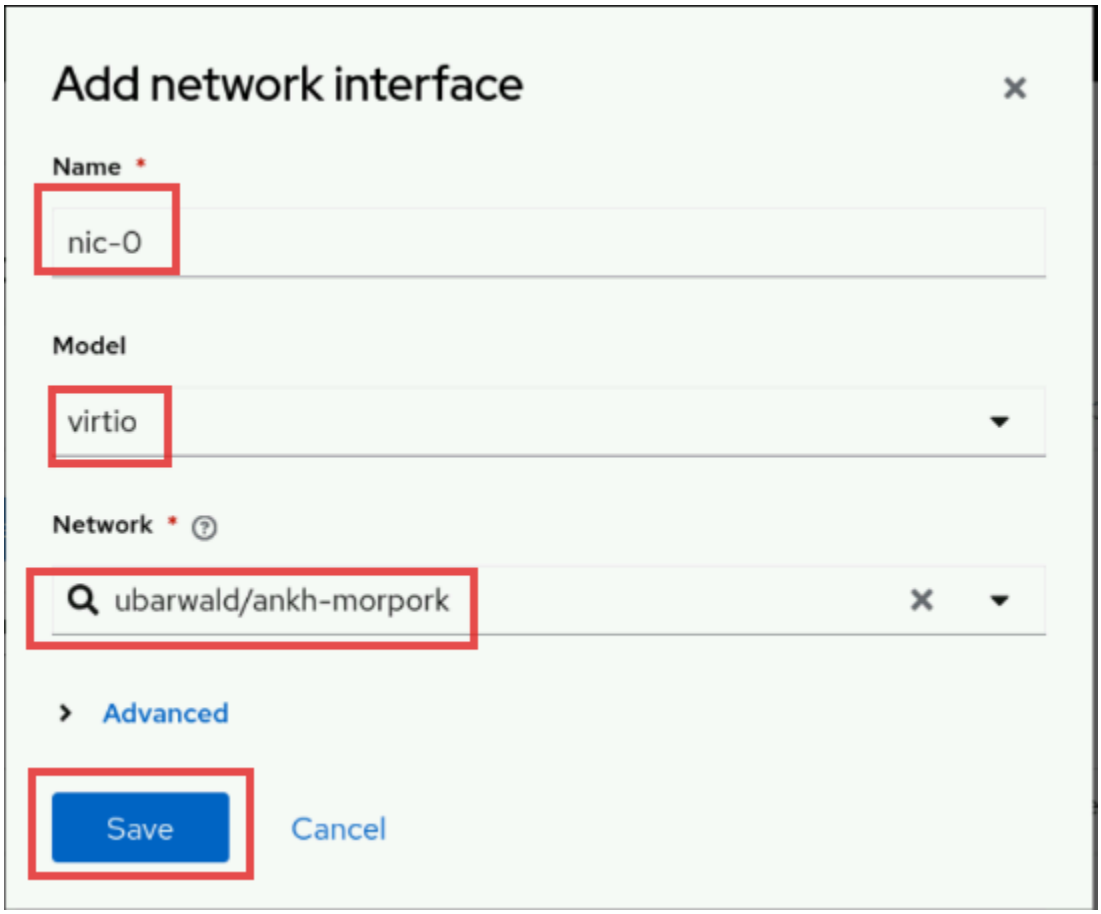
- Verify from NNCP that its available to all nodes

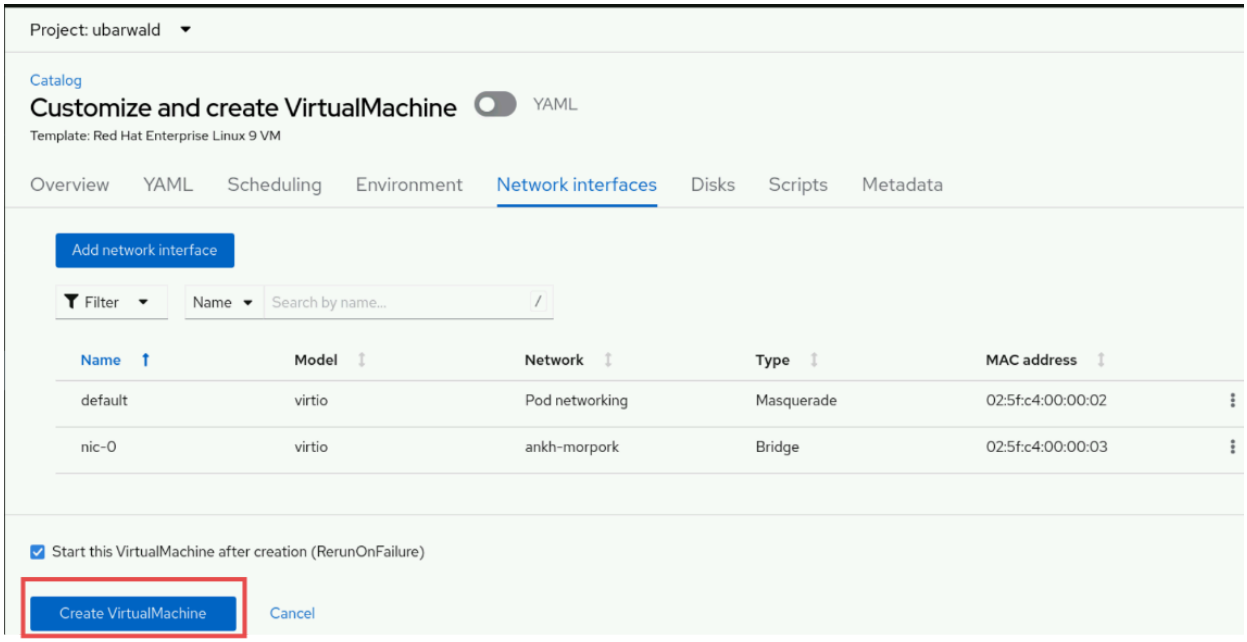


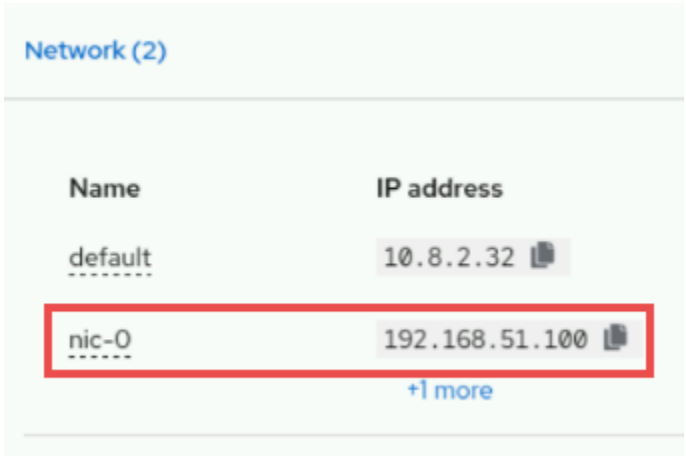

Red Hat OpenShift
7
admin
Administrator
Home
Operators
Workloads
Virtualization
Networking
Services
Routes
Ingresses
NetworkPolicies
NetworkAttachmentDefinitions
NodeNetworkConfigurationPolicy
NodeNetworkState
Project: ramtops
Network Attachment Definitions
Create Network Attachment Definition
No Network Attachment Definitions found
Create Network Attachment Definition


Project: ubarwald
Name *
ankh-morpork
Description
Network Type *
Linux bridge
Bridge name *
br0
VLAN tag number
MAC spoof check
Create
Cancel


VirtualMachines
VirtualMachine details
VM
lipwig
Running
YAML
Actions
Overview
Metrics
YAML
Configuration
Events
Console
Snapshots
Diagnostics
Details
Storage
Network
Scheduling
SSH
Network interfaces
Add network interface
Filter
Name
Search by name...
Name
Model
Network
Type
MAC address
default
virtio
Pod networking
Masquerade
02:c7:f3:00:00:04


Add network interface
Name *
nic-0
Model
virtio
Network *
ubarwald/ankh-morpork
Advanced
Save
Cancel





==========

Q-7 Manage storage for virtual machines

Create two virtual machines in the project **lonely** in the following way:

- Create a virtual machine named **dwalin** using the **dwalin-template** template

- Create a virtual machine named **fili** using the **fili-template** template
- Do not change the templates

After the virtual machines are created, perform any necessary actions on these virtual machines so that the following conditions are met:

Move the disk **gloin** from the virtual machine **dwalin** to the virtual machine **fili:**

- The disk uses the same settings as the original **gloin** on dwalin, such as storage class, access mode, etc
- The disk is a block device and is accessible via **/dev/vdc**
- The disk is mounted at **/var/www/html** on **fili** permanently

Ensure the virtual machine **dwalin** has a disk named **doril**

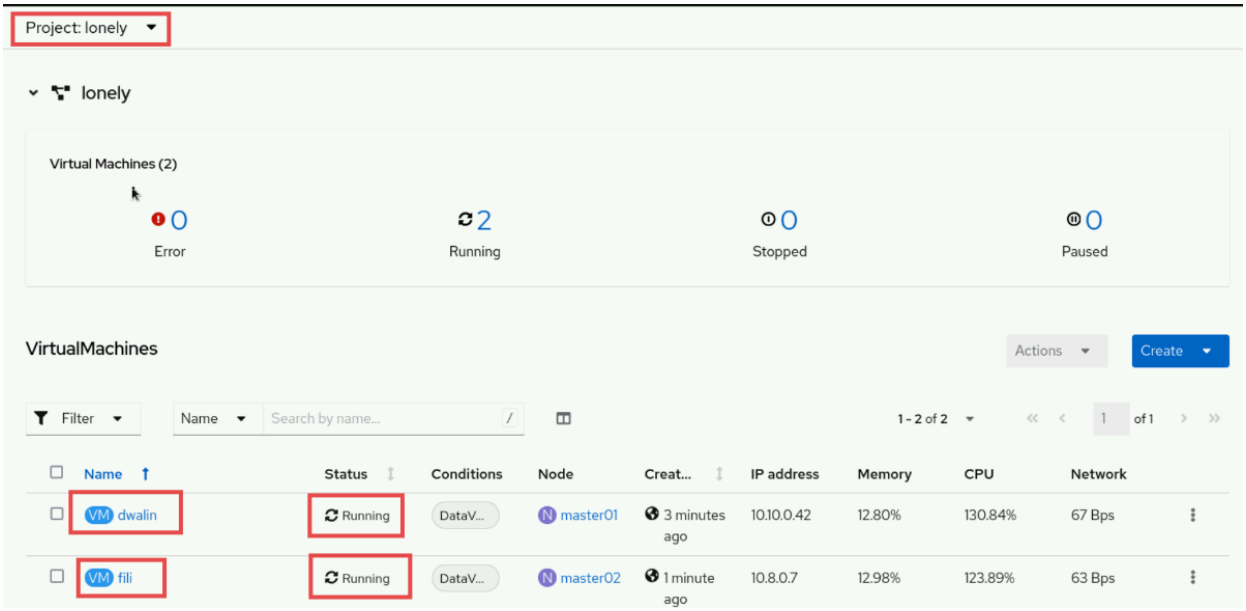
- The disk uses http://materials.domain3.example.com/images/stage-disk1-20221207.qcow2 as a source
- The disk uses the **ocs-external-storagecluster-ceph-rbd** storage class
- The disk size is **30Gi**
- The disk supports **shared access**
- The disk is a block device and is accessible via **/dev/vdc**
- The disk is mounted at **/var/www/html** on **dwalin** permanently
- Ensure the existing data on these disks is preserved
- Both virtual machines are running normally

NOTE: You can log into these virtual machines are the user **gandalf** with the password Atemorth3

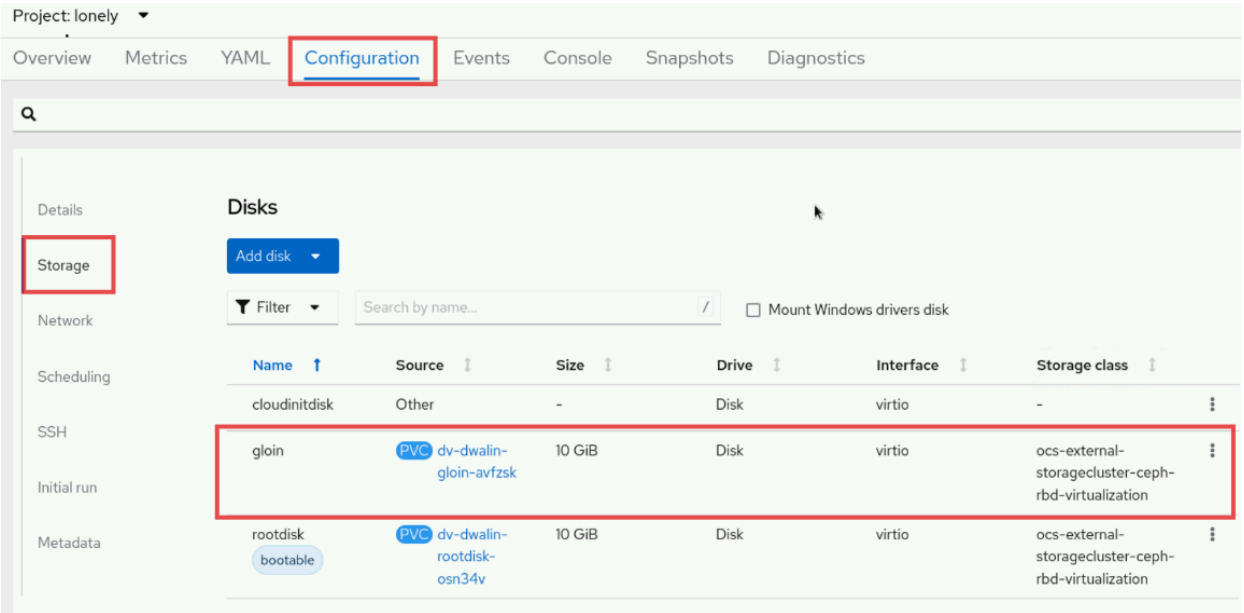
NOTE : The dialog box for the storage sizing may not function correctly , but it is default to 30 GiB

ANS:

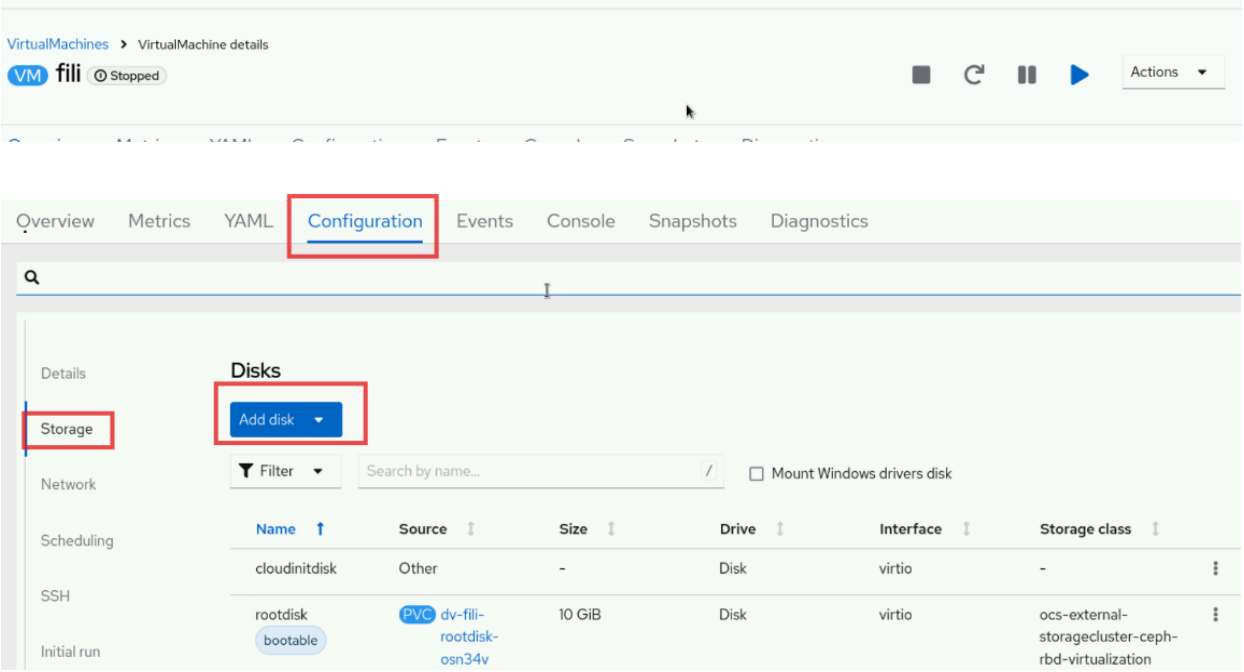
- Create vm using template

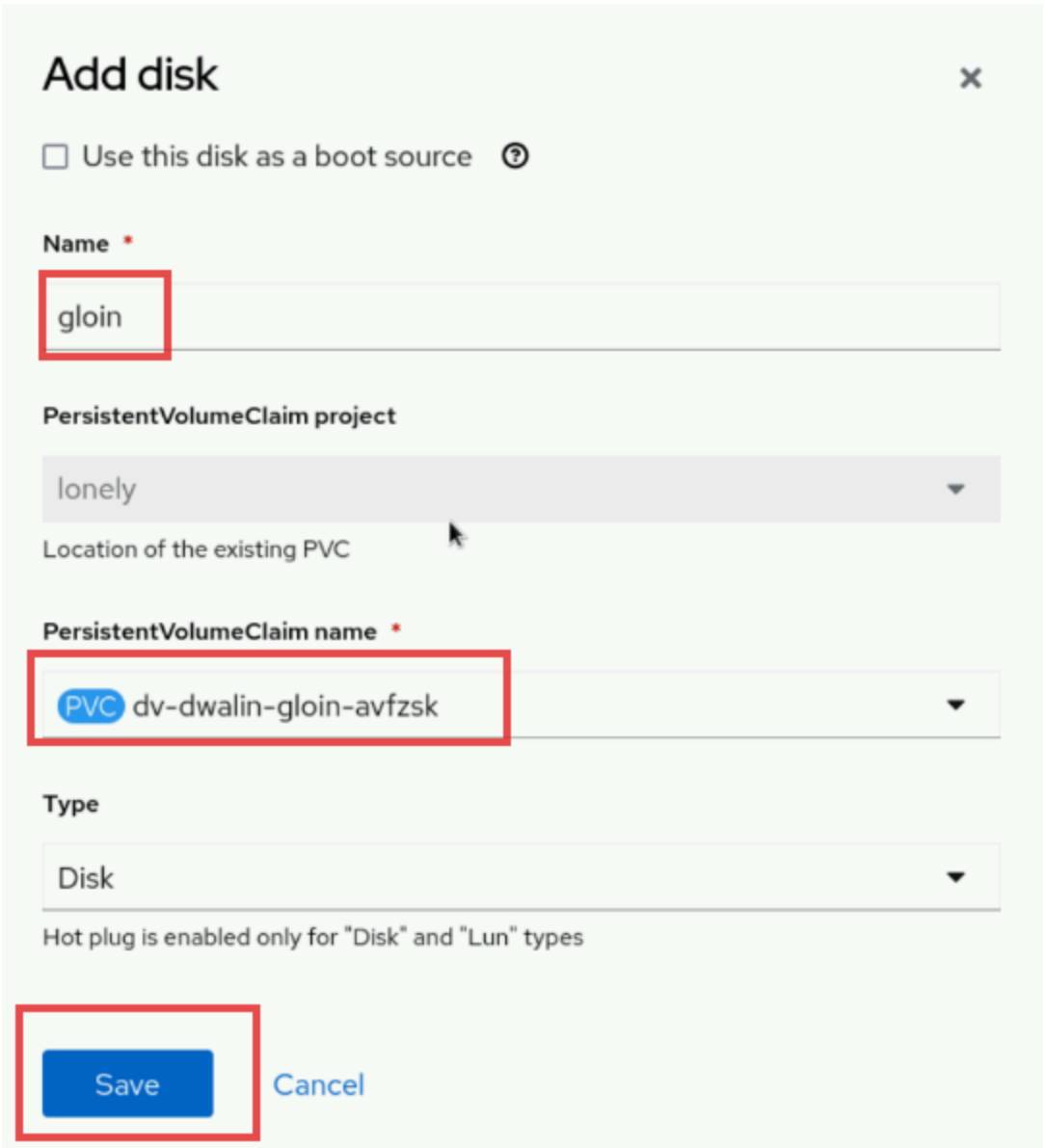


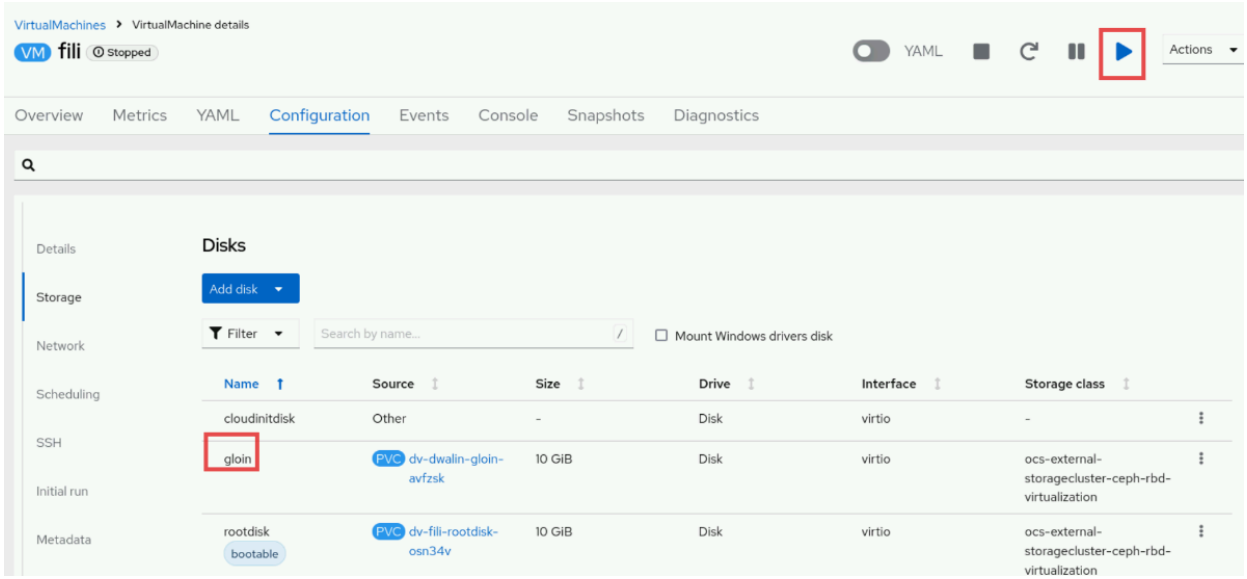
- Once both machine started, go to dwalin vm -> Configuration -> Storage
- You will see a disk named “gloin”, check the settings (storage profile, size, pvc, etc) and click on stop the machine and click on detached.



- Make sure fili, machine also stopped and go to its Configuration -> Storage and click on “Add disk” and select “volume” option and give disk name as gloin, select source as PVC and keep rest setting as it is.




Add disk
Use this disk as a boot source
Name *
gloin
PersistentVolumeClaim project
lonely
Location of the existing PVC
PersistentVolumeClaim name *
PVC dv-dwalin-gloin-avfzsk
Type
Disk
Hot plug is enabled only for "Disk" and "Lun" types
Save
Cancel



- Now access fili machine using console and check /dev/vdc disk. Now go to /etc/fstab file and make sure /dev/vdc disk mount on /var/www/html

```
[student@workstation ~]$ oc whoami
admin
[student@workstation ~]$ oc get vmi
NAME   AGE   PHASE     IP           NODENAME   READY
fili   32s   Running   10.10.0.44   master01   True
[student@workstation ~]$ virtctl console fili
```

```
[rhel@fili ~]$ sudo -i
[root@fili ~]# lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
vda    252:0    0    1M  0 disk
vdb    252:16   0   10G  0 disk
├─vdb1 252:17   0    1M  0 part
├─vdb2 252:18   0  200M  0 part /boot/efi
├─vdb3 252:19   0  600M  0 part
└─vdb4 252:20   0  9.2G  0 part
vdc    252:32   0   10G  0 disk
```

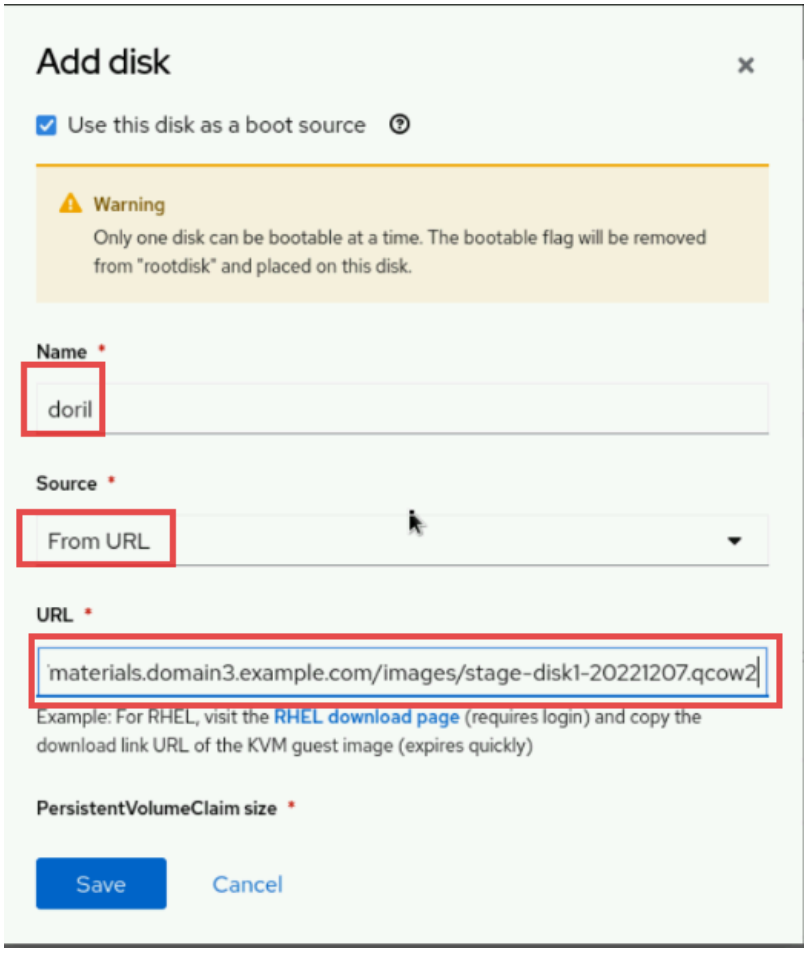
You will see in /etc/fstab, entry already mention, just remove “#” symbole and execute mount -a cmd and verify disk mounted

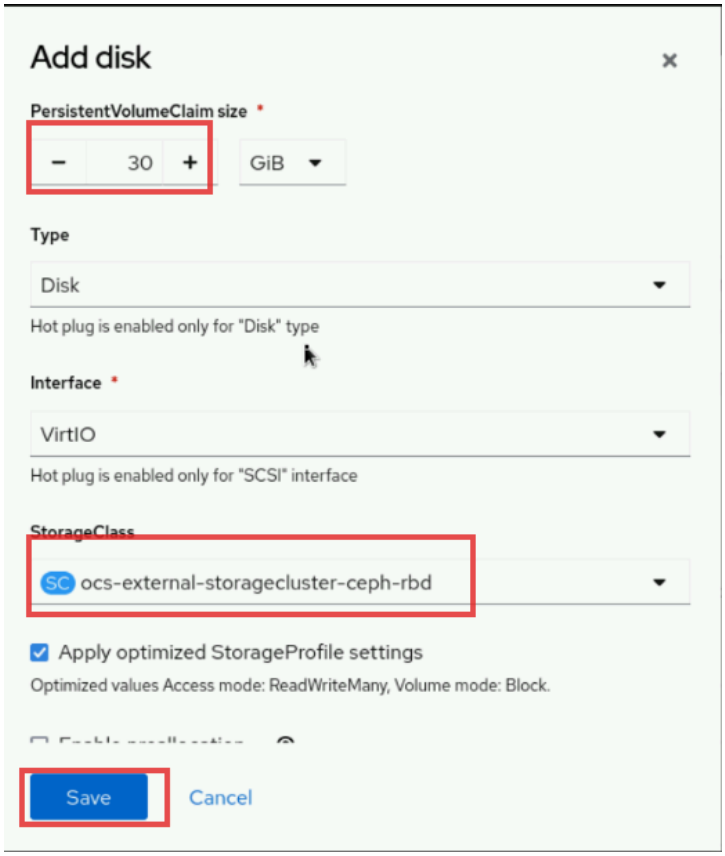
```
UUID=4b34029a-2c52-4051-bb74-af43d491bc74 /
UUID=a53019c4-0c14-499f-b4aa-10f62413b407 /b
UUID=7B77-95E7  /boot/efi vfat  defaults,uid
#/dev/vdc /var/www/html xfs defaults  0 0
```

- Make sure /var/www/html directory created and execute mount -a and if it ask execute systemctl daemon-reload
- Now go to the dwalin machine and add a disk, just use the source as “From URL” and use the given URL. In case you are not able to see the “From URL” option in the add disk, then use option-2 which is mentioned below.

OPTION-1

Mount the disk same way as you did in fili machine


Add disk
Use this disk as a boot source
Warning
Only one disk can be bootable at a time. The bootable flag will be removed from "rootdisk" and placed on this disk.
Name *
doril
Source *
From URL
URL *
materials.domain3.example.com/images/stage-disk1-20221207.qcow2
Example: For RHEL, visit the RHEL download page (requires login) and copy the download link URL of the KVM guest image (expires quickly)
PersistentVolumeClaim size *
Save
Cancel



OPTION-2

- Make sure you are in lonely project. Check existing datavolume and take output into yaml file of any existing datavolume

```
[student@workstation ~]$ oc project
Using project "lonely" on server "https://api.ocp4.example.com:6443".
[student@workstation ~]$ oc get dv
NAME                       PHASE       PROGRESS   RESTARTS   AGE
dv-dwalin-gloin-avfzsk     Succeeded   100.0%                25m
dv-dwalin-rootdisk-osn34v  Succeeded   100.0%                25m
dv-fili-rootdisk-osn34v    Succeeded   100.0%                23m
[student@workstation ~]$ oc get dv dv-dwalin-gloin-avfzsk -o yaml > dv-doril-disk.yaml
[student@workstation ~]$
```

- In the above image I have taken gloin dv output into yaml. Now remove options related to gloin and add necessary option so that dori data volume can be created. Like below

```yaml
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: doril-dv
  namespace: lonely
spec:
  source:
    http:
      url: http://utility.lab.example.com:8080/openshift4/images/rhel9.qcow2
  pvc:
    accessModes:
    - ReadWriteMany
    resources:
      requests:
        storage: 30Gi
    storageClassName: ocs-external-storagecluster-ceph-rbd-virtualization
    volumeMode: Block
```

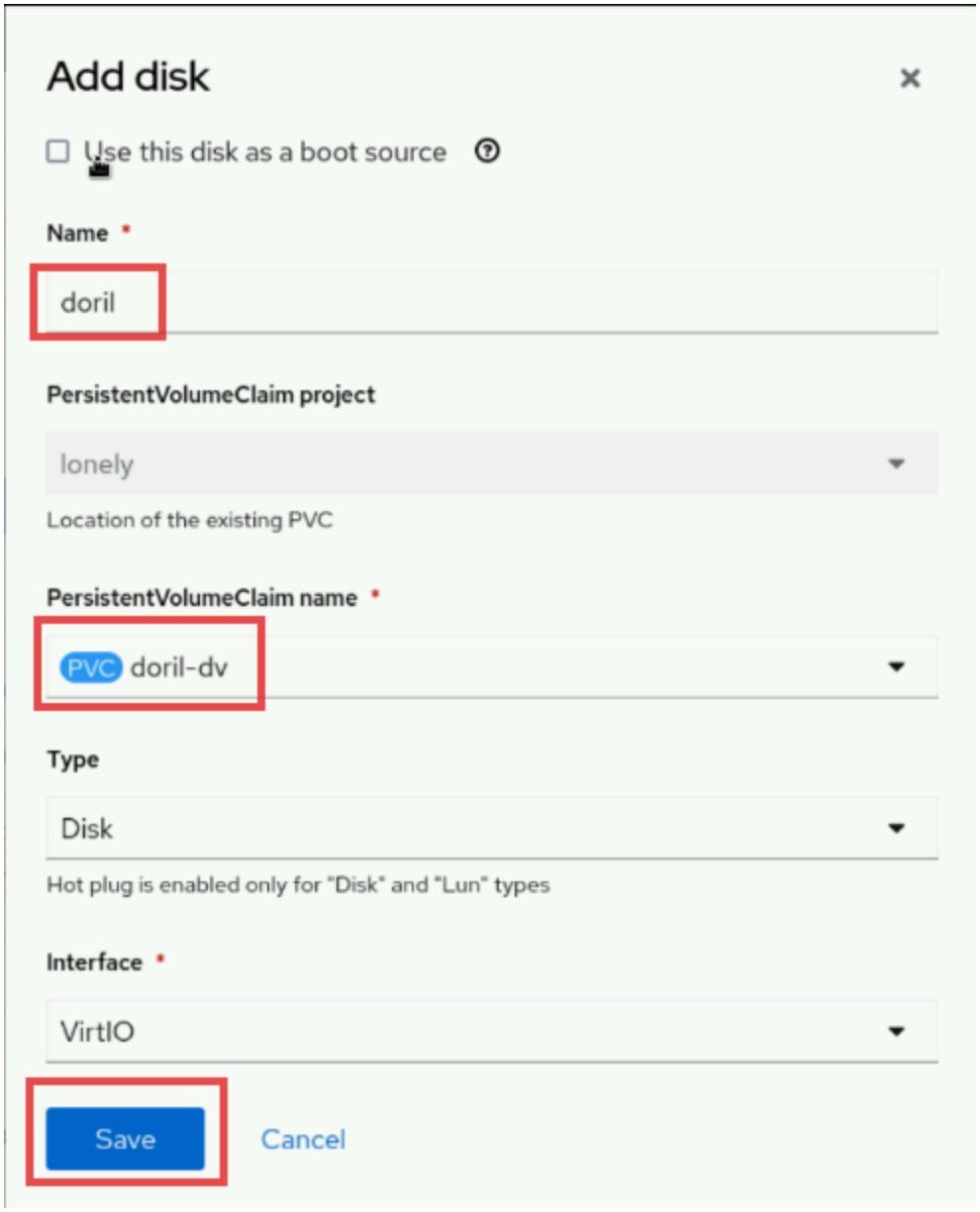
NOTE: make sure to add options accordingly which are asked in the exam. In lab to perform this, i make few changes in option like storage class

- Oc apply -f dv-doril-disk.yaml
- Verify datavolume created and wait till it got successed message and pvc is created

```
[student@workstation ~]$ oc apply -f dv-doril-disk.yaml
datavolume.cdi.kubevirt.io/doril-dv created
[student@workstation ~]$ oc get dv
NAME                       PHASE              PROGRESS   RESTARTS   AGE
doril-dv                   ImportInProgress   57.93%                24s
dv-dwalin-gloin-avfzsk     Succeeded          100.0%                48m
dv-dwalin-rootdisk-osn34v  Succeeded          100.0%                48m
dv-fili-rootdisk-osn34v    Succeeded          100.0%                46m
```

```
[student@workstation ~]$ oc get dv
NAME                        PHASE       PROGRESS   RESTARTS   AGE
doril-dv                    Succeeded   100.0%                3m11s
dv-dwalin-gloin-avfzsk      Succeeded   100.0%                50m
dv-dwalin-rootdisk-osn34v   Succeeded   100.0%                50m
dv-fili-rootdisk-osn34v     Succeeded   100.0%                49m
[student@workstation ~]$ oc get pvc
NAME                        STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAG
ECLASS                                       VOLUMEATTRIBUTESCLASS   AGE
doril-dv                    Bound    pvc-9859c3f2-2e31-4a75-b5b8-05d479f437a7   30Gi       RWX            ocs-ex
ternal-storagecluster-ceph-rbd-virtualization   <unset>                 3m13s
dv-dwalin-gloin-avfzsk      Bound    pvc-4841aa80-d465-42f9-a46d-9a741a495e45   10Gi       RWX            ocs-ex
ternal-storagecluster-ceph-rbd-virtualization   <unset>                 50m
dv-dwalin-rootdisk-osn34v   Bound    pvc-4b6a7056-0743-49bb-bea9-a00c3edfefa7   10Gi       RWX            ocs-ex
ternal-storagecluster-ceph-rbd-virtualization   <unset>                 50m
dv-fili-rootdisk-osn34v     Bound    pvc-12c05e59-9415-45a7-aac7-5a07a96ae6cd   10Gi       RWX            ocs-ex
ternal-storagecluster-ceph-rbd-virtualization   <unset>                 49m
```

- Once disk created, go to dwalin vm and add doril disk using below option and start the vm



- Verify that disk is attached and mount it permanently

```
[student@workstation ~]$ oc get vmi
NAME      AGE   PHASE     IP            NODENAME   READY
dwalin    29s   Running   10.8.0.8      master02   True
fili      45m   Running   10.10.0.44    master01   True
[student@workstation ~]$ virtctl console dwalin
```

```
[rhel@dwalin ~]$ sudo -i
[root@dwalin ~]# lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
vda    252:0    0    1M  0 disk
vdb    252:16   0   10G  0 disk
├─vdb1 252:17   0    1M  0 part
├─vdb2 252:18   0  200M  0 part /boot/efi
├─vdb3 252:19   0  600M  0 part /boot
└─vdb4 252:20   0  9.2G  0 part /
vdc    252:32   0   30G  0 disk
```

- Go to /etc/fstab
- You will line with #/dev/vdc /var/www/html…, remove # and save it and execute mount -a. If it gives daemon-reload warning then execute systemctl daemon-reload
- Verify using df -hT option
- Perform below labs to understand this question
  lab start storage-review

==========

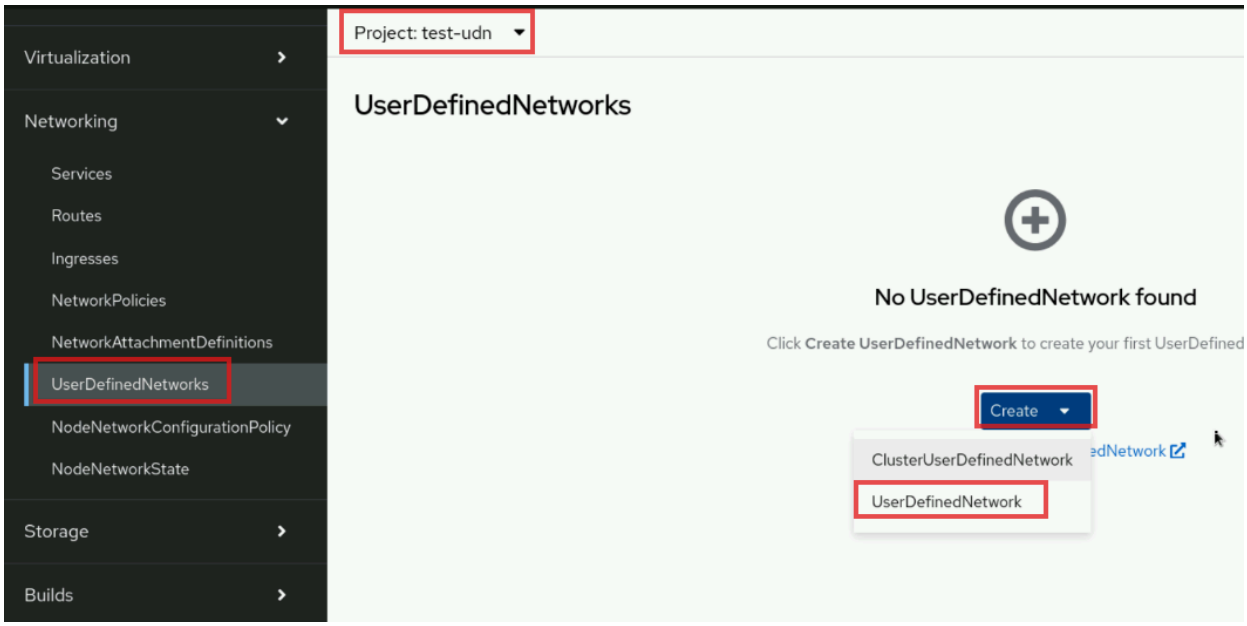
Q-8 Create UDN Network

You will be asked to create udn network on given namespace (namespace already created and subnet range give like 192.168.100.0/27)

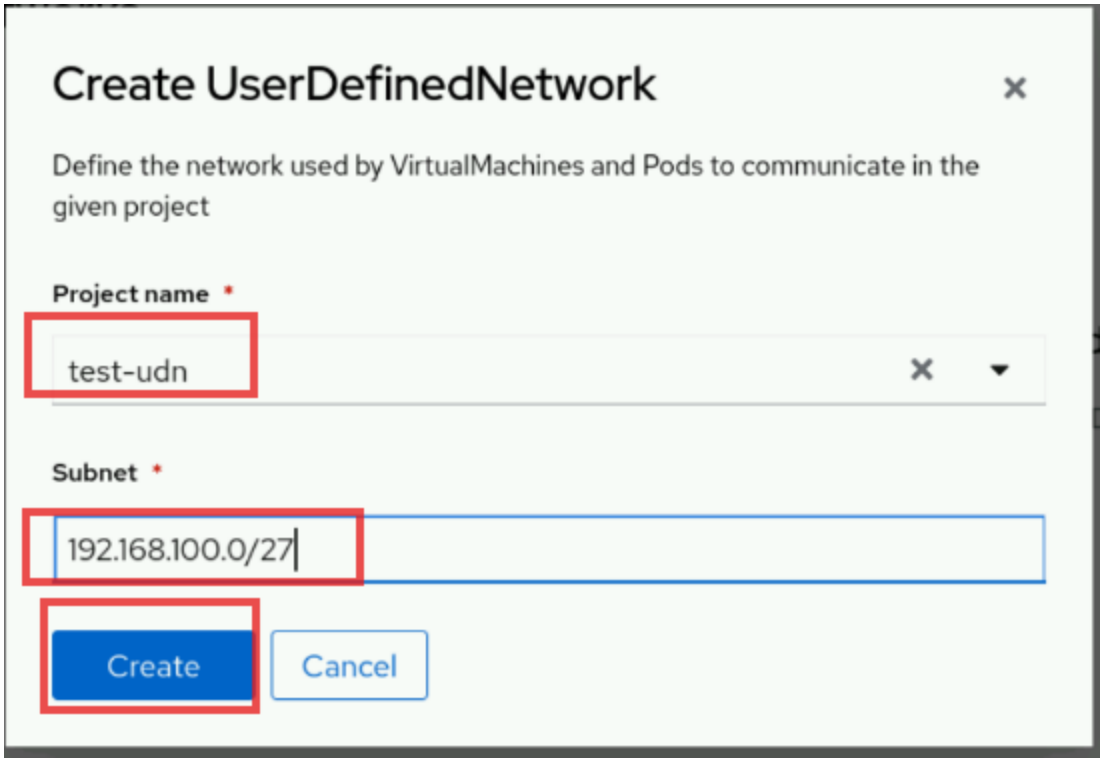
ANSWER:

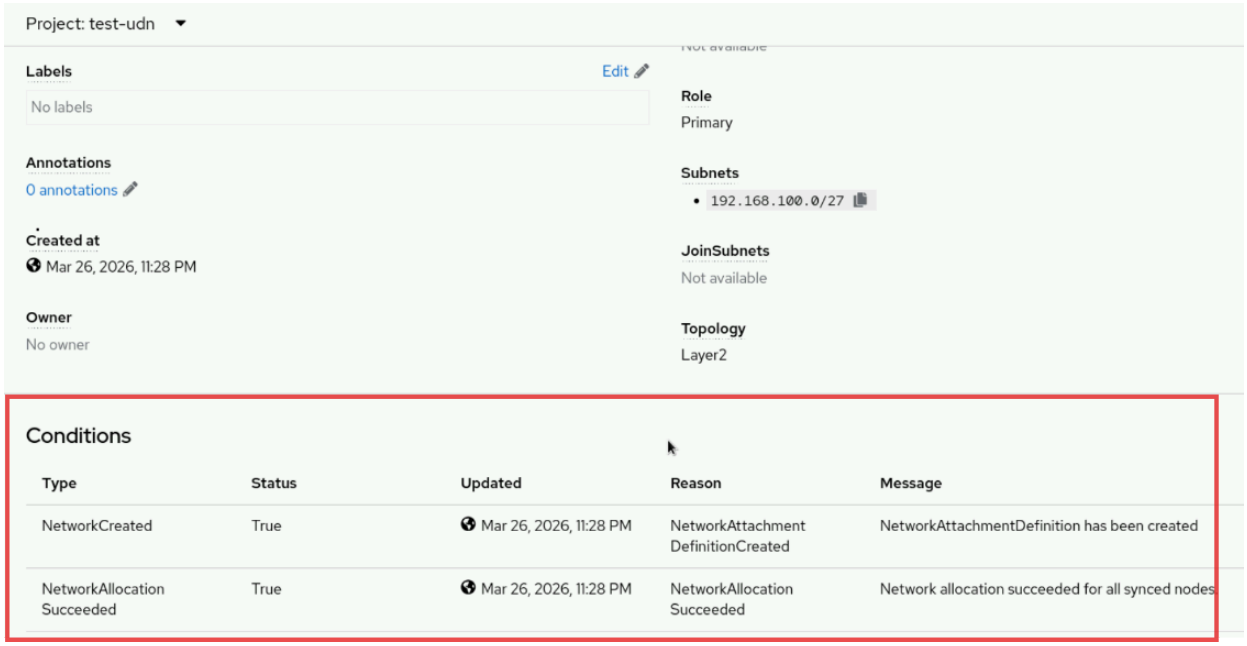
- **NOTE: namespace already exists and label "k8s.ovn.org/primary-user-defined-network"**

- Go to Network → UserDefinedNetworks → select project → click on create UDN → userdefinednetwor



- Assigned the subnet range and confirm its created via scrolling till end





==========

Q-9 Create VM on UDN Network

Go to Virtualization → select project → create vm → template → select given template → create vm

==========

Q- 10 Create a virtual machine template

Create a project and a virtual machine template according to the following requirements:

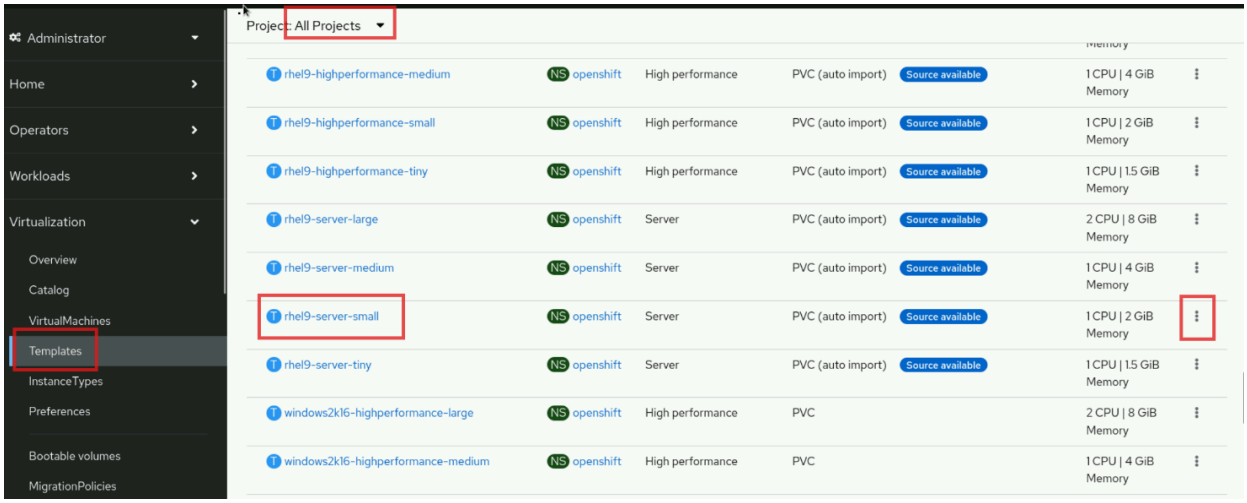
- The project is **ascifix**
- The template name is **disorurl** and it exists in the ascifix project
- The template is a clone of the built-in rhel9-server-small template
- Virtual machines created using this template use:
  - Flavor : Small, with 1 cpu , 2 Gi of RAM
  - Storage space: 15 Gi
  - Disk source: **rhel-9.2-x86_64-kvm.qcow2** provided at:
  - **http://materials.domain3.example.com/images/rhel-9.2-x86_64-kvm.qcow2**
  - StorageClassName: **ocs-external-storagecluster-ceph-rbd**
- The user **ferni** can log in to these virtual machines on the console with the password Atenorth3
- The user **ferni** can log in to these virtual machines via SSH without a password by using : /home/opsadm/.ssh/id_rsa_ex316.pub from user opsadm at **workbench.domain3.example.com**
- When instantiated, these virtual machines will have the httpd and php packages installed , and the httpd service will be running and serving the page : **http://rhpds.domain3.example.com/samples/lbtest/index.php**

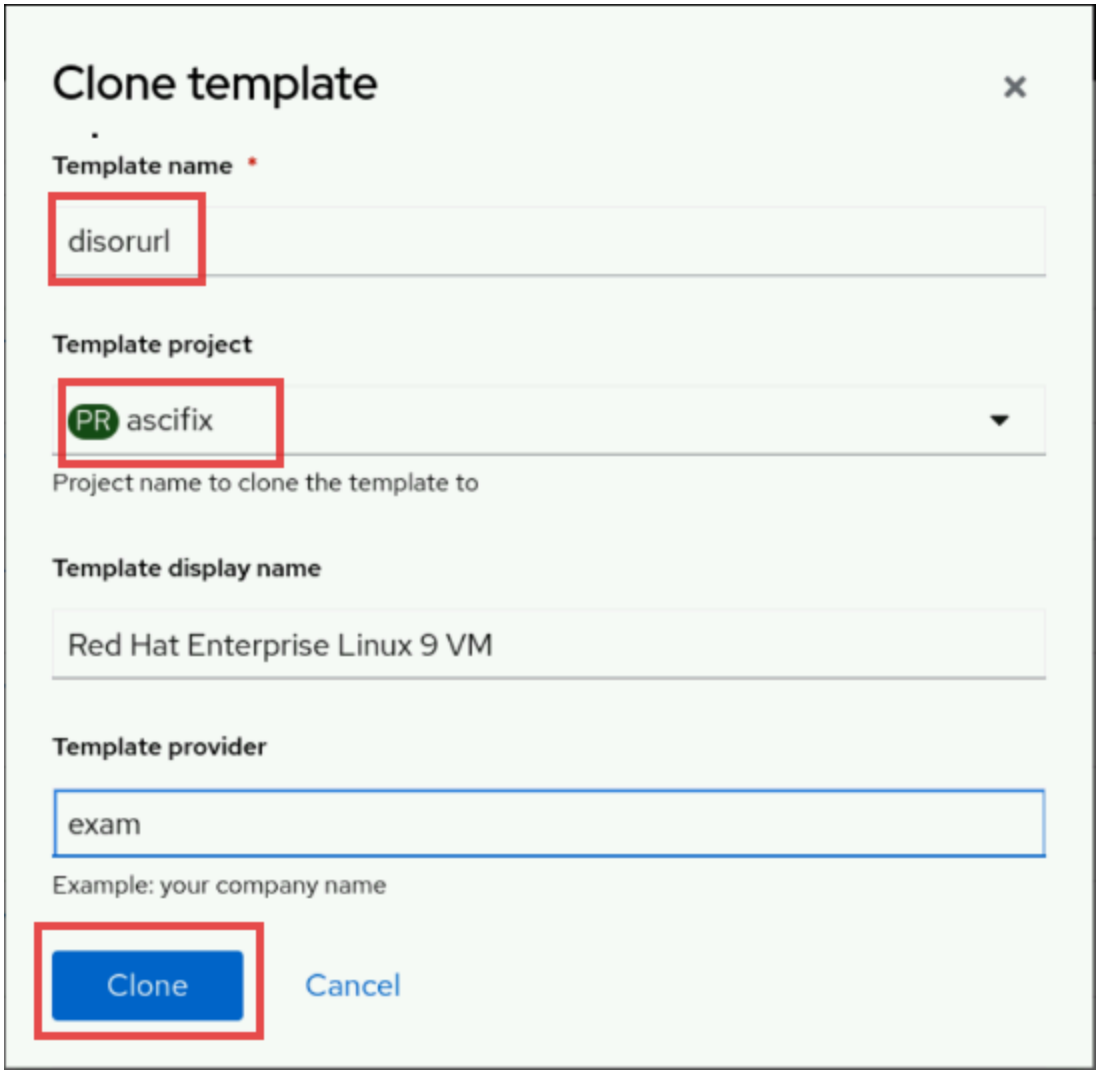
NOTE: sample YAML files are available from : **http://rhpds.domain3.example.com/samples/lbtest**

You will need to replace any strings marked for template purpose

ANS:

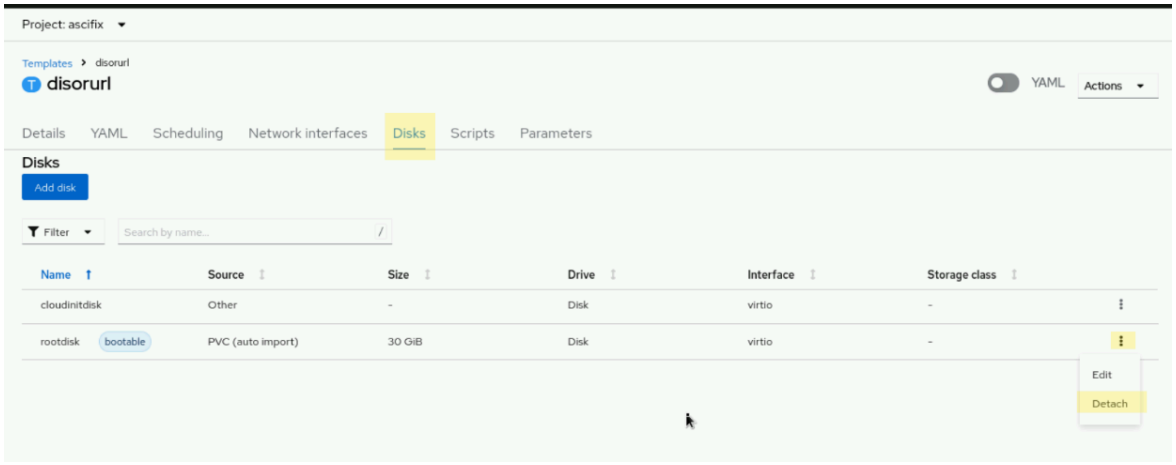
- First clone the project. Go to template, search and select “rhel9-server-small”, clic on ellipse (three dot) icon, and click on clone and give project name “ascifix” and template name as “discourl”

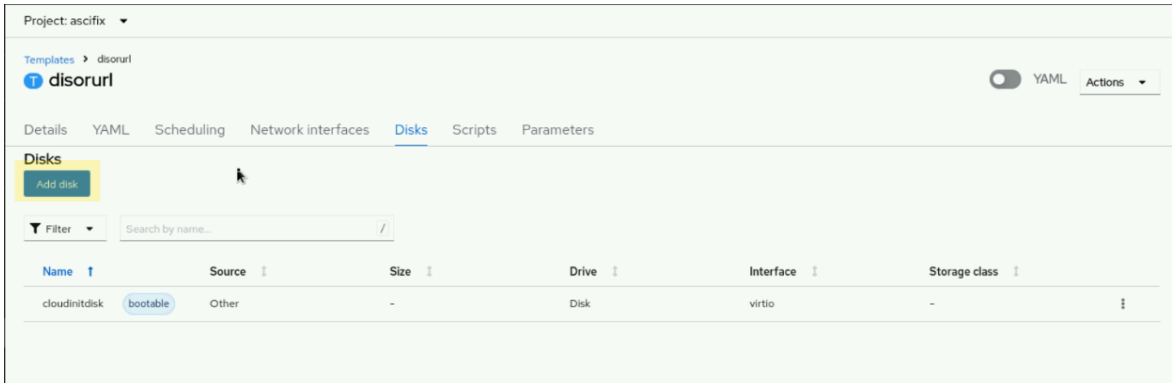


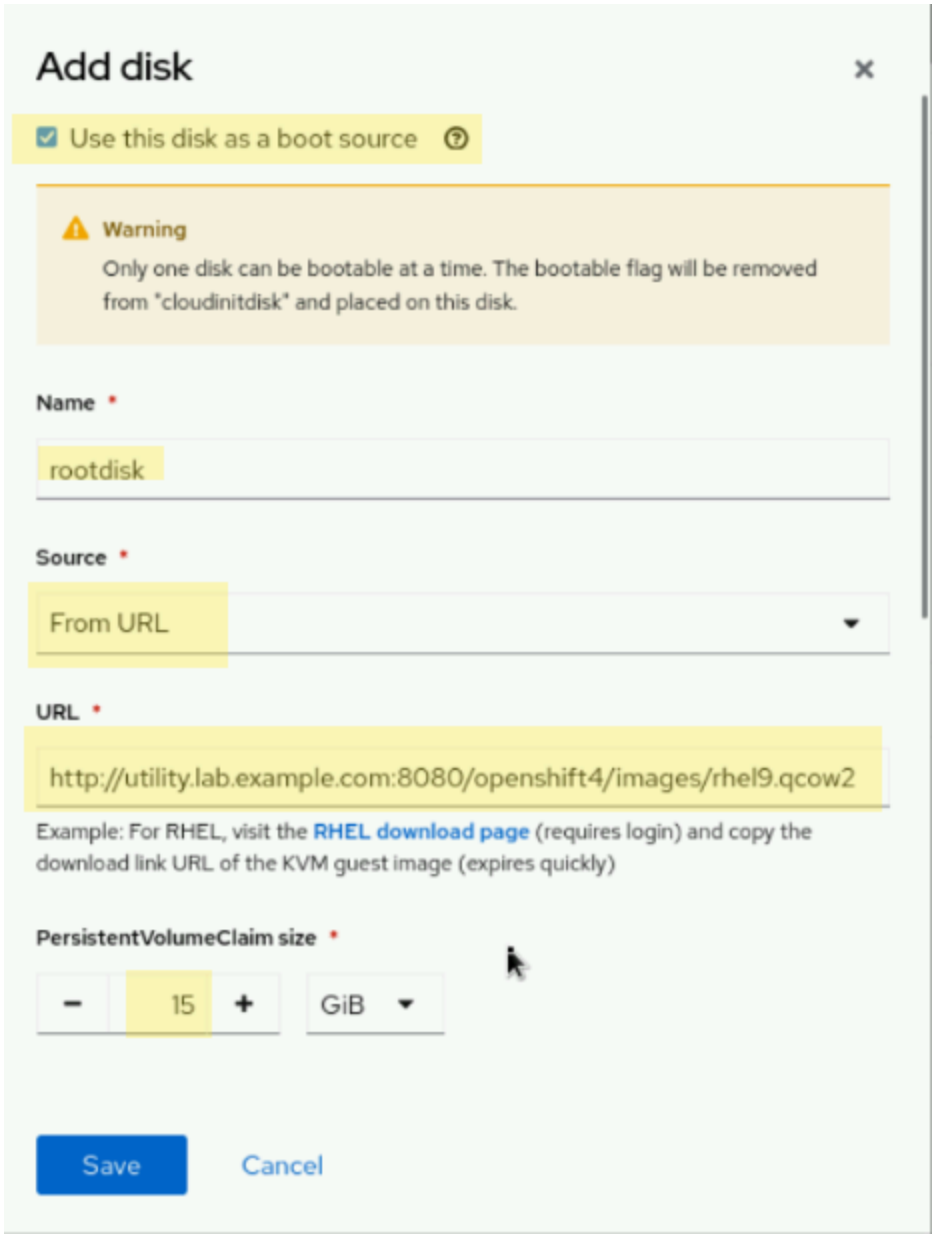


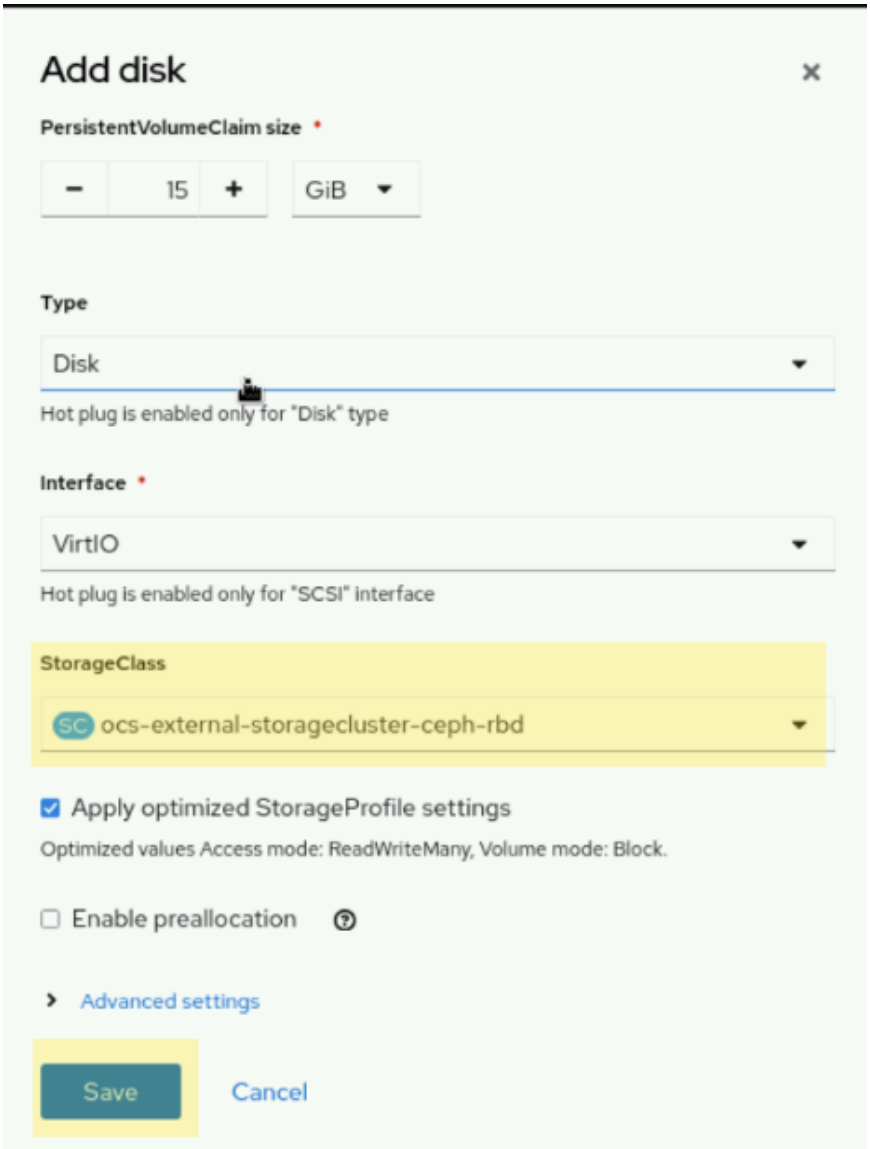
- NOTE: for adding a disk there two option
    - If you see the URL option while adding a disk then follow Option-1
    - Else go with Option-2

OPTION-1 for adding a disk


Project: ascifix
Templates › disorurl
disorurl
YAML
Actions
Details
YAML
Scheduling
Network interfaces
Disks
Scripts
Parameters
Disks
Add disk
Filter
Search by name...
Name
Source
Size
Drive
Interface
Storage class
cloudinitdisk
Other
-
Disk
virtio
-
rootdisk
bootable
PVC (auto import)
30 GiB
Disk
virtio
-
Edit
Detach


Project: ascifix
Templates › disorurl
disorurl
YAML
Actions
Details
YAML
Scheduling
Network interfaces
Disks
Scripts
Parameters
Disks
Add disk
Filter
Search by name...
Name
Source
Size
Drive
Interface
Storage class
cloudinitdisk
bootable
Other
-
Disk
virtio
-


Add disk
Use this disk as a boot source
Warning
Only one disk can be bootable at a time. The bootable flag will be removed from "cloudinitdisk" and placed on this disk.
Name *
rootdisk
Source *
From URL
URL *
http://utility.lab.example.com:8080/openshift4/images/rhel9.qcow2
Example: For RHEL, visit the RHEL download page (requires login) and copy the download link URL of the KVM guest image (expires quickly)
PersistentVolumeClaim size *
15
GiB
Save
Cancel



OPTION-2

- go to the terminal, make sure you are in the “ascifix” project and verify the template. And edit into project

```
[student@workstation ~]$ oc project ascifix
Now using project "ascifix" on server "https://api.ocp4.example.com:6443".
[student@workstation ~]$ oc get template
NAME       DESCRIPTION
    PARAMETERS        OBJECTS
disorurl   Template for Red Hat Enterprise Linux 9 VM or newer. A PVC with the RHEL disk..
.   4 (2 generated)   1
[student@workstation ~]$ oc edit template disorurl
```

- Now edit the disk section into the project like below using oc edit template disourl cmd

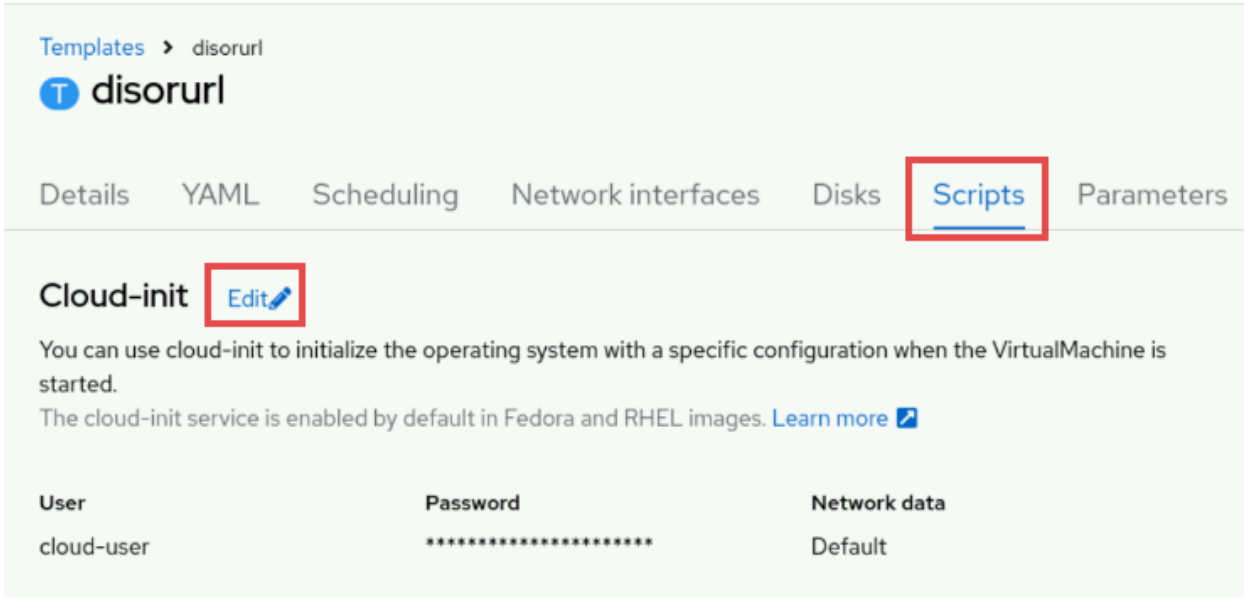
```
spec:
  dataVolumeTemplates:
  - apiVersion: cdi.kubevirt.io/v1beta1
    kind: DataVolume
    metadata:
      name: ${NAME}
    spec:
      pvc:
        accessModes:
        - ReadWriteMany
        resources:
          requests:
            storage: 15Gi
        storageClassName: ocs-external-storagecluster-ceph-rbd-virtualization
        volumeMode: Block
      source:
        http:
          url: http://utility.lab.example.com:8080/openshift4/images/rhel9.qcow2
  runStrategy: Halted
```

- Also make sure you will add app:tube label to the vm template. This setting will be useful for next question

```
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    labels:
      app: tube
      kubevirt.io/domain: ${NAME}
      kubevirt.io/size: small
  spec:
```

- Save it

- Add cloud-init script option from the terminal or you can edit from webUI as well. Like below



NOTE: you need to perform below steps in the lab only not in the exam. We will use this repo to download the packages

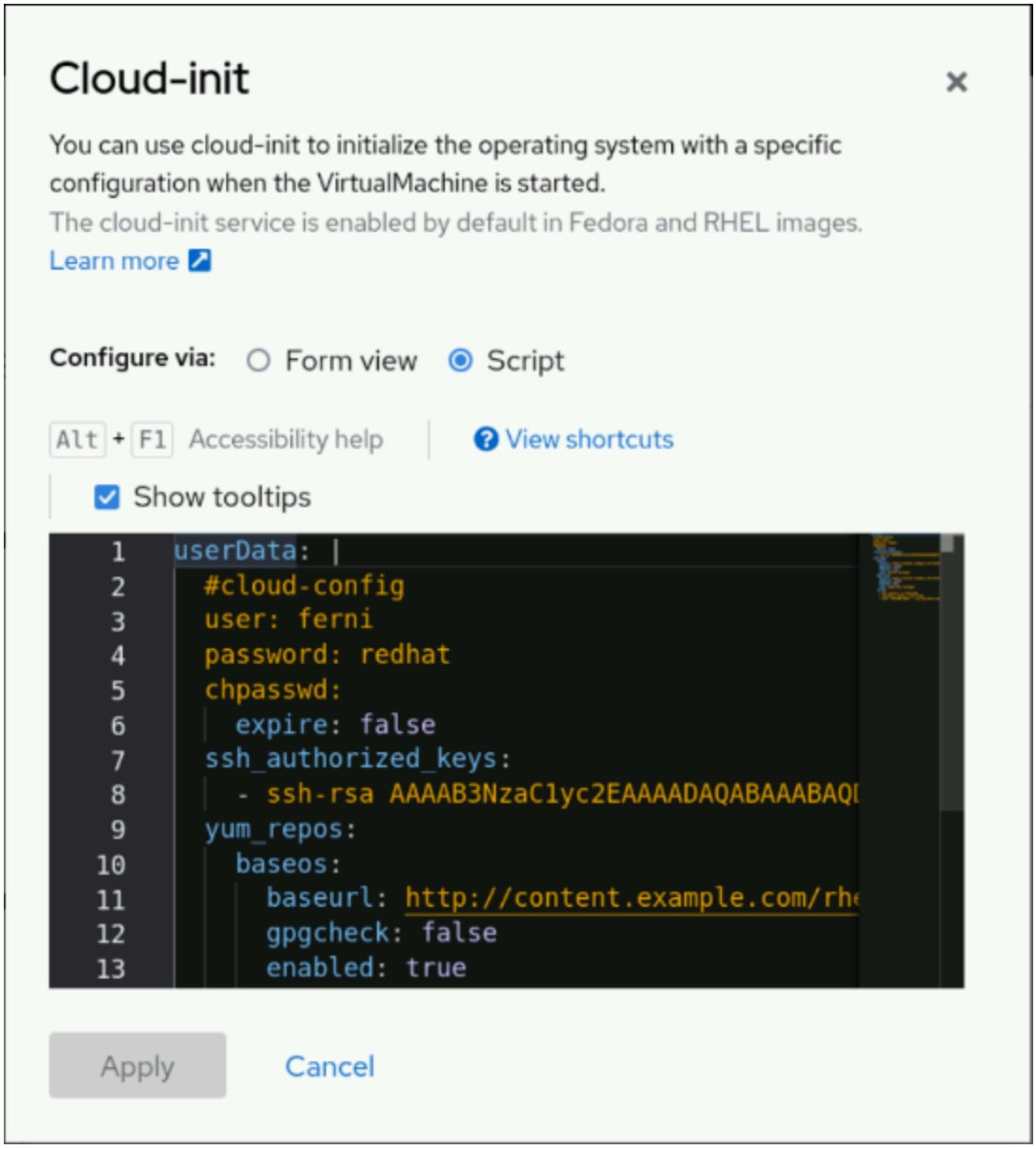
- Ssh to utility machine and check the repo

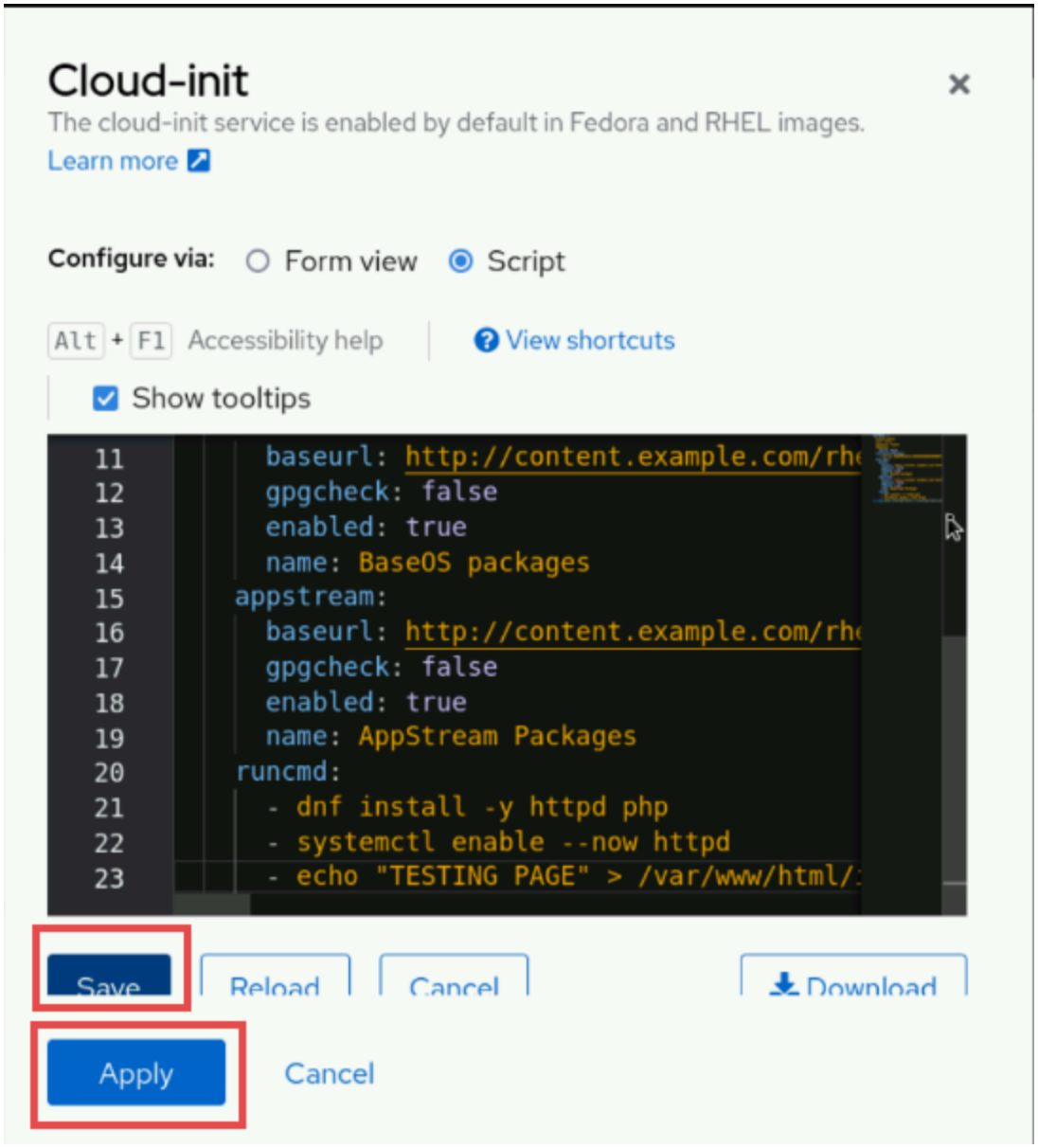
```
[student@workstation ~]$ ssh lab@utility
Warning: Permanently added 'utility' (ED25519) to the list of known hosts.
Register this system with Red Hat Insights: rhc connect
```

```
[lab@utility ~]$ cat /etc/yum.repos.d/rhel_dvd.repo
[rhel-9.5-for-x86_64-baseos-rpms]
baseurl = http://content.example.com/rhel9.5/x86_64/dvd/BaseOS
enabled = true
gpgcheck = false
name = Red Hat Enterprise Linux 9.5 BaseOS (dvd)

[rhel-9.5-for-x86_64-appstream-rpms]
baseurl = http://content.example.com/rhel9.5/x86_64/dvd/AppStream
enabled = true
gpgcheck = false
name = Red Hat Enterprise Linux 9.5 AppStream (dvd)
[lab@utility ~]$
```

- Now above information you need to add in cloud-init-script


Cloud-init
You can use cloud-init to initialize the operating system with a specific configuration when the VirtualMachine is started.
The cloud-init service is enabled by default in Fedora and RHEL images.
Learn more
Configure via:
Form view
Script
Alt + F1 Accessibility help
View shortcuts
Show tooltips
userData: |
#cloud-config
user: ferni
password: redhat
chpasswd:
expire: false
ssh_authorized_keys:
- ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQ
yum_repos:
baseos:
baseurl: http://content.example.com/rh
gpgcheck: false
enabled: true
Apply
Cancel



- Or you can edit the same info in template using oc edit template “disorurl” from terminal

```
    - cloudInitNoCloud:
        userData: |
          #cloud-config
          user: ferni
          password: redhat
          chpasswd:
            expire: false
          ssh_authorized_keys:
            - ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDGtUW3ismHyuCW4CDdTVOOOq6aySdtYenXFWWx7HJa4VTepkG00aaLId9ocra10hc+MB0GTJMCyabDv3
i8NKdi6GDH/aOLVsp/Ewy8DEzZMBlJDCt4v2i4/wU4liw6KgEFkZs+5hnqU8d4QzldyGJ5onr+AGvFOKG68CS0BBl40Z1twf1HhCyx8k6nzD2ovlkxWRFZKPAFrtPCBVvQDkOfVF
ZF+lwzaSztgAjbFZ4A9jqQyUYx4kOJ5DtRef36ucdUdVQale0+8lICl7/gb142SPpYfhxe88/BJScLPRjvVNeu1TxRmoHtVazqnAoRxQYAn2MoI6AG+w6QuZf8f7aL LabGradin
gKey
          yum_repos:
            baseos:
              baseurl: http://content.example.com/rhel9.5/x86_64/dvd/BaseOS
              gpgcheck: false
              enabled: true
              name: BaseOS packages
            appstream:
              baseurl: http://content.example.com/rhel9.5/x86_64/dvd/AppStream
              gpgcheck: false
              enabled: true
              name: AppStream Packages
          runcmd:
            - dnf install -y httpd php
            - systemctl enable --now httpd
            - echo "TESTING PAGE" > /var/www/html/index.html
      name: cloudinitdisk
```

- Please note in exam you have to edit only below option:
  - userData: |
  - #cloud-config
  - user: ferni
  - password: Atenorth3
  - chpasswd: { expire: False }
  - ssh_authorized_keys:
  - - ssh-rsa <copt from the given file>
  - runcmd:
  - - curl -o /etc/yum.repos.d/rhel.repo <link copy from previous/letter question>
  - - dnf install -y httpd php
  - - systemctl enable --now httpd
  - - curl -o /var/www/html/index.php http://rhpds.domain3.example.com/samples/lbtest/index.php

==========

Q-11 Create a service using virtual machine load balancing

In the project name **asciinix** , create the following objects:

- Using the template **dioscuri** which you created previously instantiated two identical virtual machines named **castor** and **pollux.**

Create a service named **flix-service:**

- Type : ClusterIP
- Pod selector: **app:tube**
- TCP port: 80

Create a route named **flix-route** so that the above service is made available at: **tube.asciiinix.apps.domain3.example.com**

Running the following command line on **workbench.domain3.example.com**:

- curl **http://tube.asciiinix.apps.domain3.example.com**
  Effectively results in a page being displayed with an ASCII figure and a reference to the VM where it is running .

NOTE: Sample YAML files are available from: **http://rghls.domain3.example.com/samples/lbtest** You will need to replace any strings marked with @ with the correct value for the purpose.

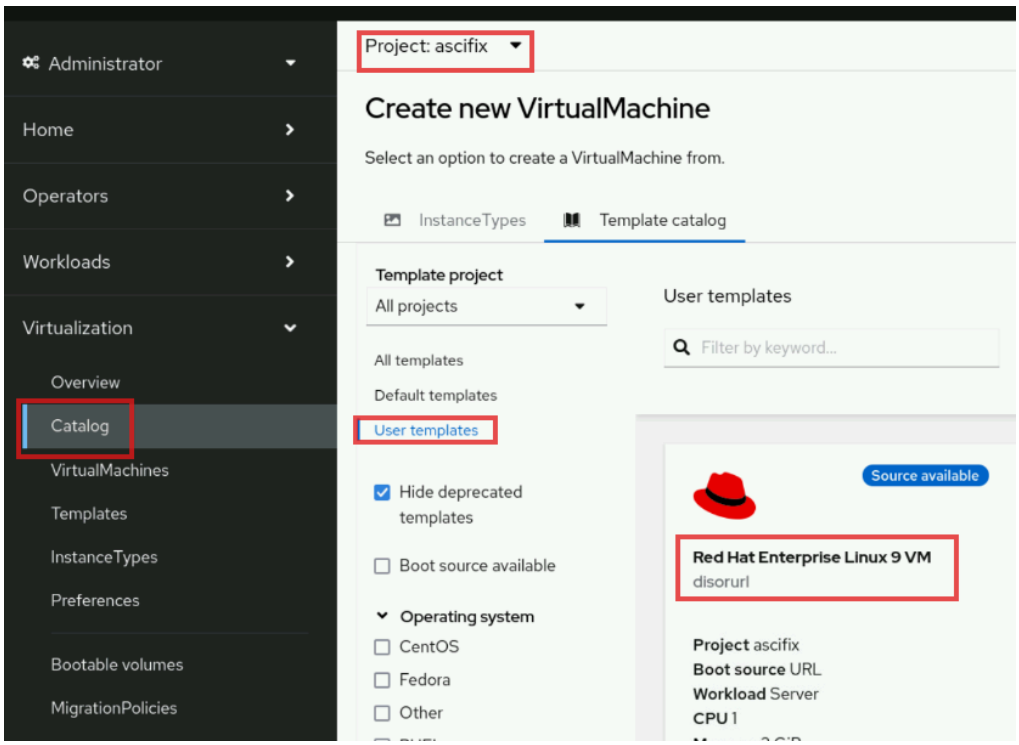
ANSWER:

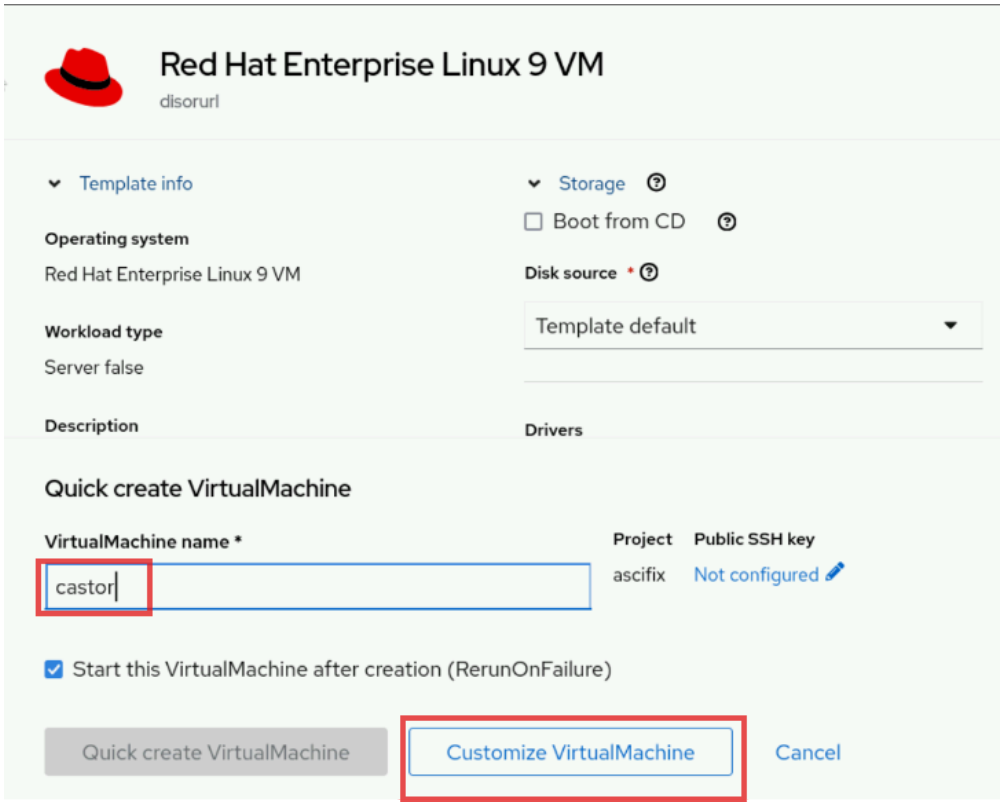
- Create vm using the webUI or you can create using cmd line using below method

  METHOD-1

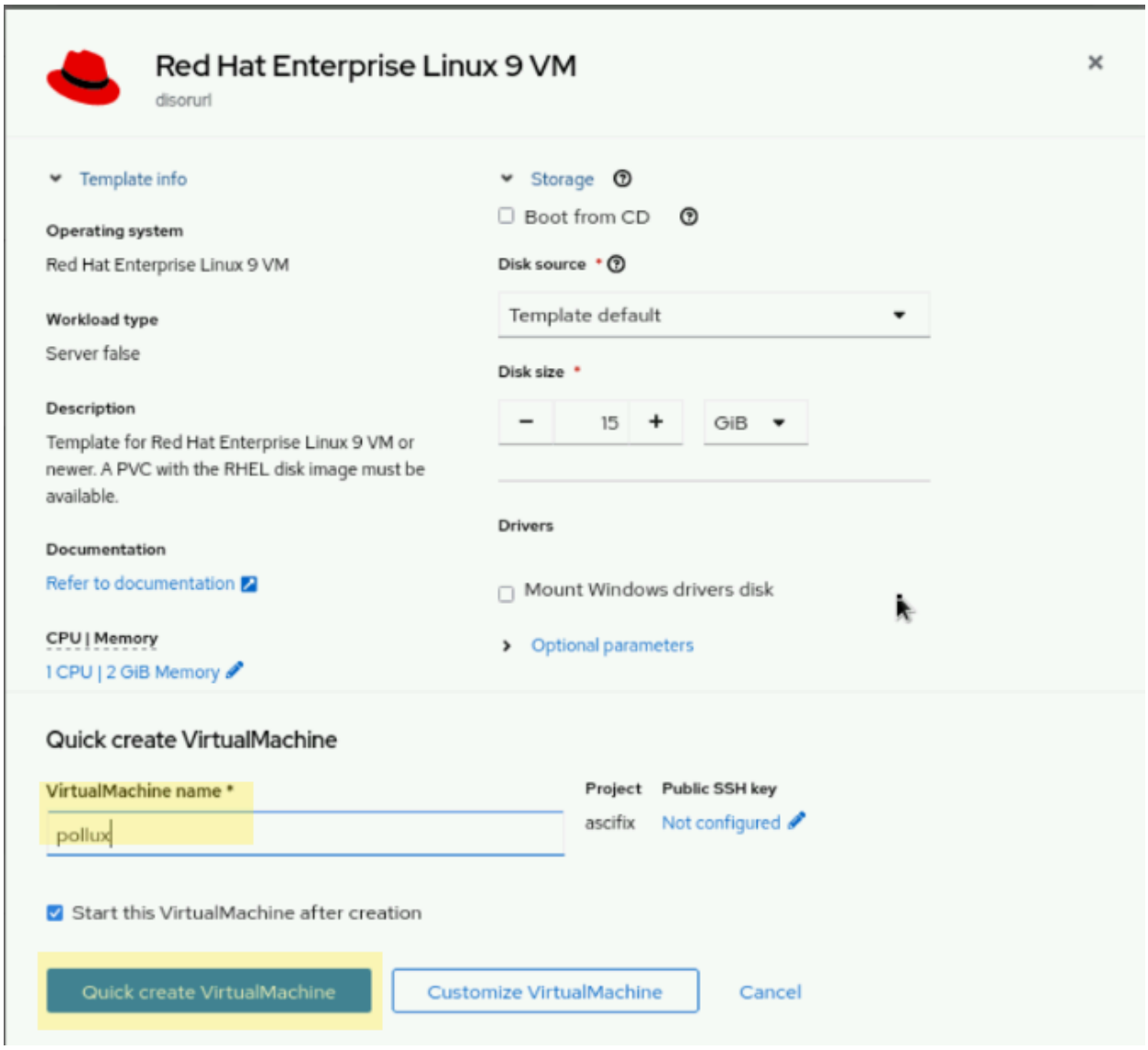
```
[student@workstation ~]$ oc project
Using project "ascifllix" on server "https://api.ocp4.example.com:6443".
[student@workstation ~]$ oc get template
NAME       DESCRIPTION
    PARAMETERS        OBJECTS
diisourl   Template for Red Hat Enterprise Linux 9 VM or newer. A PVC with the RHEL disk..
.   4 (2 generated)   1
[student@workstation ~]$ oc get vm
No resources found in ascifllix namespace.
[student@workstation ~]$ oc process template/diisourl -p NAME=castor | oc apply -f -
virtualmachine.kubevirt.io/castor created
[student@workstation ~]$ oc process template/diisourl -p NAME=pollux | oc apply -f -
virtualmachine.kubevirt.io/pollux created
[student@workstation ~]$
```

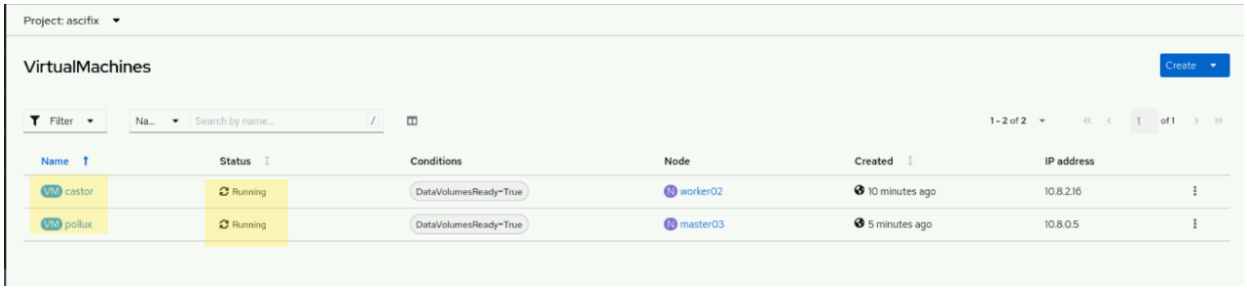
METHOD-2




Red Hat Enterprise Linux 9 VM
disorurl
Template info
Operating system
Red Hat Enterprise Linux 9 VM
Workload type
Server false
Description
Storage
Boot from CD
Disk source *
Template default
Drivers
Quick create VirtualMachine
VirtualMachine name *
castor
Project
ascifix
Public SSH key
Not configured
Start this VirtualMachine after creation (RerunOnFailure)
Quick create VirtualMachine
Customize VirtualMachine
Cancel


Project: ascifix
VirtualMachines
Create
From InstanceType
From template
With YAML
Filter
Name
Status
Conditions
Node
Created
IP address
castor
Running
DataVolumesReady=True
worker02
2 minutes ago
10.8.2.16


Red Hat Enterprise Linux 9 VM
disorurl
Template info
Operating system
Red Hat Enterprise Linux 9 VM
Workload type
Server false
Description
Template for Red Hat Enterprise Linux 9 VM or newer. A PVC with the RHEL disk image must be available.
Documentation
Refer to documentation
CPU | Memory
1 CPU | 2 GiB Memory
Storage
Boot from CD
Disk source *
Template default
Disk size *
15
GiB
Drivers
Mount Windows drivers disk
Optional parameters
Quick create VirtualMachine
VirtualMachine name *
pollux
Project
ascifix
Public SSH key
Not configured
Start this VirtualMachine after creation
Quick create VirtualMachine
Customize VirtualMachine
Cancel


Project: ascifix
VirtualMachines
Create
Filter
Name
Status
Conditions
Node
Created
IP address
castor
Running
DataVolumesReady=True
worker02
10 minutes ago
10.8.2.16
pollux
Running
DataVolumesReady=True
master03
5 minutes ago
10.8.0.5

```
[student@workstation ~]$ oc get vmi --show-labels
NAME     AGE   PHASE     IP          NODENAME   READY   LABELS
castor   64s   Running   10.8.2.20   worker02   True    app=tube,kubevirt.io/domain=castor,ku
bevirt.io/nodeName=worker02,kubevirt.io/size=small,network.kubevirt.io/headlessService=headle
ss
pollux   14s   Running   10.8.0.7    master03   True    app=tube,kubevirt.io/domain=pollux,ku
bevirt.io/nodeName=master03,kubevirt.io/size=small,network.kubevirt.io/headlessService=headle
ss
[student@workstation ~]$ oc get pods --show-labels
NAME                          READY   STATUS    RESTARTS   AGE   LABELS
virt-launcher-castor-vgxqn    1/1     Running   0          73s   app=tube,kubevirt.io/created-
by=ab3ee929-2179-4484-9170-d444b2c6a6b1,kubevirt.io/domain=castor,kubevirt.io/nodeName=worker
02,kubevirt.io/size=small,kubevirt.io=virt-launcher,network.kubevirt.io/headlessService=headl
ess,vm.kubevirt.io/name=castor
virt-launcher-pollux-967qv    1/1     Running   0          23s   app=tube,kubevirt.io/created-
by=a355d115-1208-483f-b12f-0beaf81fd069,kubevirt.io/domain=pollux,kubevirt.io/nodeName=master
03,kubevirt.io/size=small,kubevirt.io=virt-launcher,network.kubevirt.io/headlessService=headl
ess,vm.kubevirt.io/name=pollux
```

```yaml
apiVersion: v1
kind: Service
metadata:
  labels:
    app: flix-service
  name: flix-service
spec:
  ports:
  - port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: tube
  type: ClusterIP
```

```
[student@workstation ~]$ oc apply -f flix-service.yaml
service/flix-service created
[student@workstation ~]$ oc get svc
NAME           TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
flix-service   ClusterIP   172.30.243.177   <none>        80/TCP     23s
headless       ClusterIP   None             <none>        5434/TCP   38m
[student@workstation ~]$ oc get vmi -o wide
NAME     AGE     PHASE     IP          NODENAME   READY   LIVE-MIGRATABLE   PAUSED
castor   6m18s   Running   10.8.2.20   worker02   True    True
pollux   5m28s   Running   10.8.0.7    master03   True    True
[student@workstation ~]$ oc get endpoints
NAME           ENDPOINTS                      AGE
flix-service   10.8.0.7:80,10.8.2.20:80       34s
headless       10.8.0.7:5434,10.8.2.20:5434   38m
[student@workstation ~]$
```

```
[student@workstation ~]$ oc expose service flix-service --name flix-route --hostname tube.asc
iiinix.apps.domain3.example.com
route/flix-route exposed
[student@workstation ~]$ oc get route
NAME         HOST/PORT                                PATH   SERVICES       PORT   TERMINATI
ON   WILDCARD
flix-route   tube.asciiinix.apps.domain3.example.com          flix-service   80
     None
[student@workstation ~]$
```

==========

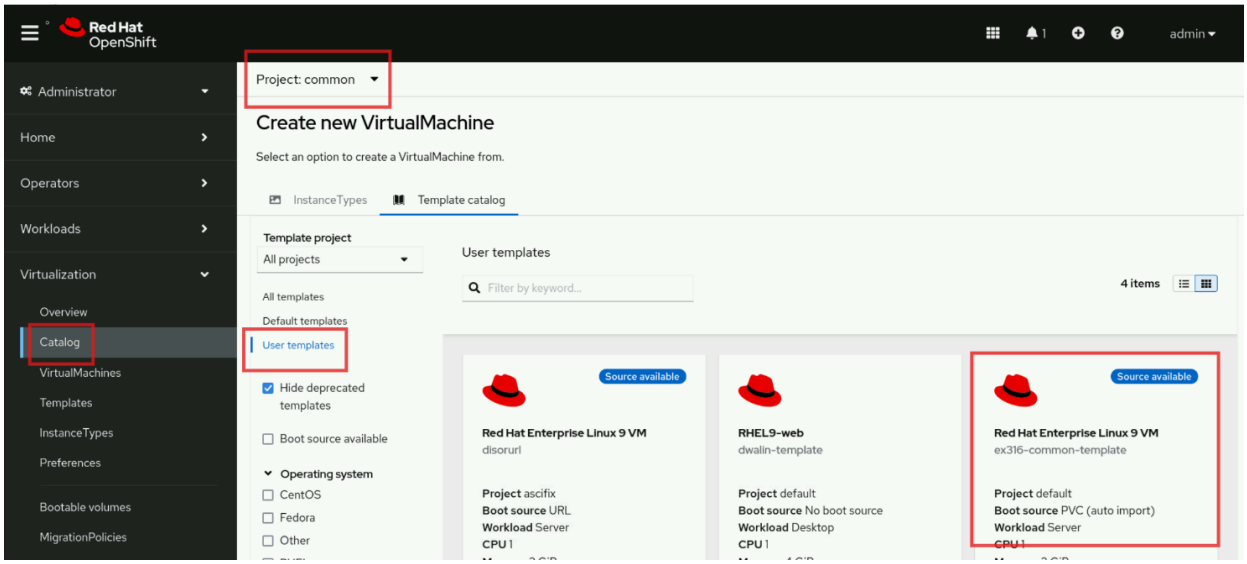
Q-12 Create a virtual machine snapshot

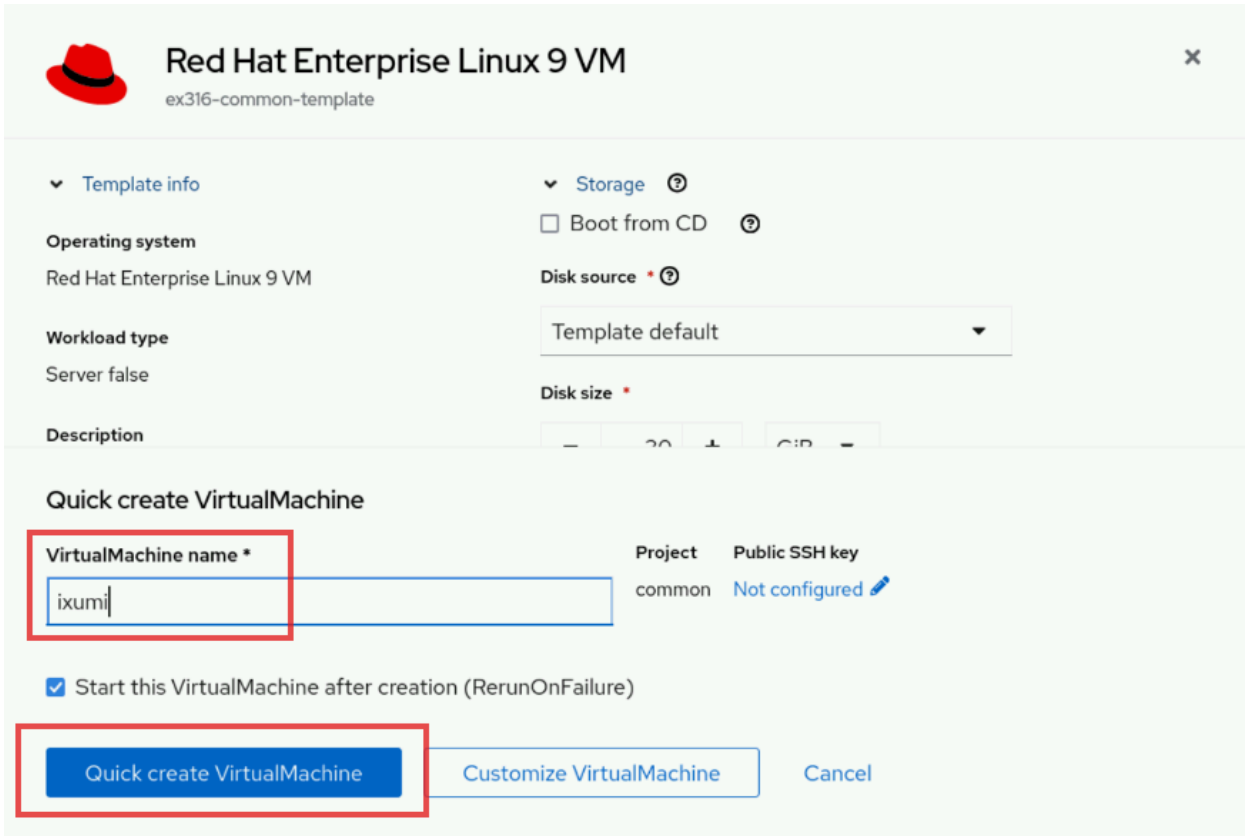
Create a snapshot of a virtual machine **ixumi** as per the following requirements:

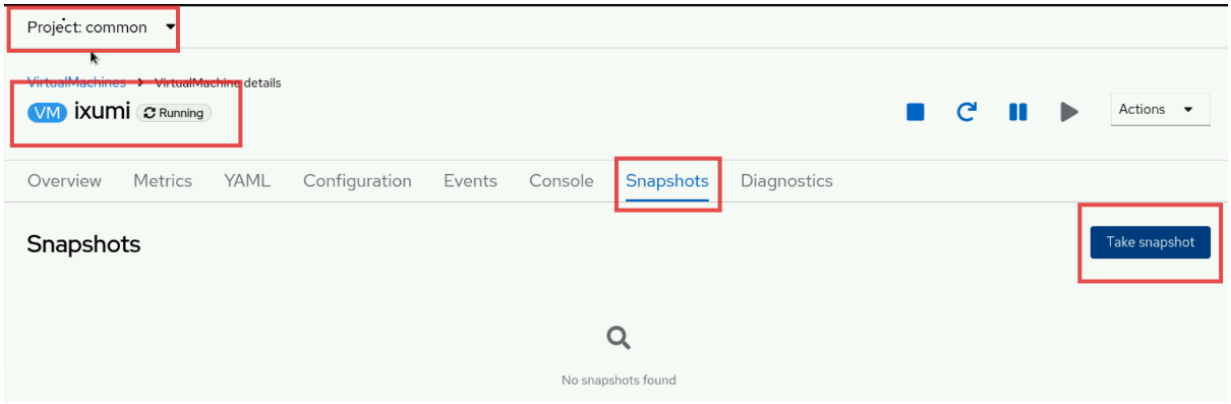
- Create the virtual machine **ixumi** from a template
  ex316-common-template in the project **common**
  NOTE: Do not change the templates
- The snapshot name is **ixumi-freeze**

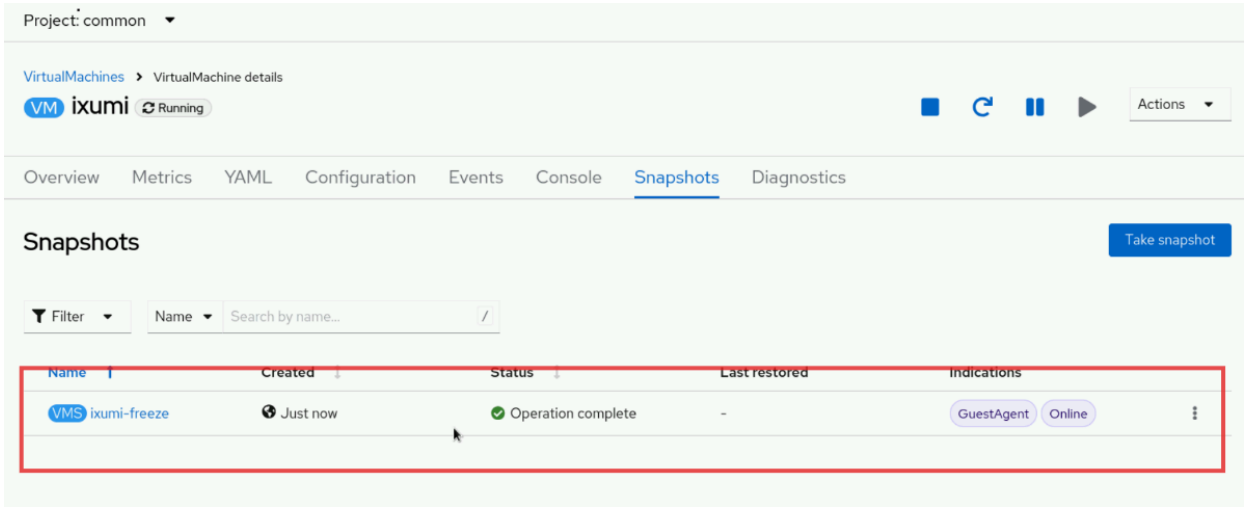
NOTE: You can log in to the virtual as the user **lloyd** with the password: Atenorth3

ANS:


Red Hat OpenShift
admin
Administrator
Home
Operators
Workloads
Virtualization
Overview
Catalog
VirtualMachines
Templates
InstanceTypes
Preferences
Bootable volumes
MigrationPolicies
Project: common
Create new VirtualMachine
Select an option to create a VirtualMachine from.
InstanceTypes
Template catalog
Template project
All projects
All templates
Default templates
User templates
Hide deprecated templates
Boot source available
Operating system
CentOS
Fedora
Other
User templates
Filter by keyword...
4 items
Source available
Red Hat Enterprise Linux 9 VM
disorurl
Project ascifix
Boot source URL
Workload Server
CPU 1
RHEL9-web
dwalin-template
Project default
Boot source No boot source
Workload Desktop
CPU 1
Source available
Red Hat Enterprise Linux 9 VM
ex316-common-template
Project default
Boot source PVC (auto import)
Workload Server
CPU 1


Red Hat Enterprise Linux 9 VM
ex316-common-template
Template info
Operating system
Red Hat Enterprise Linux 9 VM
Workload type
Server false
Description
Storage
Boot from CD
Disk source *
Template default
Disk size *
Quick create VirtualMachine
VirtualMachine name *
ixumi
Project
common
Public SSH key
Not configured
Start this VirtualMachine after creation (RerunOnFailure)
Quick create VirtualMachine
Customize VirtualMachine
Cancel





==========

Q-13 Configure virtual machine migration

Configure node placement for virtual machine migration so that the following conditions are true:

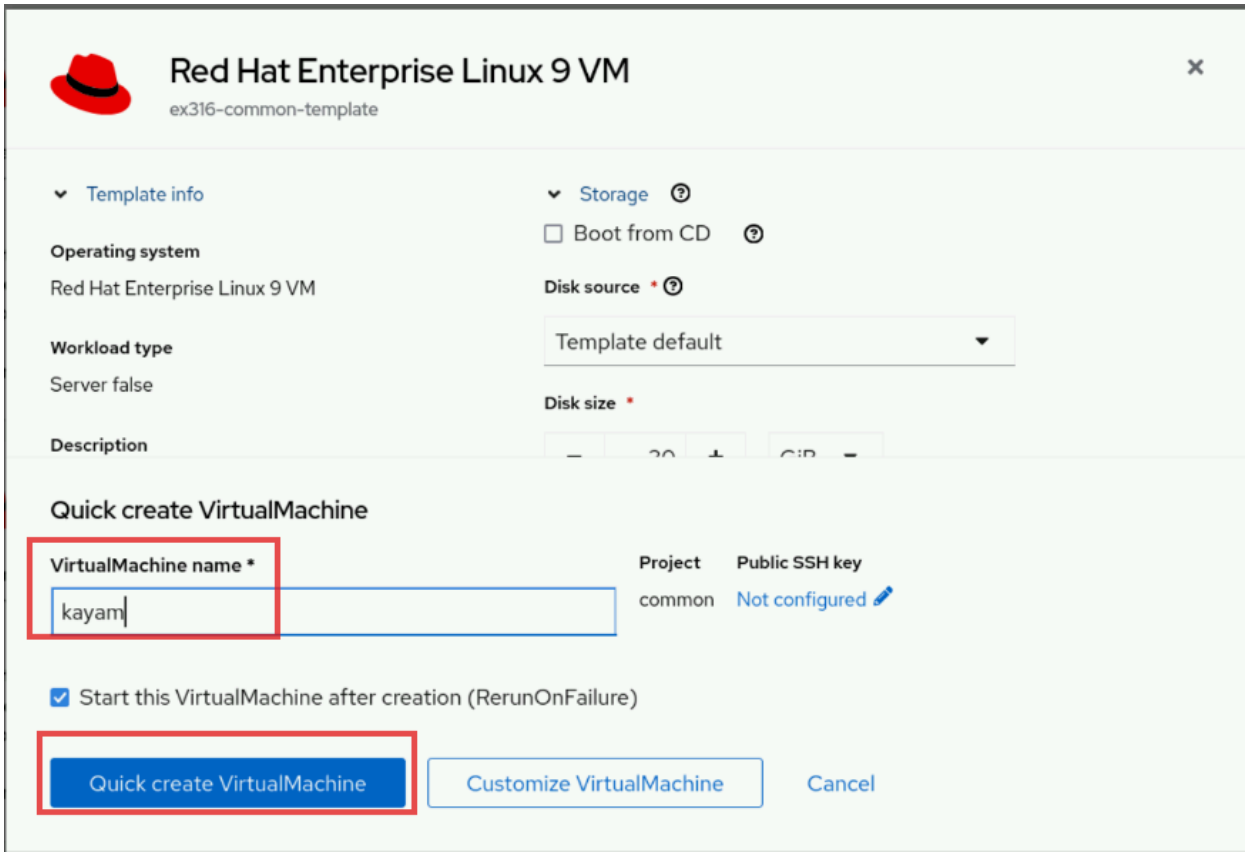
- Create the virtual machine **kayam** from a **templat**
  ex316-common-template in the project **common**
  NOTE: Do not change the template
- The virtual machine **kayam** is able to migrate between the nodes with the label **vsempire=twobats**

- The virtual machine **kayam** in the project **common** is not able to migrate to any other nodes

NOTE 1: the virtual machine will be rebooted before grading in order to test migrations

NOTE 2: you can log in to the virtual machine as the user lloyd with password : **Atenorth3**

ANS:



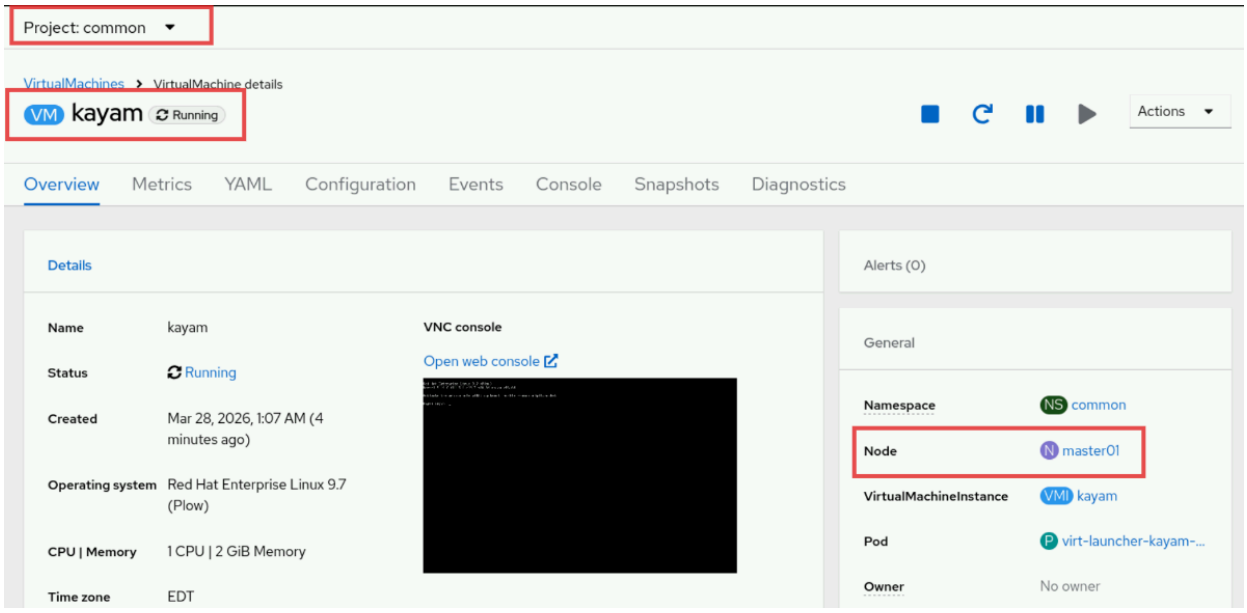
NOTE: in exam the nodes have lable assigned "vsempire=twobats". In lab you need to assigned it, so perform below steps only in lab not in exam

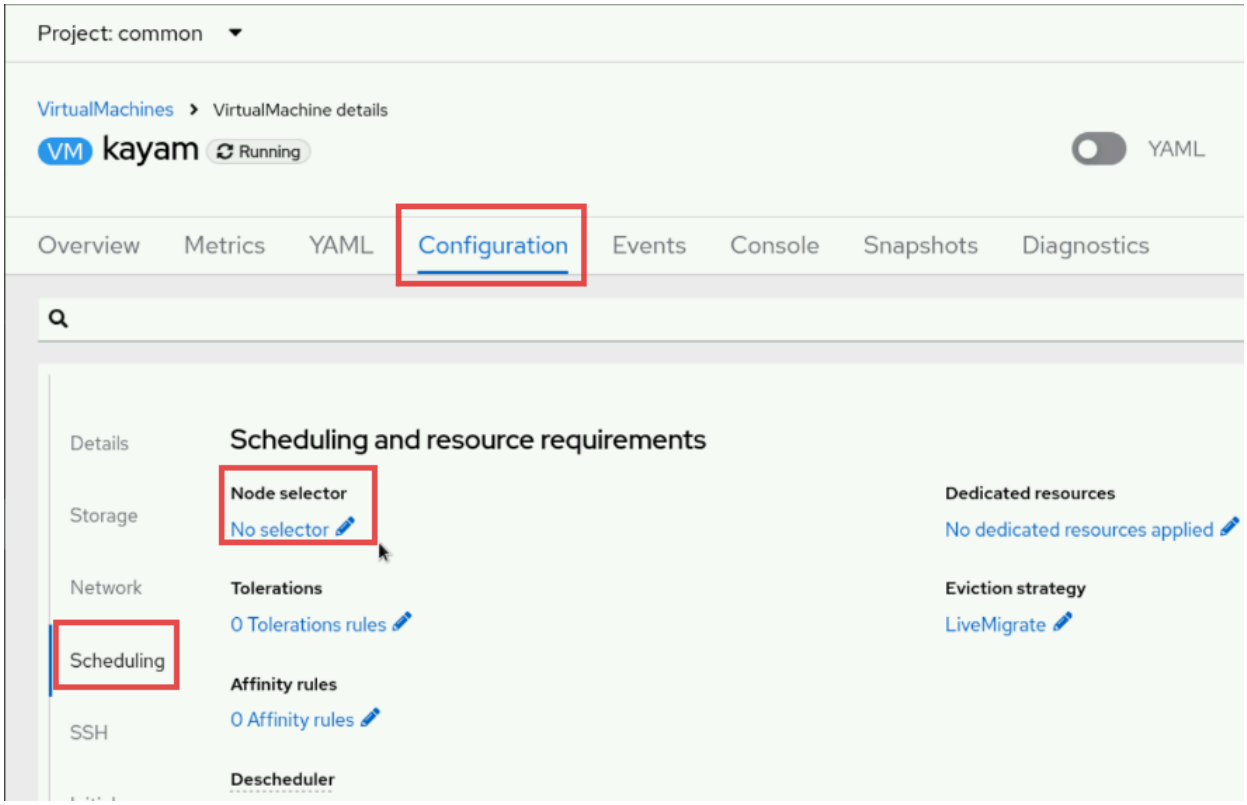
```
[student@workstation ~]$ oc get nodes
\NAME       STATUS   ROLES                          AGE    VERSION
master01   Ready    control-plane,master,worker   304d   v1.31.6
master02   Ready    control-plane,master,worker   304d   v1.31.6
master03   Ready    control-plane,master,worker   304d   v1.31.6
[student@workstation ~]$ oc get nodes -l vsempire=twobats
No resources found
[student@workstation ~]$
```

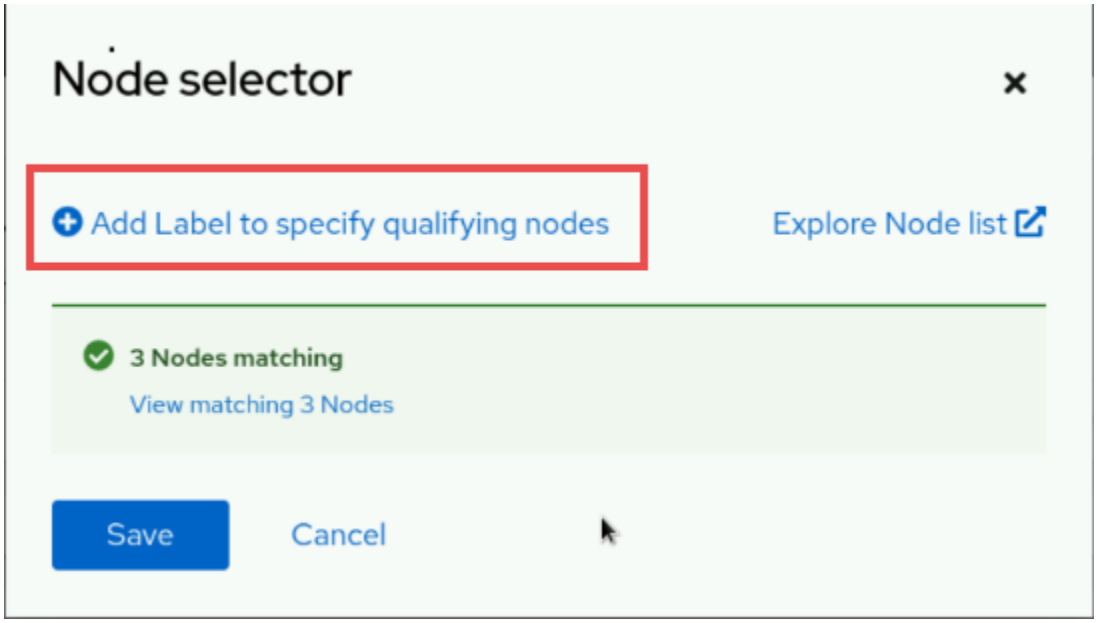
```
[student@workstation ~]$ oc label node master02 vsempire=twobats
node/master02 labeled
[student@workstation ~]$ oc label node master03 vsempire=twobats
node/master03 labeled
[student@workstation ~]$ oc get nodes -l vsempire=twobats
NAME       STATUS   ROLES                          AGE    VERSION
master02   Ready    control-plane,master,worker   304d   v1.31.6
master03   Ready    control-plane,master,worker   304d   v1.31.6
[student@workstation ~]$
```

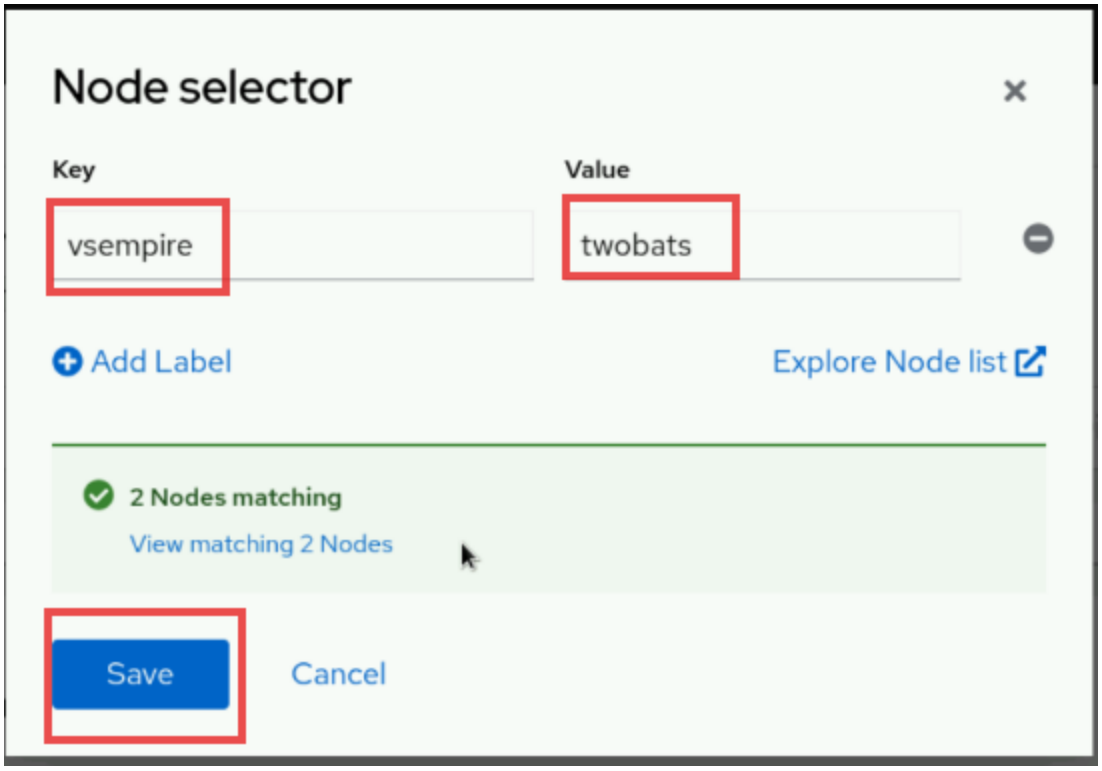
NOW below option you need to perform in exam

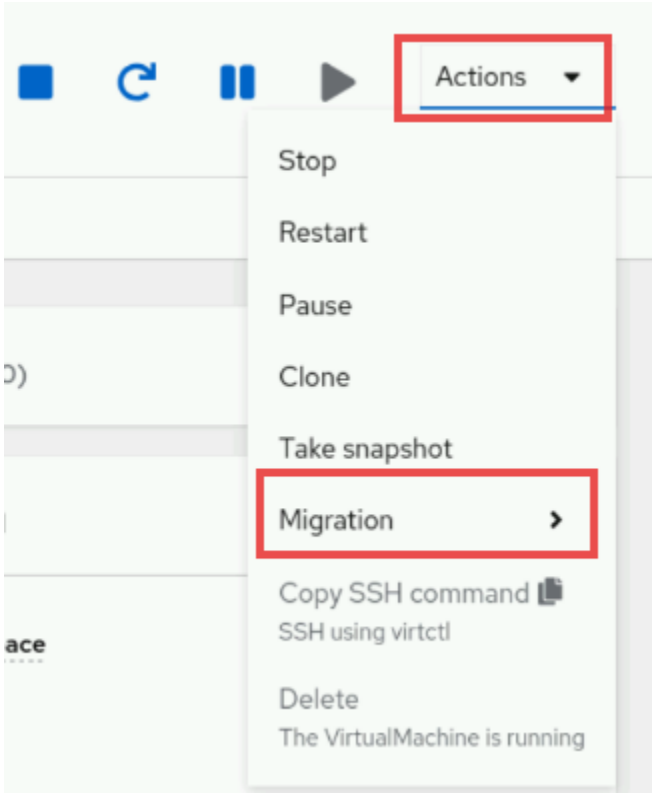
First check existing node where machine is running




Project: common
VirtualMachines
VirtualMachine details
VM
kayam
Running
YAML
Overview
Metrics
YAML
Configuration
Events
Console
Snapshots
Diagnostics
Details
Storage
Network
Scheduling
SSH
Scheduling and resource requirements
Node selector
No selector
Tolerations
0 Tolerations rules
Affinity rules
0 Affinity rules
Descheduler
Dedicated resources
No dedicated resources applied
Eviction strategy
LiveMigrate

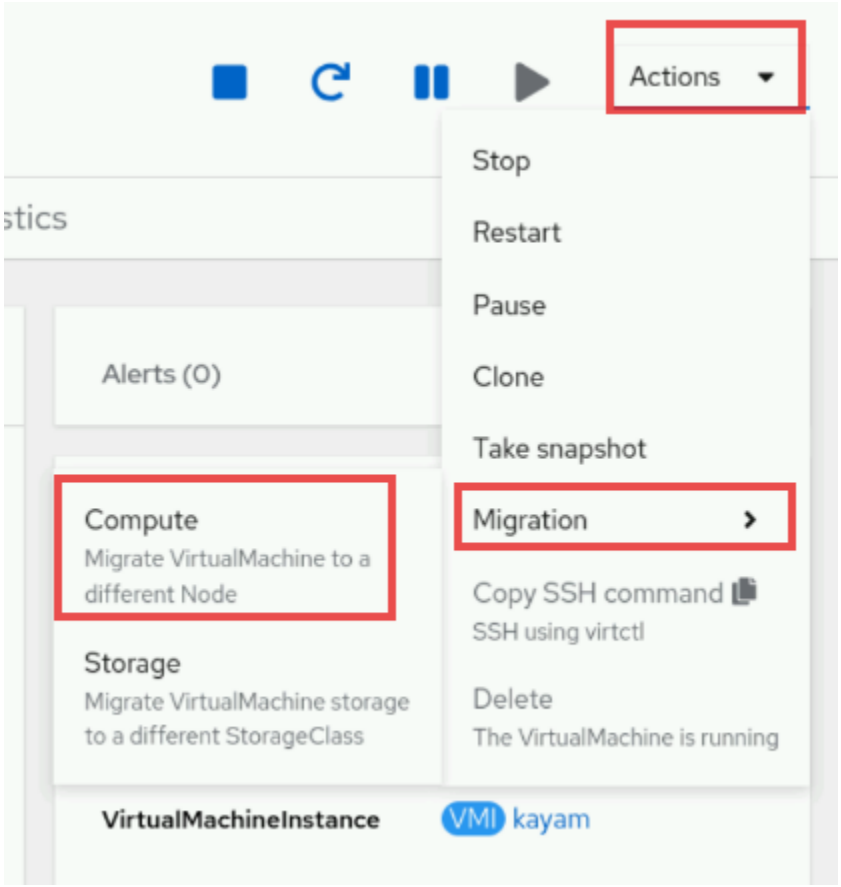

Node selector
Add Label to specify qualifying nodes
Explore Node list
3 Nodes matching
View matching 3 Nodes
Save
Cancel


Node selector
Key
vsempire
Value
twobats
Add Label
Explore Node list
2 Nodes matching
View matching 2 Nodes
Save
Cancel


Actions
Stop
Restart
Pause
Clone
Take snapshot
Migration
Copy SSH command
SSH using virtctl
Delete
The VirtualMachine is running

Post start, verify node placement, it should be between those nodes where label applied

NOTE: in 4.18 you will see "compute" option in migration, so you need to select option → migration → compute.



==========

Q-14 Clone the root disk of a virtual machine

Clone the root disk of virtual machine named **satos** using Data volumes with the following requirements:

- Create the virtual machine **satos** from a template **ex316-common-template** in the project **common**
  NOTE: Do not change the template

- The clone name: **satos-copy**
- The **satos-copy** is the only one clone for the virtual machine **satos**

NOTE 1: Sealing the virtual machine is not necessary

NOTE 2: there are sample files at:

http://rhgls.domain3.example.com/samples/clone

NOTE 3: you can log into the virtual machine as the user **lloyd**

With the password **Atenorth3**

ANS:

- **Create vm and once it running stop it.**

```
[student@workstation ~]$ oc get dv
NAME     PHASE       PROGRESS   RESTARTS   AGE
ixumi    Succeeded   100.0%                28m
kayam    Succeeded   100.0%                21m
satos    Succeeded   100.0%                6m8s
[student@workstation ~]$ oc get dv satos -o yaml > satos-copy.yaml
```

```
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: satos-copy
  namespace: common
spec:
  source:
    pvc:
      namespace: common
      name: satos
  pvc:
    volumeMode: Block
    accessModes:
    - ReadWriteMany
    resources:
      requests:
        storage: 30Gi
    storageClassName: ocs-external-storagecluster-ceph-rbd-virtualization
```

Oc apply -f satos-copy.yaml

```
[student@workstation ~]$ oc get dv
NAME          PHASE       PROGRESS   RESTARTS   AGE
ixumi         Succeeded   100.0%                39m
kayam         Succeeded   100.0%                31m
satos         Succeeded   100.0%                16m
satos-copy    Succeeded   100.0%                3s
```

```
[student@workstation ~]$ oc get pvc
NAME         STATUS   VOLUME                                     CAPACITY   ACCESS MODES
 STORAGECLASS                                    VOLUMEATTRIBUTESCLASS   AGE
ixumi        Bound    pvc-ec20411b-4516-451d-a88c-b1d7f793c36c   30Gi       RWX
 ocs-external-storagecluster-ceph-rbd-virtualization   <unset>                 39m
kayam        Bound    pvc-0522950d-a9a9-4057-8e74-c3f65bdb9c6e   30Gi       RWX
 ocs-external-storagecluster-ceph-rbd-virtualization   <unset>                 32m
satos        Bound    pvc-771bf0bb-98da-4f73-a6de-491d2d6ae2bc   30Gi       RWX
 ocs-external-storagecluster-ceph-rbd-virtualization   <unset>                 17m
satos-copy   Bound    pvc-76827c13-e345-47d3-8473-84f5e7067a8a   30Gi       RWX
 ocs-external-storagecluster-ceph-rbd-virtualization   <unset>                 48s
[student@workstation ~]$
```

==========

Q-15 Prepare the OpenShift cluster for OADP backups

Prepare your Openshift cluster so that it will be able to handle backup using the OADP operator using the following instructions:

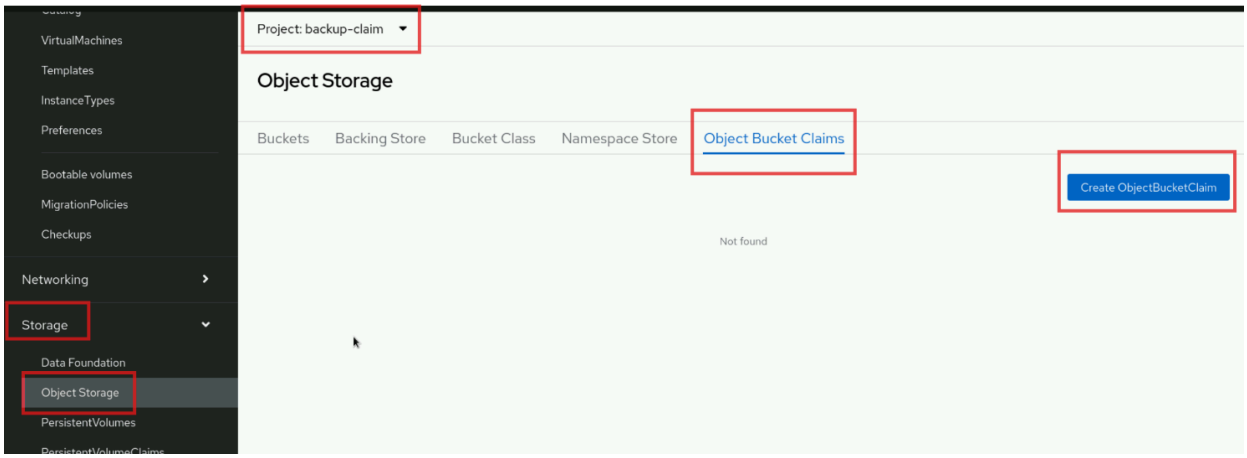
- Install the OADP Operator using defaults for versions and namespace
- Create an **ObjectBucketClaim** in the same namespace , named **backup-claim** using the **openshift-storage.noobaa.io** storage class to create an **ObjectBucket** named **backed**
- Create a configuration file named **.s3cfg** in the home directory of the openshift user so that it will be possible to use the **s3cmd** command to operate on the **S3** bucket successfully
- Create a **BackupStorageLocation** object in the same namespace and make sure it is in the **Available** state
- For each volume snapshot class, set the **velero.io/csi-volumesnapshot-class** label to true

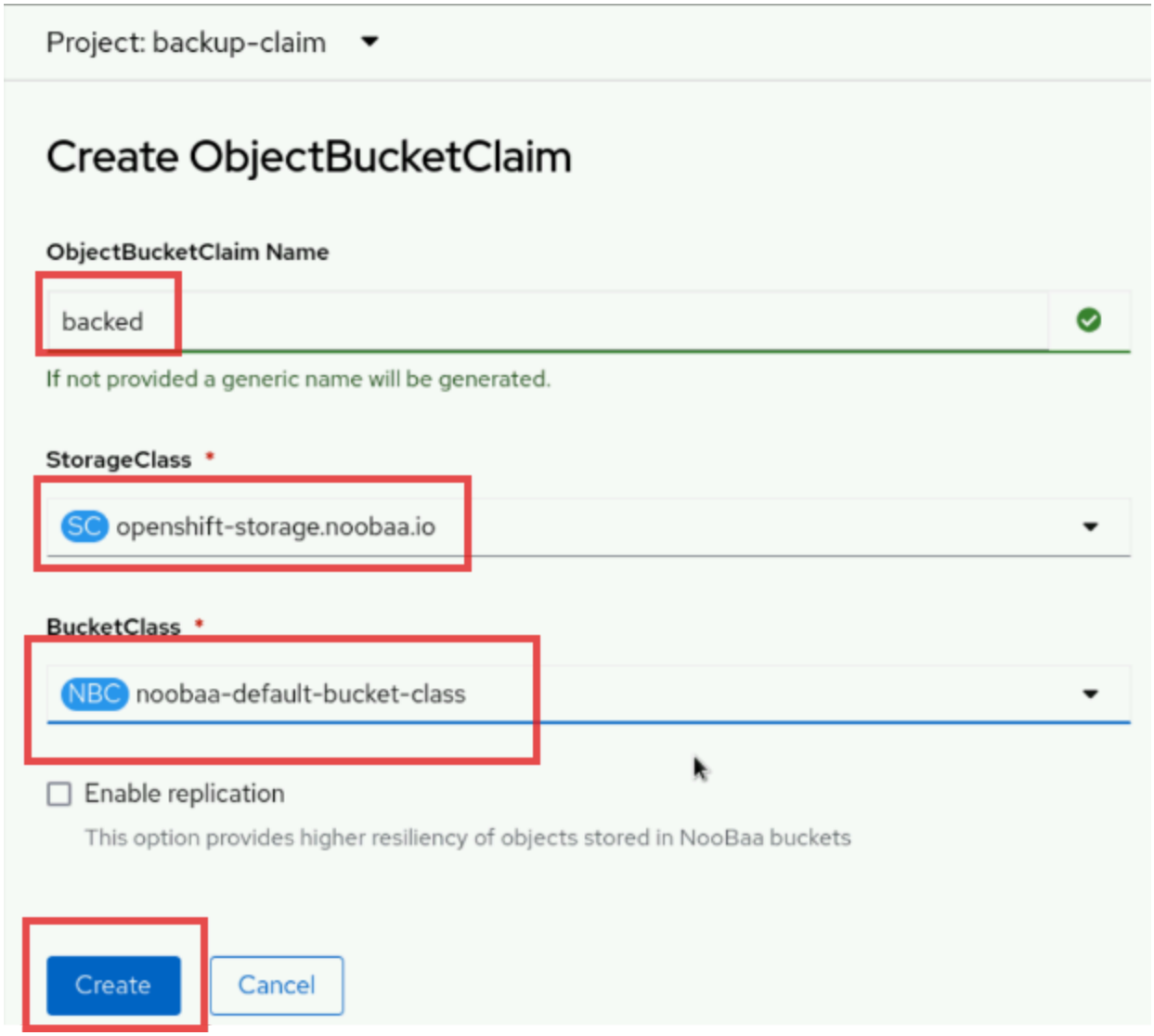
Sample files are available from :
http://rhgls.domain3.example.com/samples/oadp

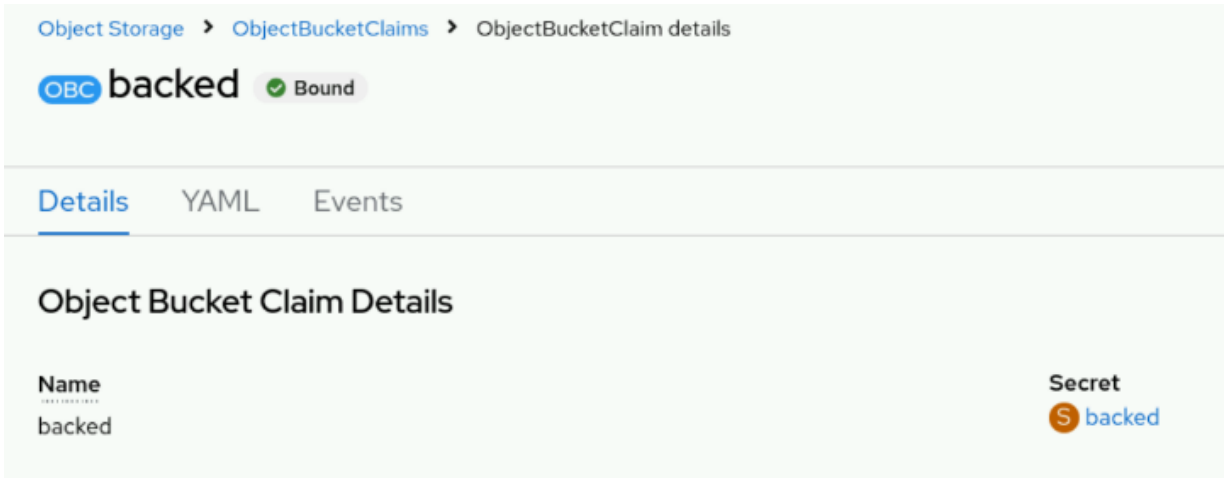
ANS:
NOTE: in lab only perform below cmd to create necessary arrangements
In 4.16 its - lab start -t do316 backup-oadp
In 4.18 its - lab start oadp-operator

Steps:

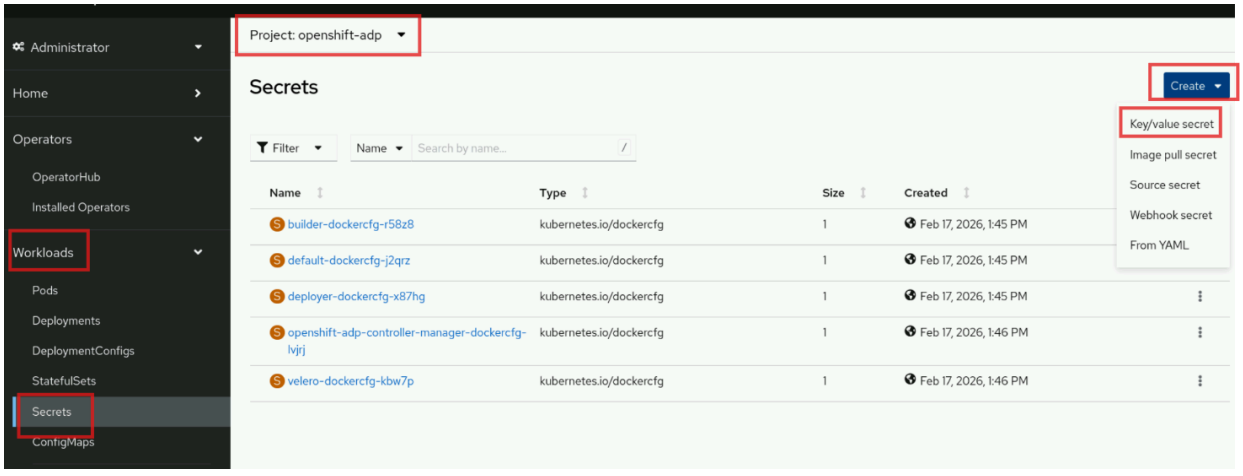
- **Create object bucket claim**
  - **Storage > object bucket claim > select namespace (backup-claim) > Click on Object Bucket Claims > Create ObjectBucketClaim**
  - **Add the name as backed and storage class as "openshift-storage.noobaa.io" and bucket class as "noobaa-default-bucket-class" as defined and click on create**
  - **Scroll down and click on "Revel Values" to see all details and keep that page open in tab**
  - **Now open "workload" page on another tab and click on secret and select namespace "openshift-adp" and click on create and select "Key/Value" secret**
  - **Keep name of secret as cloud-credentials and "cloud" as key**
  - **Mention below detail under value section**

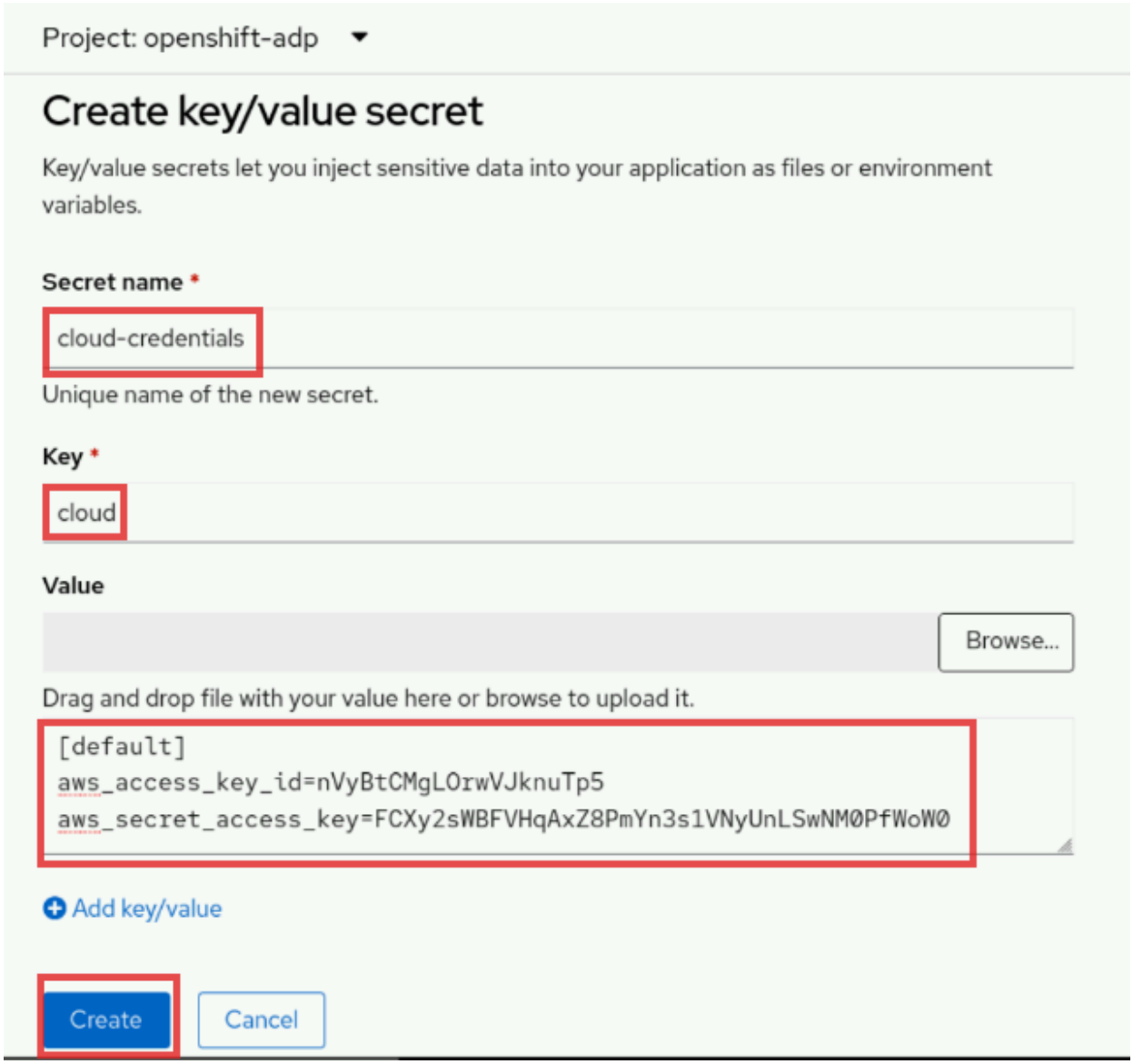

VirtualMachines
Templates
InstanceTypes
Preferences
Bootable volumes
MigrationPolicies
Checkups
Networking
Storage
Data Foundation
Object Storage
PersistentVolumes
Project: backup-claim
Object Storage
Buckets
Backing Store
Bucket Class
Namespace Store
Object Bucket Claims
Create ObjectBucketClaim
Not found

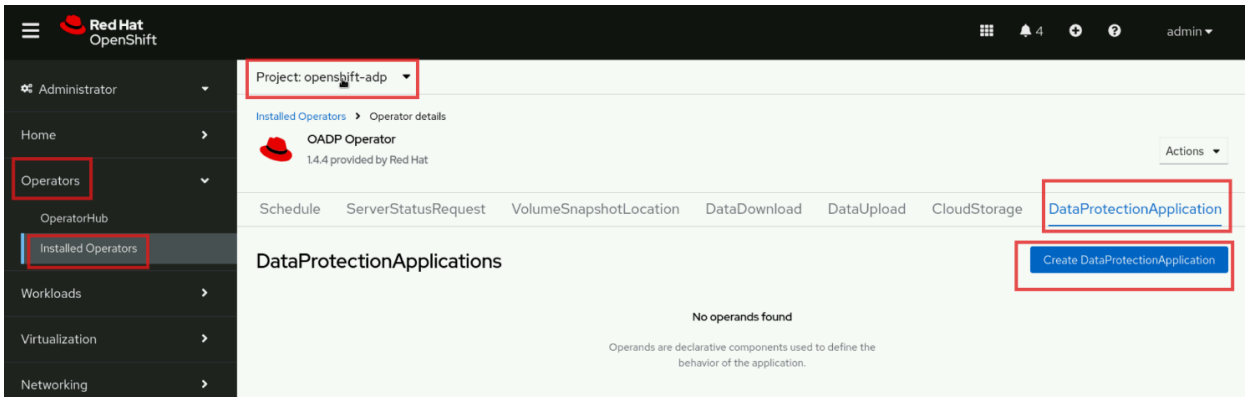

Project: backup-claim
Create ObjectBucketClaim
ObjectBucketClaim Name
backed
If not provided a generic name will be generated.
StorageClass *
SC openshift-storage.noobaa.io
BucketClass *
NBC noobaa-default-bucket-class
Enable replication
This option provides higher resiliency of objects stored in NooBaa buckets
Create
Cancel


Object Storage
ObjectBucketClaims
ObjectBucketClaim details
OBC backed
Bound
Details
YAML
Events
Object Bucket Claim Details
Name
backed
Secret
backed


Object Bucket Claim Data
Hide Values
Endpoint
s3.openshift-storage.svc:443
Bucket Name
backed-47bf7642-3fa6-42c6-ba4c-e35a29dd41e0
Access Key
nVyBtCMgLOrwVJknuTp5
Secret Key
FCXy2sWBFVHqAxZ8PmYn3s1VNyUnLSwNM0PfWoW0


Administrator
Home
Operators
OperatorHub
Installed Operators
Workloads
Pods
Deployments
DeploymentConfigs
StatefulSets
Secrets
ConfigMaps
Project: openshift-adp
Secrets
Create
Key/value secret
Image pull secret
Source secret
Webhook secret
From YAML
Filter
Name
Search by name...
Name
Type
Size
Created
builder-dockercfg-r58z8
kubernetes.io/dockercfg
1
Feb 17, 2026, 1:45 PM
default-dockercfg-j2qrz
kubernetes.io/dockercfg
1
Feb 17, 2026, 1:45 PM
deployer-dockercfg-x87hg
kubernetes.io/dockercfg
1
Feb 17, 2026, 1:45 PM
openshift-adp-controller-manager-dockercfg-lvjrj
kubernetes.io/dockercfg
1
Feb 17, 2026, 1:46 PM
velero-dockercfg-kbw7p
kubernetes.io/dockercfg
1
Feb 17, 2026, 1:46 PM


Project: openshift-adp
Create key/value secret
Key/value secrets let you inject sensitive data into your application as files or environment variables.
Secret name *
cloud-credentials
Unique name of the new secret.
Key *
cloud
Value
Browse...
Drag and drop file with your value here or browse to upload it.
[default]
aws_access_key_id=nVyBtCMgLOrwVJknuTp5
aws_secret_access_key=FCXy2sWBFVHqAxZ8PmYn3s1VNyUnLSwNM0PfWoW0
Add key/value
Create
Cancel


Red Hat OpenShift
admin
Administrator
Home
Operators
OperatorHub
Installed Operators
Workloads
Virtualization
Networking
Project: openshift-adp
Installed Operators
Operator details
OADP Operator
1.4.4 provided by Red Hat
Actions
Schedule
ServerStatusRequest
VolumeSnapshotLocation
DataDownload
DataUpload
CloudStorage
DataProtectionApplication
DataProtectionApplications
Create DataProtectionApplication
No operands found
Operands are declarative components used to define the behavior of the application.

Create DataProtectionApplication

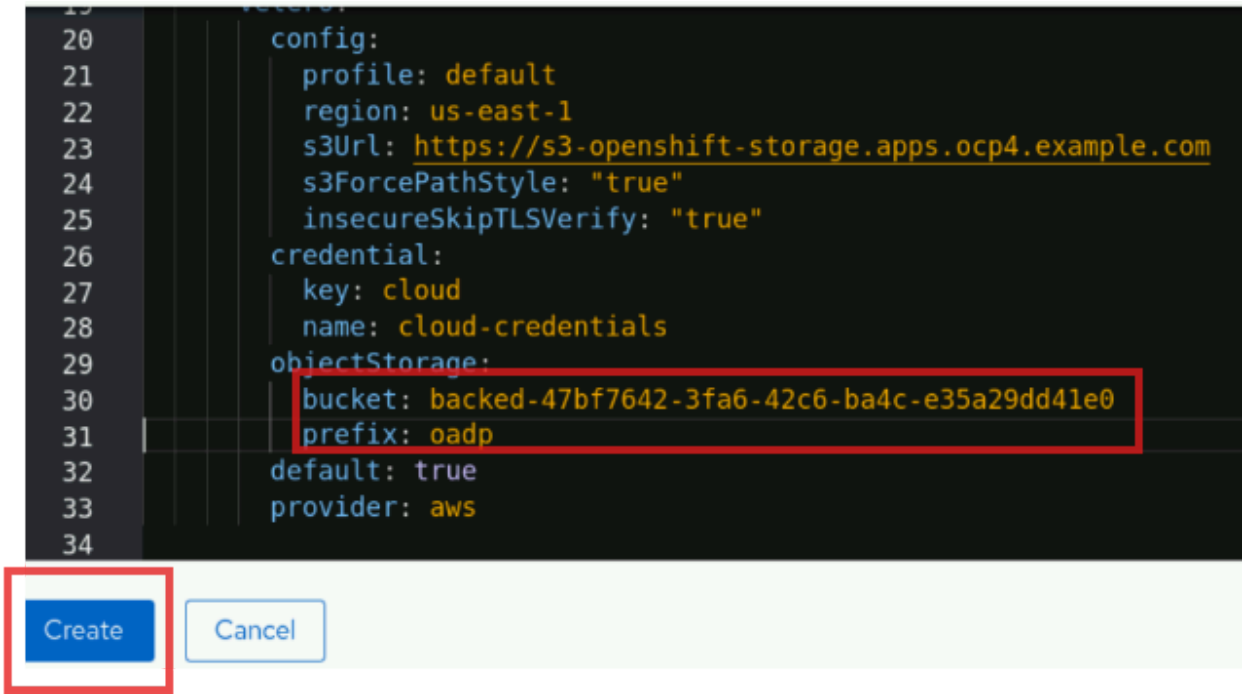
Create by manually entering YAML or JSON definitions, or by dragging and dropping a file into the editor.

Configure via: ○ Form view ◉ YAML view

Alt + F1 Accessibility help | View shortcuts | ☑ Show tooltips

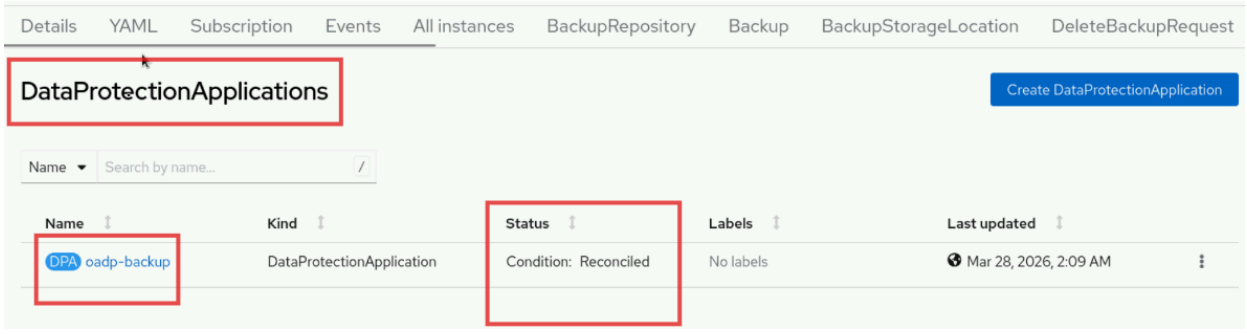
```yaml
apiVersion: oadp.openshift.io/v1alpha1
kind: DataProtectionApplication
metadata:
  name: oadp-backup
  namespace: openshift-adp
spec:
  configuration:
    nodeAgent:
      enable: true
      uploaderType: kopia
    velero:
      defaultPlugins:
        - openshift
        - aws
        - kubevirt
```

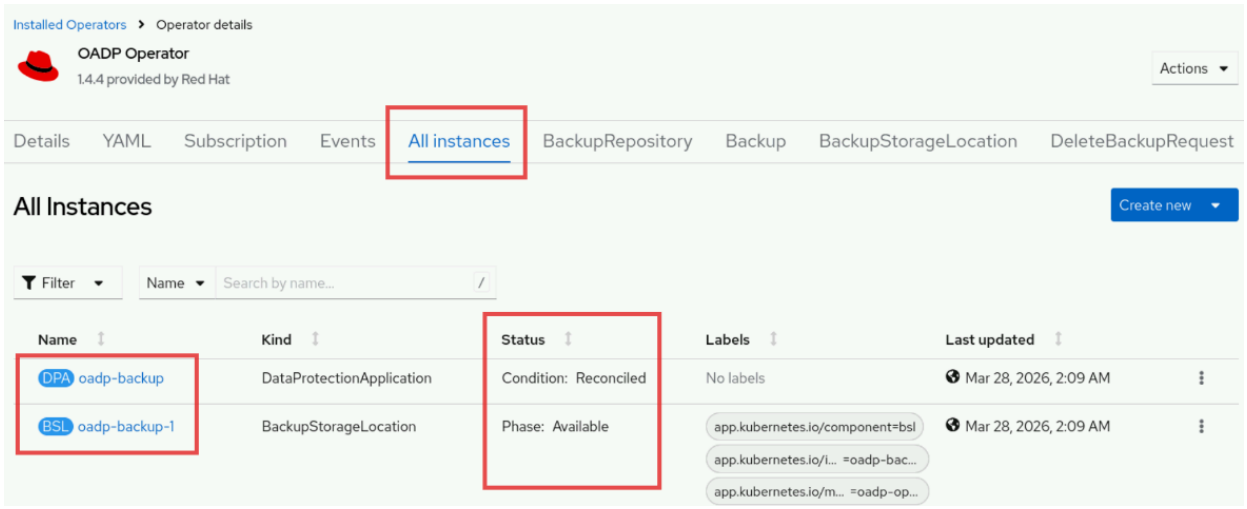
```yaml
      defaultPlugins:
        - openshift
        - aws
        - csi
        - kubevirt
      defaultSnapshotMoveData: true
  backupLocations:
    - velero:
        config:
          profile: default
          region: us-east-1
          s3Url: https://s3-openshift-storage.apps.ocp4.example.com
          s3ForcePathStyle: "true"
          insecureSkipTLSVerify: "true"
        credential:
```



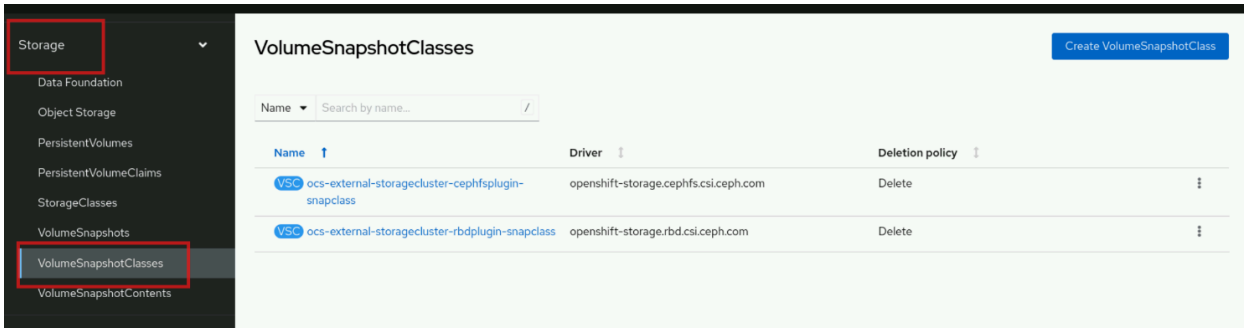
FYI below is the only for reference of data i have added in dataprotectionapplication (DPA)

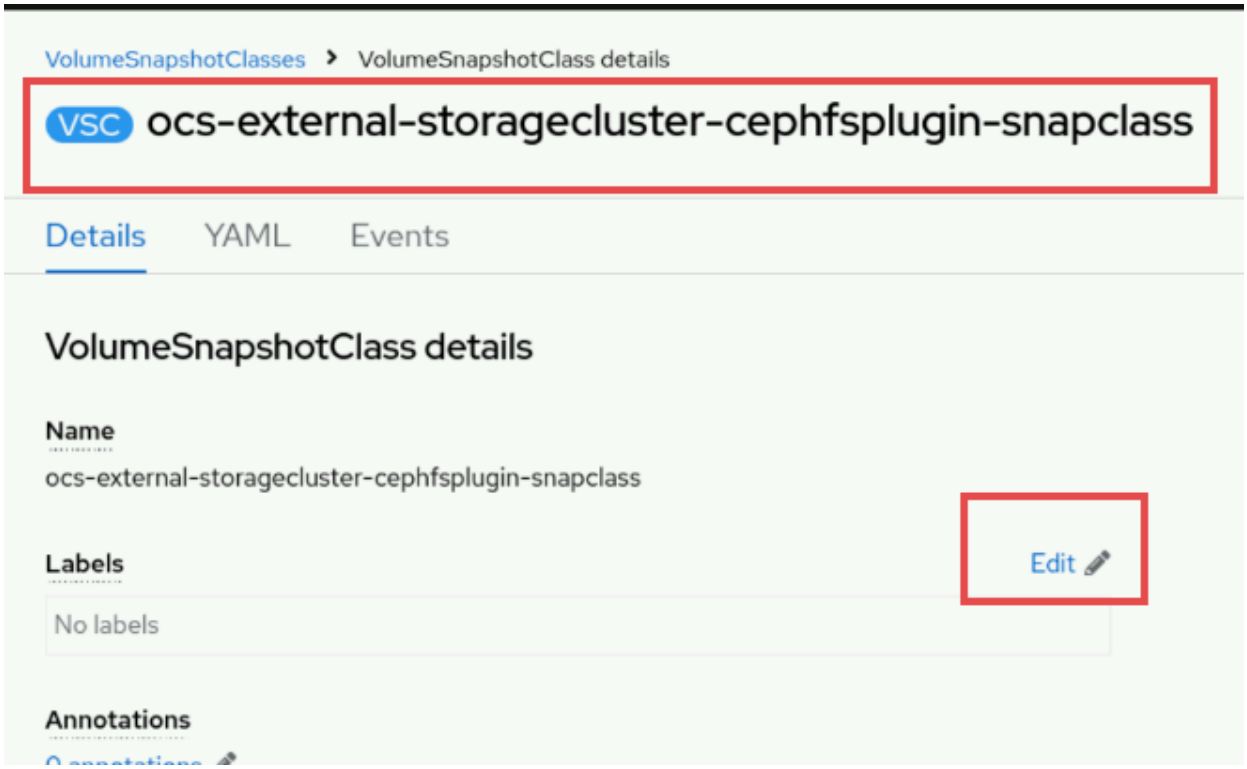
```
spec:
  backupLocations:
  - velero:
      config:
        insecureSkipTLSVerify: "true"
        profile: default
        region: us-east-1
        s3ForcePathStyle: "true"
        s3Url: https://s3-openshift-storage.apps.ocp4.example.com
      credential:
        key: cloud
        name: cloud-credentials
      default: true
      objectStorage:
        bucket: backed-47bf7642-3fa6-42c6-ba4c-e35a29dd41e0
        prefix: oadp
      provider: aws
  configuration:
    nodeAgent:
      enable: true
      uploaderType: kopia
    velero:
      defaultPlugins:
      - openshift
      - aws
      - csi
      - kubevirt
      defaultSnapshotMoveData: true
```

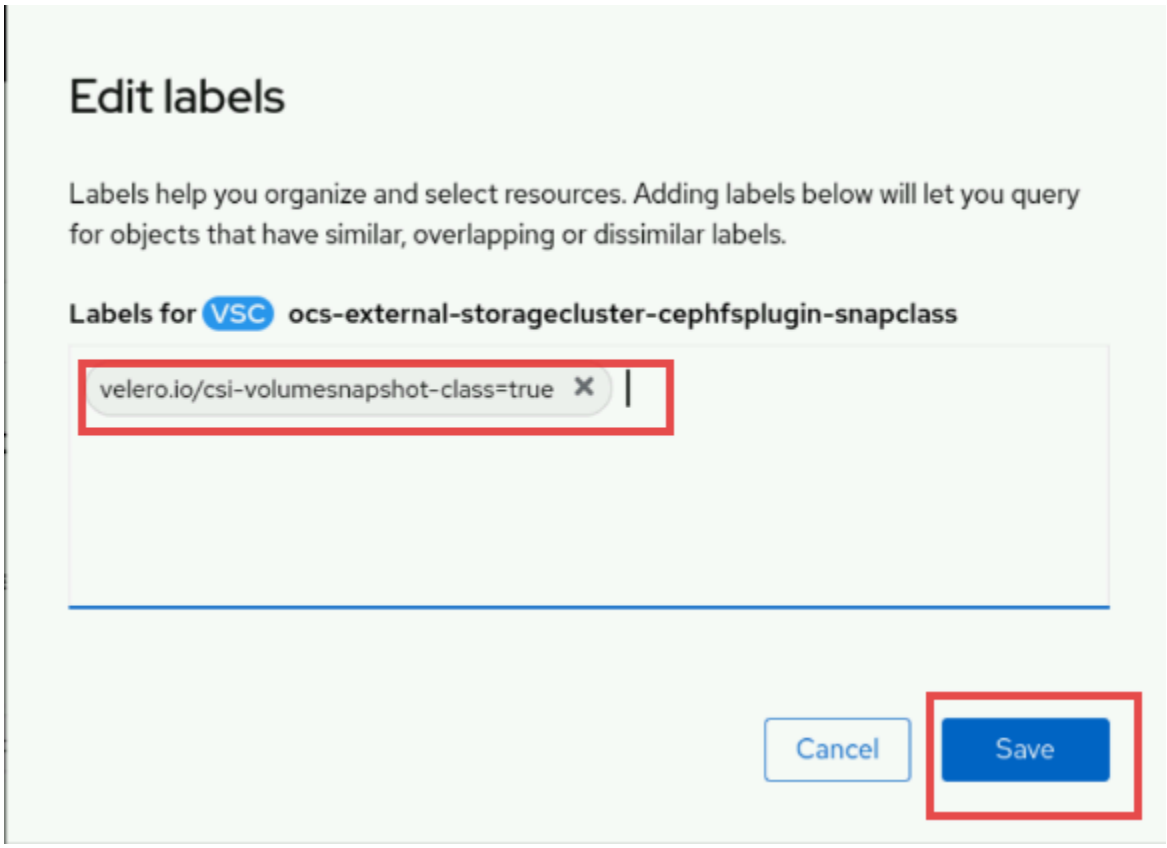


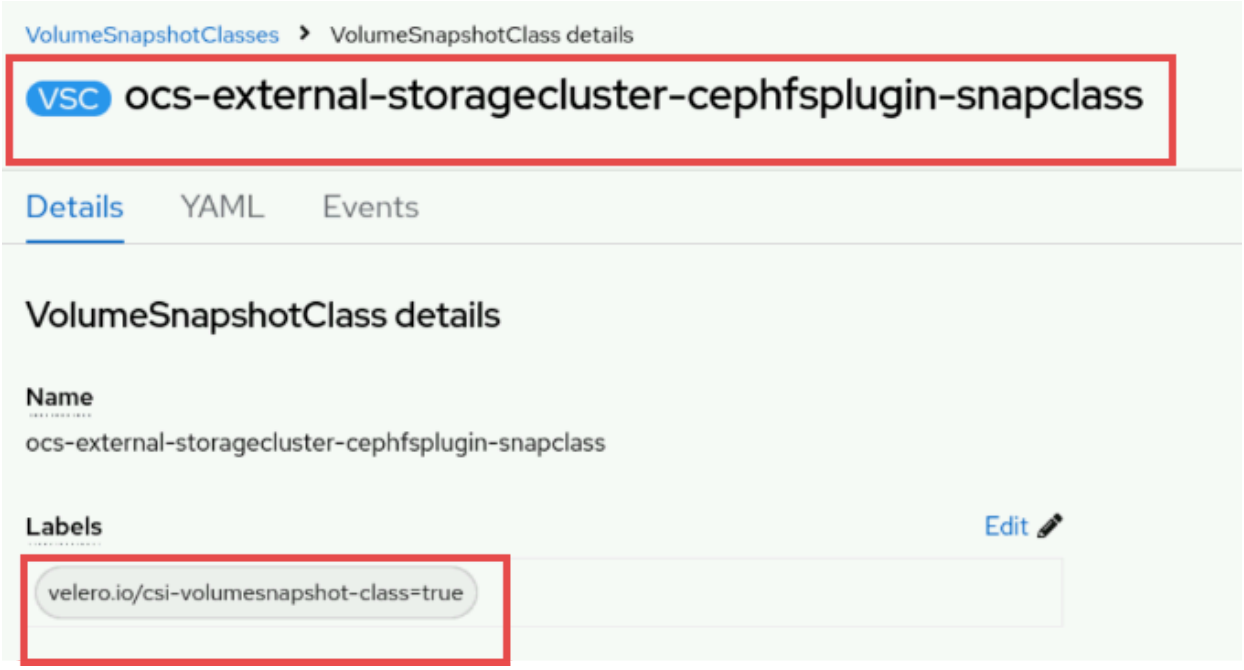


Add storage label to volumesnapshotclass




VolumeSnapshotClasses
VolumeSnapshotClass details
VSC
ocs-external-storagecluster-cephfsplugin-snapclass
Details
YAML
Events
VolumeSnapshotClass details
Name
ocs-external-storagecluster-cephfsplugin-snapclass
Labels
Edit
No labels
Annotations


Edit labels
Labels help you organize and select resources. Adding labels below will let you query for objects that have similar, overlapping or dissimilar labels.
Labels for VSC ocs-external-storagecluster-cephfsplugin-snapclass
velero.io/csi-volumesnapshot-class=true
Cancel
Save



Do the same for other volumesnapshotclass

NOTE: you can do the same using oc lable cmd as well

```
[student@workstation ~]$ oc get volumesnapshotclasses.snapshot.storage.k8s.io
NAME                                                DRIVER
   DELETIONPOLICY   AGE
ocs-external-storagecluster-cephfsplugin-snapclass   openshift-storage.cephfs.csi.ceph.com
   Delete           38d
ocs-external-storagecluster-rbdplugin-snapclass      openshift-storage.rbd.csi.ceph.com
   Delete           38d
[student@workstation ~]$ oc label volumesnapshotclasses.snapshot.storage.k8s.io ocs-extern
al-storagecluster-rbdplugin-snapclass velero.io/csi-volumesnapshot-class=true
volumesnapshotclass.snapshot.storage.k8s.io/ocs-external-storagecluster-rbdplugin-snapclas
s labeled
```

==========

Q-16 Backup VMs using OADP

Using the OADP infrastructure that you configuerd in a previous item, create a consistent backup named: **italy-pr-do-backup** that contain the virtual machines and related objects .

Sample files are available from : **http://rhgls.domain3.example.com/samples/oadp-backup**

NOTE: During evaluation, the virtual machine in the **rampops** namespace will be deleted and then restored from the **italy-pr-do-backup** backup to ensure the backup is consistent.

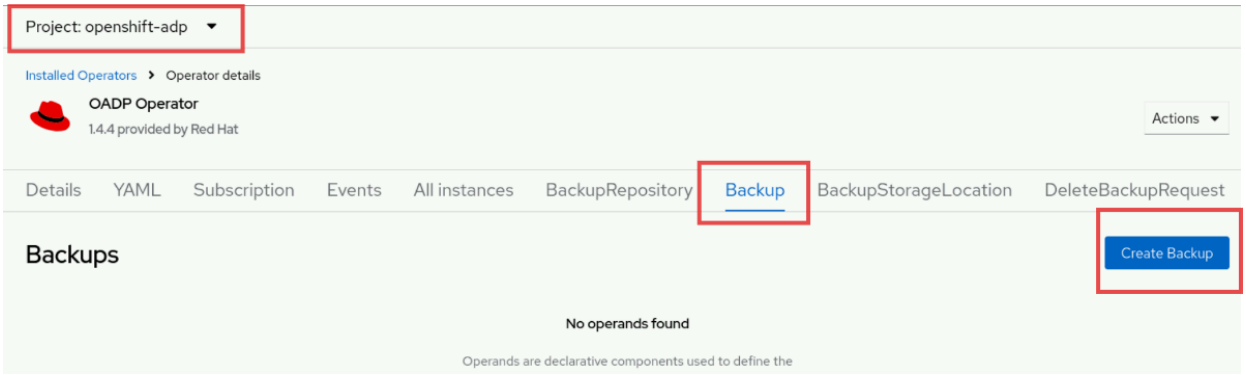
ANS:

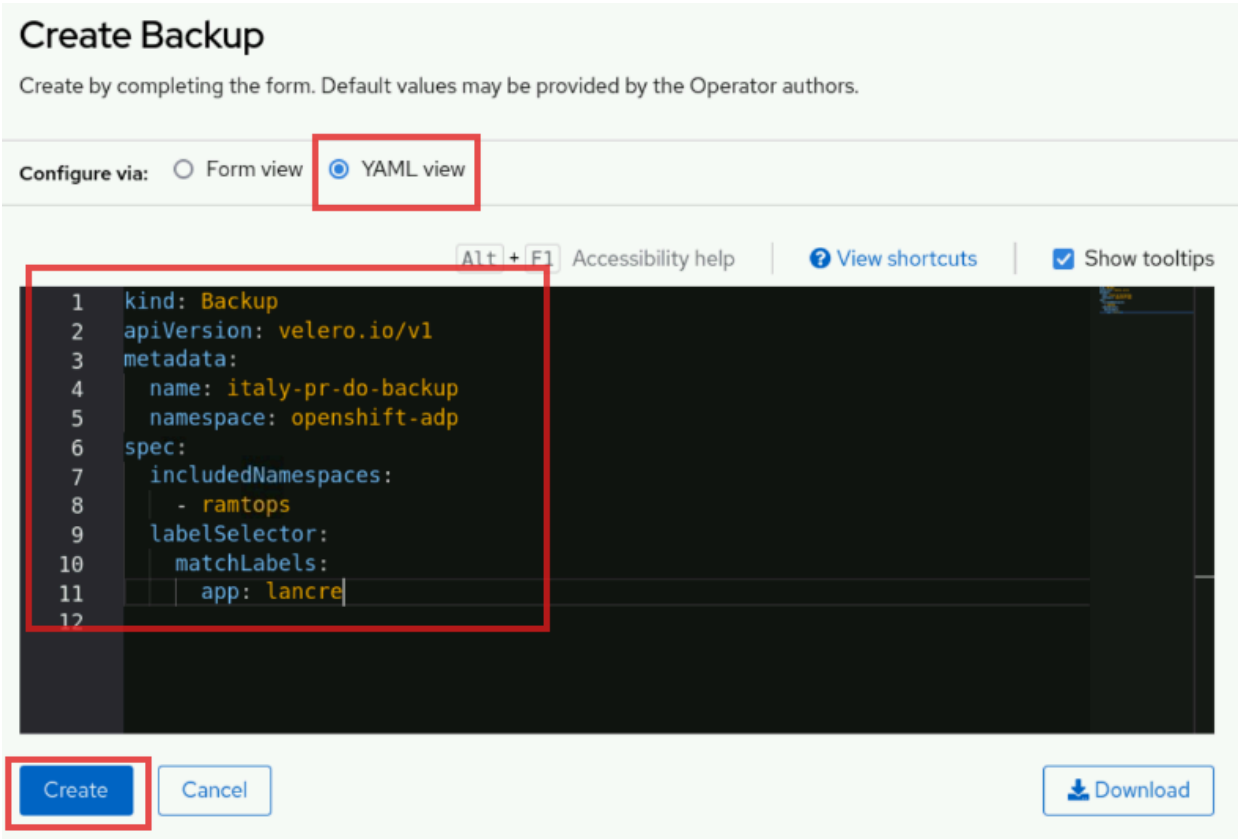
- Change to project ramtops and verify labels

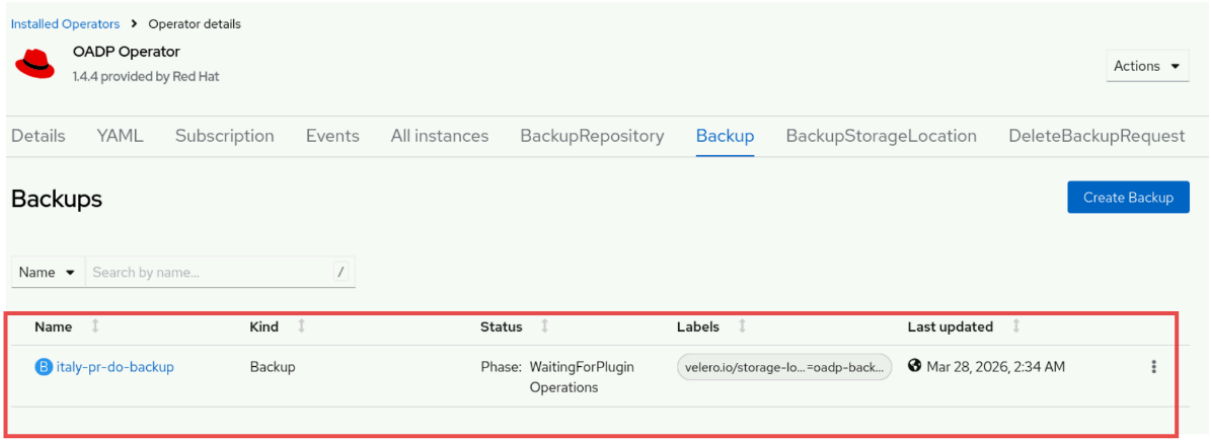
```
[student@workstation ~]$ oc get vmi --show-labels
NAME     AGE   PHASE     IP          NODENAME   READY   LABELS
lancre   85m   Running   10.10.0.7   master01   True    app=lancre,kubevirt.io/domain=lenc
re,kubevirt.io/nodeName=master01,kubevirt.io/size=small,network.kubevirt.io/headlessServic
e=headless
[student@workstation ~]$ oc label svc lancre-svc app=lancre
service/lancre-svc labeled
[student@workstation ~]$ oc get svc lancre-svc --show-labels
NAME         TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)   AGE    LABELS
lancre-svc   ClusterIP   172.30.190.197   <none>        80/TCP    2d1h   app=lancre
[student@workstation ~]$
```

```
[student@workstation ~]$ oc get ns ramtops --show-labels
NAME      STATUS   AGE    LABELS
ramtops   Active   2d2h   kubernetes.io/metadata.name=ramtops,pod-security.kubernetes.io/a
udit-version=latest,pod-security.kubernetes.io/audit=restricted,pod-security.kubernetes.io
/warn-version=latest,pod-security.kubernetes.io/warn=restricted
[student@workstation ~]$
```

Now create backup object







You can verify backup progress, backup and restore using velero cmd. Do below configuration

```
[student@workstation ~]$ alias velero='oc exec -it -n openshift-adp deployments/velero -c
velero -- ./velero'
[student@workstation ~]$ velero get backup
NAME                STATUS      ERRORS   WARNINGS   CREATED                         EXPIR
ES   STORAGE LOCATION   SELECTOR
italy-pr-do-backup   Completed   0        0          2026-03-28 06:34:35 +0000 UTC   29d
     oadp-backup-1      app=lancre
[student@workstation ~]$
```

```
[student@workstation ~]$ velero describe backup italy-pr-do-backup --details --insecure-skip-tls-verify
Name:         italy-pr-do-backup
Namespace:    openshift-adp
Labels:       velero.io/storage-location=oadp-backup-1
Annotations:  velero.io/resource-timeout=10m0s
              velero.io/source-cluster-k8s-gitversion=v1.31.6
              velero.io/source-cluster-k8s-major-version=1
              velero.io/source-cluster-k8s-minor-version=31

Phase:  Completed


Namespaces:
  Included:  ramtops
  Excluded:  <none>

Resources:
  Included:        *
  Excluded:        <none>
  Cluster-scoped:  auto

Label selector:  app=lancre
```

==========

Q-17 Configure a liveness probe

Configure a liveness probe for the **kagura** virtual machine by testing the liveness of a database server so that the following conditions are true:

- Create the virtual machine kagura from a template **ex316-common-template** in the project **common**
  NOTE: Do not change the templates
- The **mariadb-server** packages that provides the database server is installed in the virtual machine
- The **mariadb** service is enabled and started automatically during the boot process of the virtual machine

- The database server uses the default configuration at **TCP port 3306**
- The liveness probe tests the database server at **TCP port 3306**
- The time, in seconds, after the virtual machine instance starts before the liveness probe is initiated is **120**
- The delay, in seconds , between performing probes is **5**

NOTE: The virtual machine will be rebooted before grading .

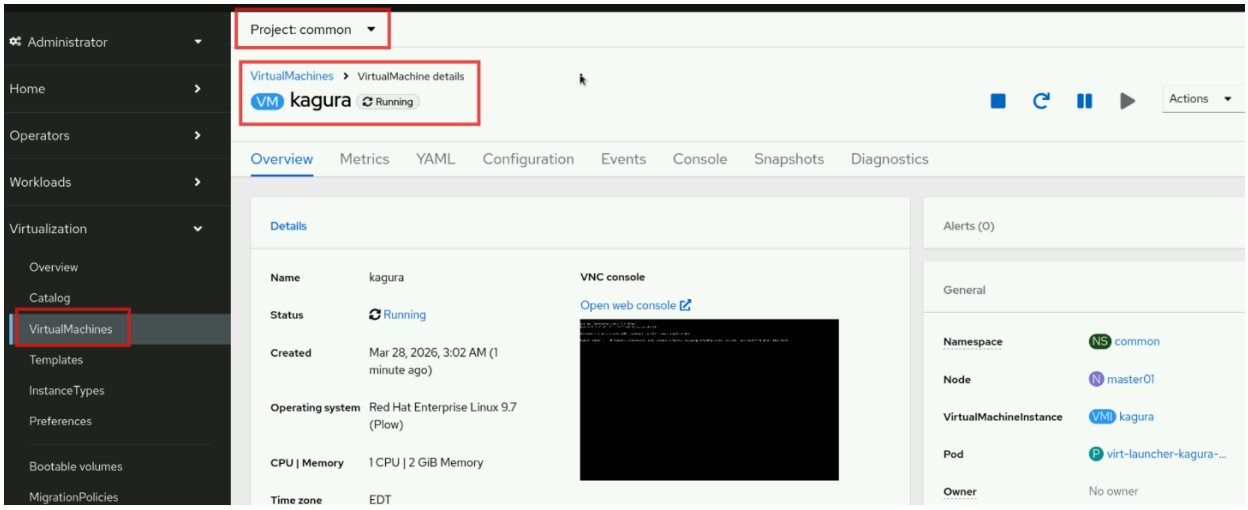
The file: **http://materials.domain3.example.com/files/rhel.repo**

Can be used to configure repositories in the virtual machine by downloading the files into the **/etc/yum.repos.d** directory.

NOTE: you can log into the virtual machine as the user **lloyd** with password **Atenorth3**

ANS:

- **Create machine using given template**
-



```
[student@workstation ~]$ oc project common
Now using project "common" on server "https://api.ocp4.example.com:6443".
[student@workstation ~]$ oc get vmi
NAME      AGE     PHASE     IP           NODENAME   READY
ixumi     124m    Running   10.9.0.9     master03   True
kagura    2m30s   Running   10.10.0.50   master01   True
kayam     117m    Running   10.9.0.12    master03   True
[student@workstation ~]$ virtctl console kagura
```

```
Successfully connected to kagura console. The escape sequence is ^]

kagura login: lloyd
Password:
[lloyd@kagura ~]$
```

- **Switch to root using sudo -i cmd**
- **Change to /etc/yum.repos.d directory**
- **And download the repo file using curl -lo http://materials.domain3.example.com/files/rhel.repo cmd**

```
[root@kagura yum.repos.d]#http://utility.lab.example.com:8080/openshift4/rhel_dvd.repol_dv
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   366  100   366    0     0   5462      0 --:--:-- --:--:-- --:--:--  5462
[root@kagura yum.repos.d]# ls
rhel_dvd.repo
[root@kagura yum.repos.d]#
```

```
[root@kagura yum.repos.d]# ls
rhel_dvd.repo
[root@kagura yum.repos.d]# cat rhel_dvd.repo
[rhel-9.5-for-x86_64-baseos-rpms]
baseurl = http://content.example.com/rhel9.5/x86_64/dvd/BaseOS
enabled = true
gpgcheck = false
name = Red Hat Enterprise Linux 9.5 BaseOS (dvd)

[rhel-9.5-for-x86_64-appstream-rpms]
baseurl = http://content.example.com/rhel9.5/x86_64/dvd/AppStream
enabled = true
gpgcheck = false
name = Red Hat Enterprise Linux 9.5 AppStream (dvd)
[root@kagura yum.repos.d]#
```

```
[root@kagura yum.repos.d]# yum install mariadb* -y
Red Hat Enterprise Linux 9.5 BaseOS (dvd)        14 MB/s | 2.0 MB     00:00
Red Hat Enterprise Linux 9.5 AppStream (dvd)     30 MB/s | 7.2 MB     00:00
Dependencies resolved.
================================================================================
=========
 Package                      Arch    Version             Repository
     Size
================================================================================
=========
Installing:
 mariadb                      x86_64  3:10.5.22-1.el9_2   rhel-9.5-for-x86_64-appstream-rp
ms  1.6 M
 mariadb-backup               x86_64  3:10.5.22-1.el9_2   rhel-9.5-for-x86_64-appstream-rp
ms  6.5 M
```

```
[root@kagura yum.repos.d]# systemctl enable --now mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.servic
e.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.servi
```

```
[root@kagura yum.repos.d]# systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
     Active: active (running) since Sat 2026-03-28 03:13:00 EDT; 1min 0s ago
       Docs: man:mariadbd(8)
```

Now exit from machine and type “oc edit vm kagura” and go to spec.template.spec and livenessprobe setting as below

Save it and restart it using virtctl restart kagura and wait till time vm is running again.

Verify using oc describe vm kagura cmd you should see something like below

```
livenessProbe:
  initialDelaySeconds: 120
  periodSeconds: 5
  tcpSocket:
    port: 3306
```

```
Liveness Probe:
  Initial Delay Seconds:  120
  Period Seconds:         5
  Tcp Socket:
    Port:  3306
```

==========

Q-18 Prepare a virtual machine for a node failure

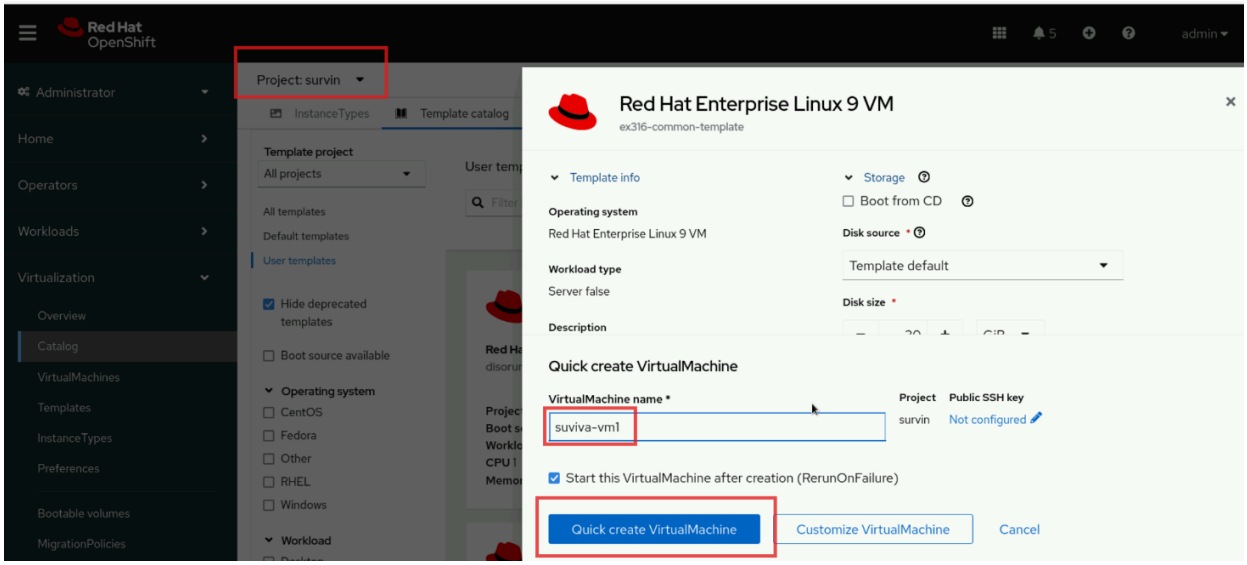
Create and configure a virtual machine to survive a node failure so that the following conditions are true:

- The virtual machine **suviva-vm1** is deployed using the template **ex316-common-template** in the project survin
  NOTE: Do not change the templates
- The virtual machine is automatically scheduled on either:
  - **master1.domain3.example.com or master2.domain3.example.com**
- In the event of a failure of either node,the virtual machine is automatically started on the other node

NOTE: The virtual machine will be rebooted before grading.
NOTE: you can log into the virtual machine as the user **lloyd** with the password **Aetnorth3**

ANS:

- **Change to project survin and create vm**

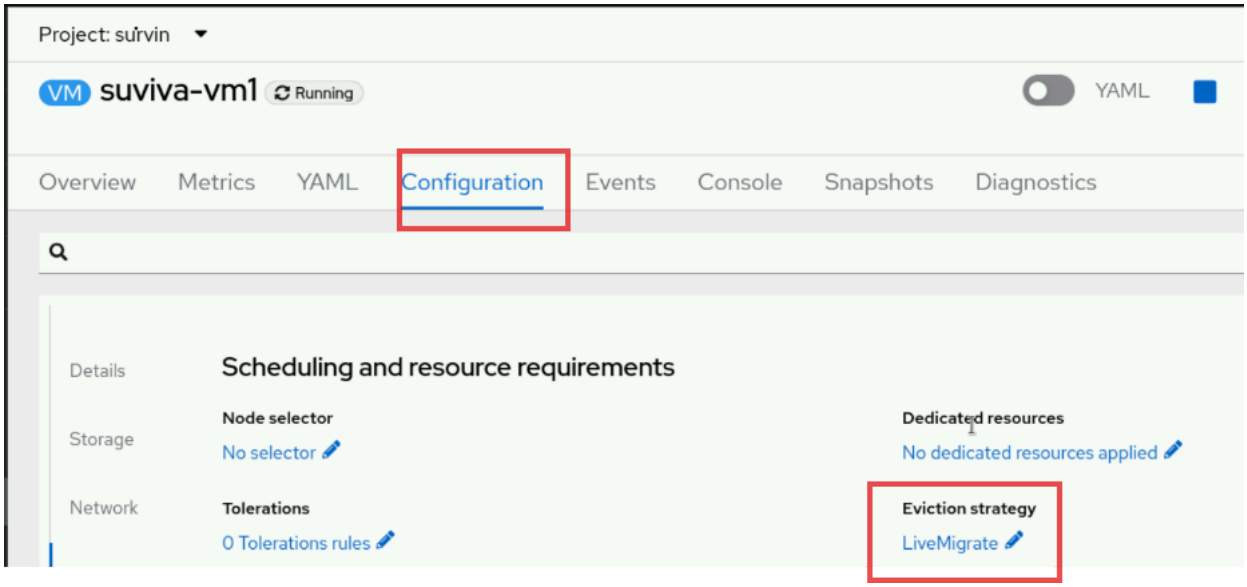


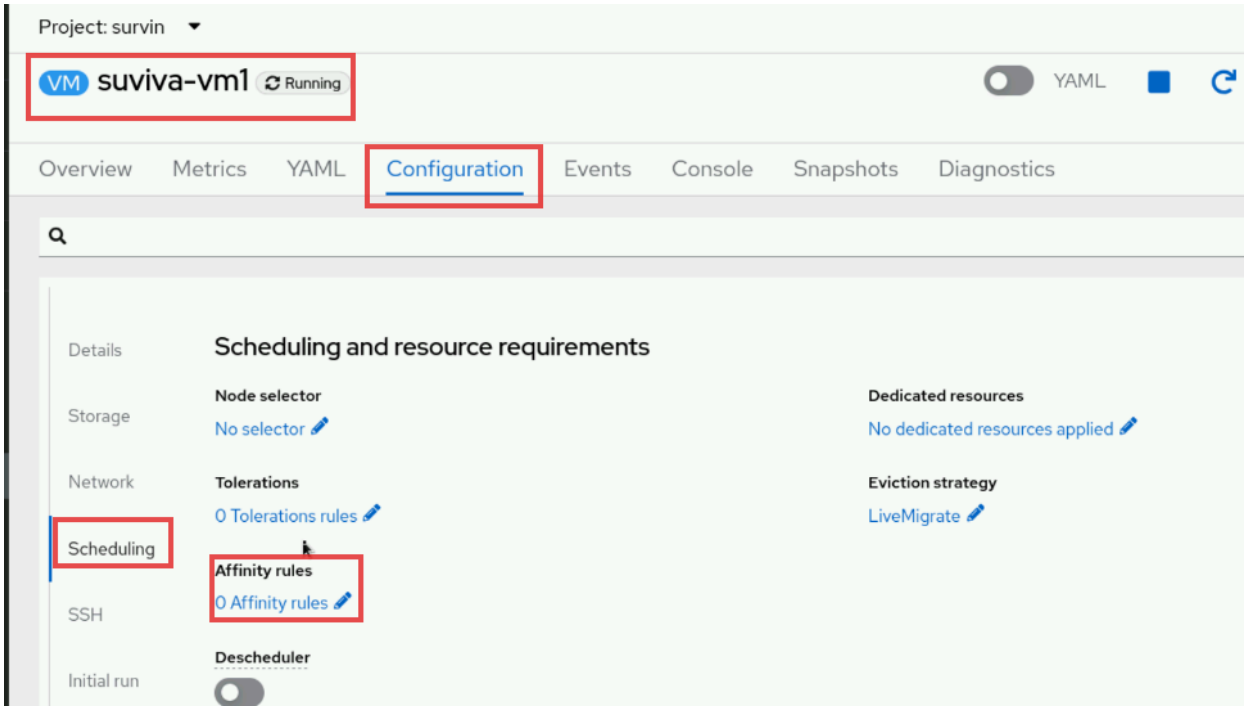
- **Now verify the label assigned to both master01 and master02. In reality, both have one common label key which value is different**

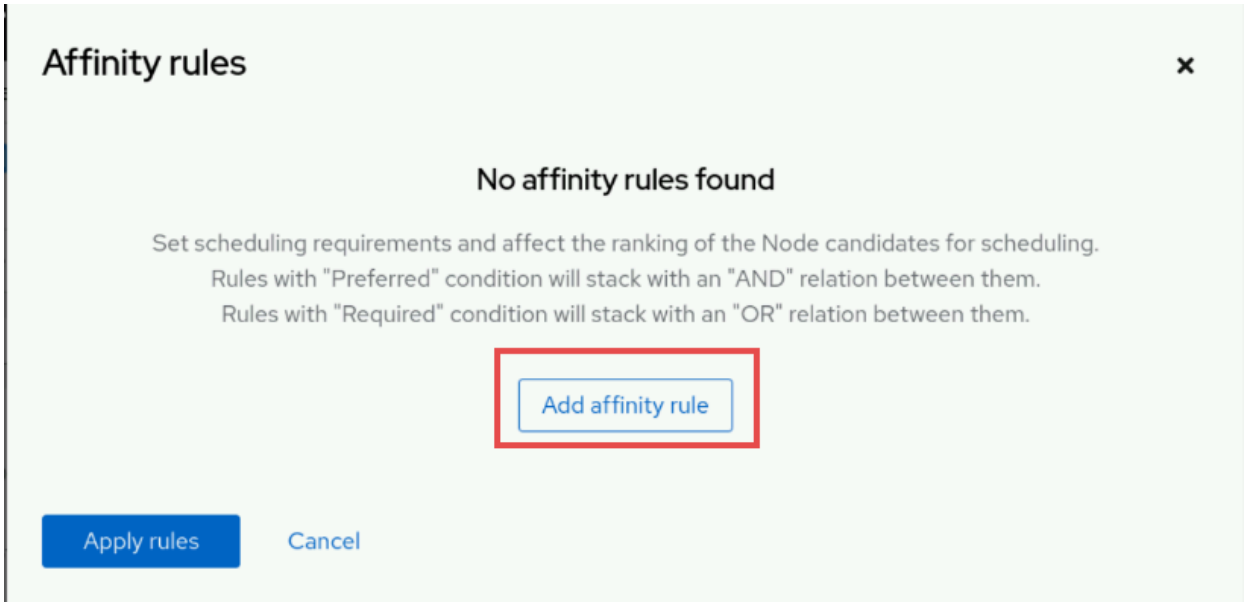
```
[student@workstation ~]$ oc get nodes master01 --show-labels | grep master01
```

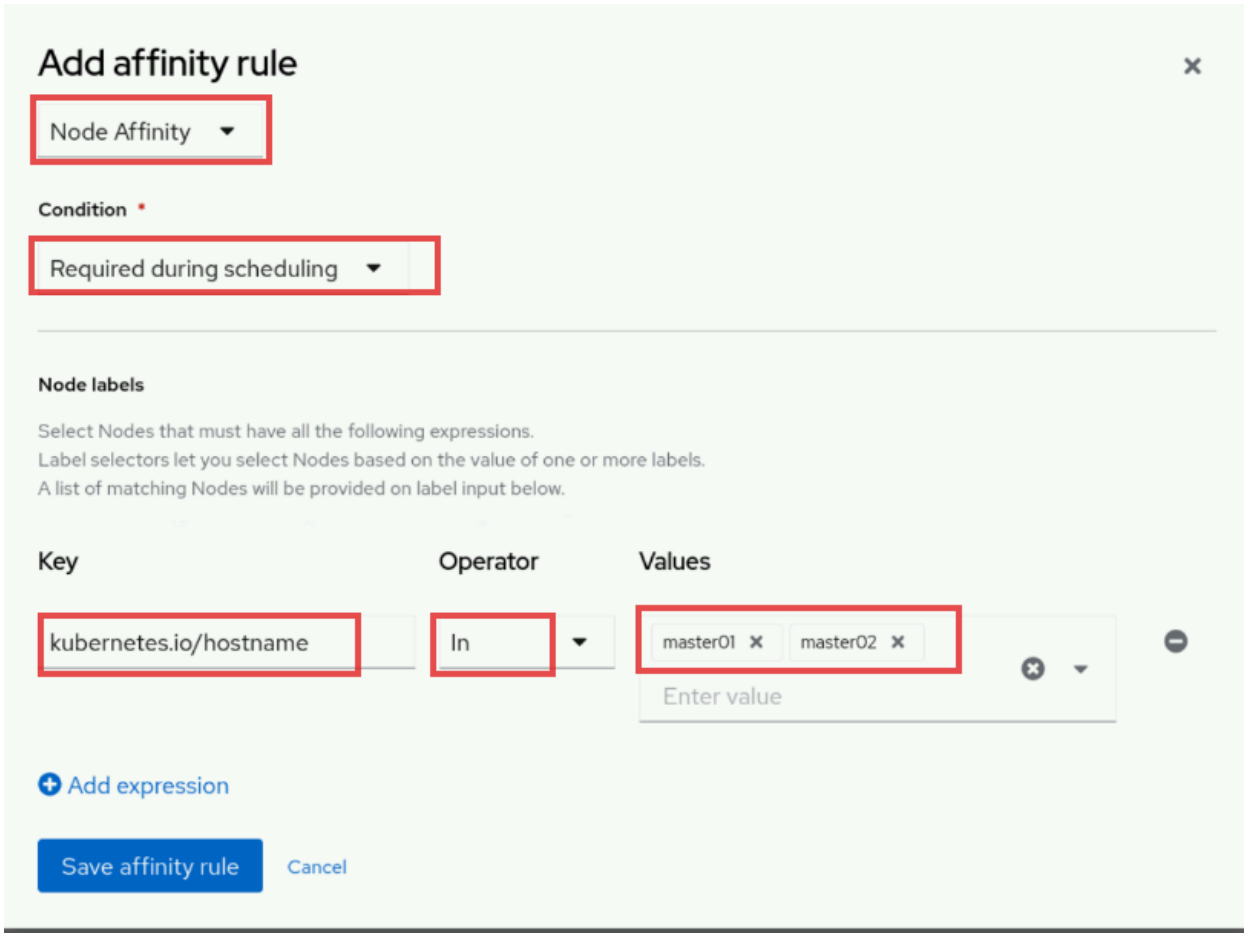
```
e,hyperv.node.kubevirt.io/reset=true,hyperv.node.kubevirt.io/runtime=true,hyperv.node.kube
virt.io/synic2=true,hyperv.node.kubevirt.io/synic=true,hyperv.node.kubevirt.io/synictimer=
true,hyperv.node.kubevirt.io/time=true,hyperv.node.kubevirt.io/tlbflush=true,hyperv.node.k
ubevirt.io/vpindex=true,kubernetes.io/arch=amd64,kubernetes.io/hostname=master01,kubernete
s.io/os=linux,kubevirt.io/ksm-enabled=false,kubevirt.io/schedulable=true,node-role.kuberne
tes.io/control-plane=,node-role.kubernetes.io/master=,node-role.kubernetes.io/worker=,node
.openshift.io/os_id=rhcos
```

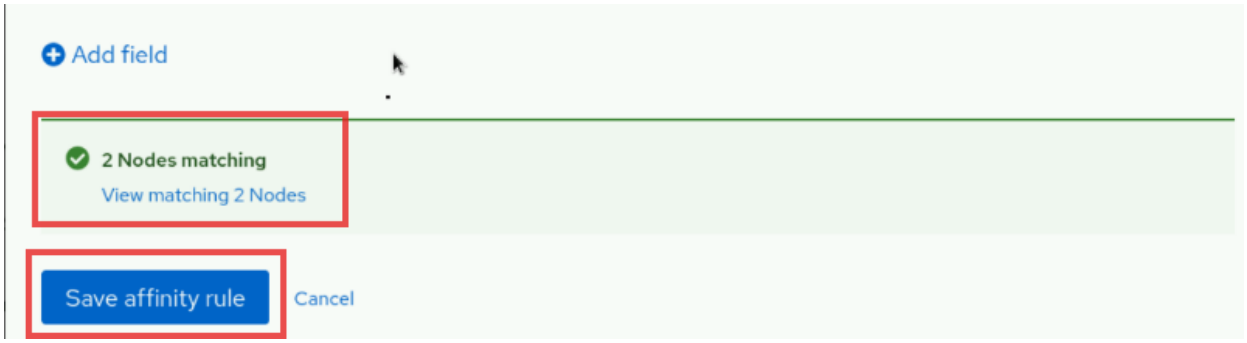
Both nodes having the label key as kubernetes.io/hostname and key as master01 and master02 respectively

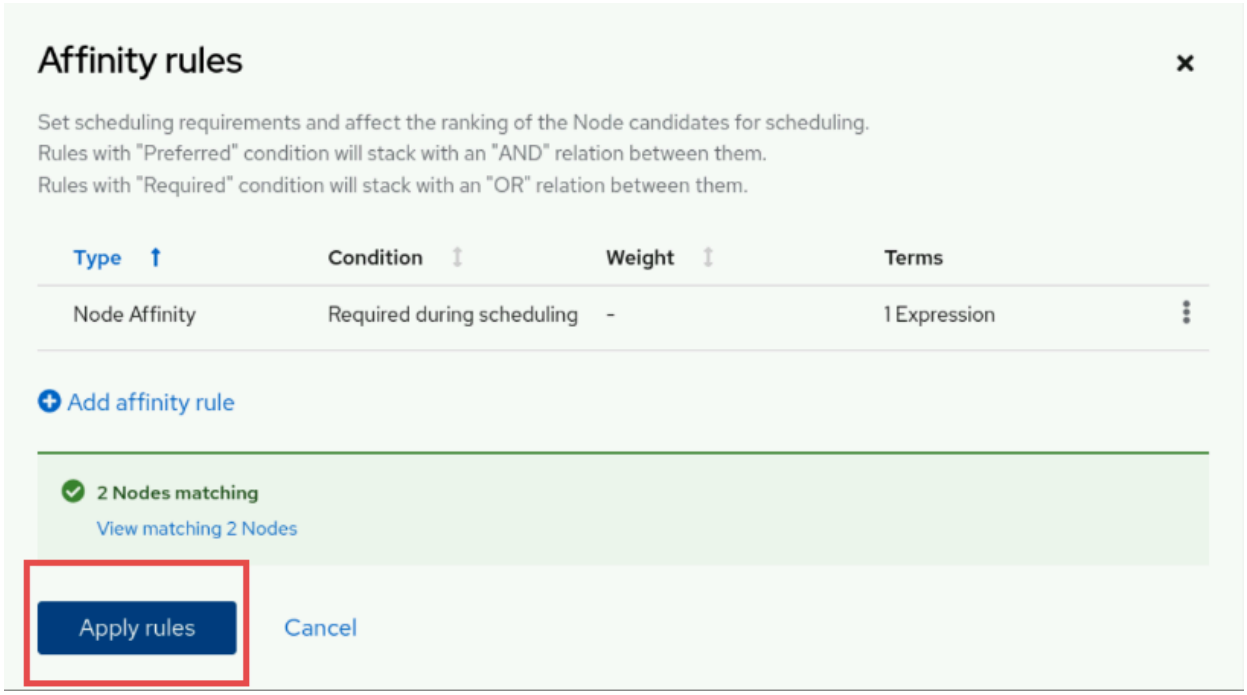

Project: survin
VM suviva-vm1 Running
YAML
Overview
Metrics
YAML
Configuration
Events
Console
Snapshots
Diagnostics
Details
Storage
Network
Scheduling and resource requirements
Node selector
No selector
Tolerations
0 Tolerations rules
Dedicated resources
No dedicated resources applied
Eviction strategy
LiveMigrate


Project: survin
VM suviva-vm1 Running
YAML
Overview
Metrics
YAML
Configuration
Events
Console
Snapshots
Diagnostics
Details
Storage
Network
Scheduling
SSH
Initial run
Scheduling and resource requirements
Node selector
No selector
Tolerations
0 Tolerations rules
Affinity rules
0 Affinity rules
Descheduler
Dedicated resources
No dedicated resources applied
Eviction strategy
LiveMigrate


Affinity rules
No affinity rules found
Set scheduling requirements and affect the ranking of the Node candidates for scheduling.
Rules with "Preferred" condition will stack with an "AND" relation between them.
Rules with "Required" condition will stack with an "OR" relation between them.
Add affinity rule
Apply rules
Cancel


Add affinity rule
Node Affinity
Condition *
Required during scheduling
Node labels
Select Nodes that must have all the following expressions.
Label selectors let you select Nodes based on the value of one or more labels.
A list of matching Nodes will be provided on label input below.
Key
Operator
Values
kubernetes.io/hostname
In
master01
master02
Enter value
Add expression
Save affinity rule
Cancel





Now migrate the machine and verify its running between master01 and master02 only

==========

Q-19 VM Migration

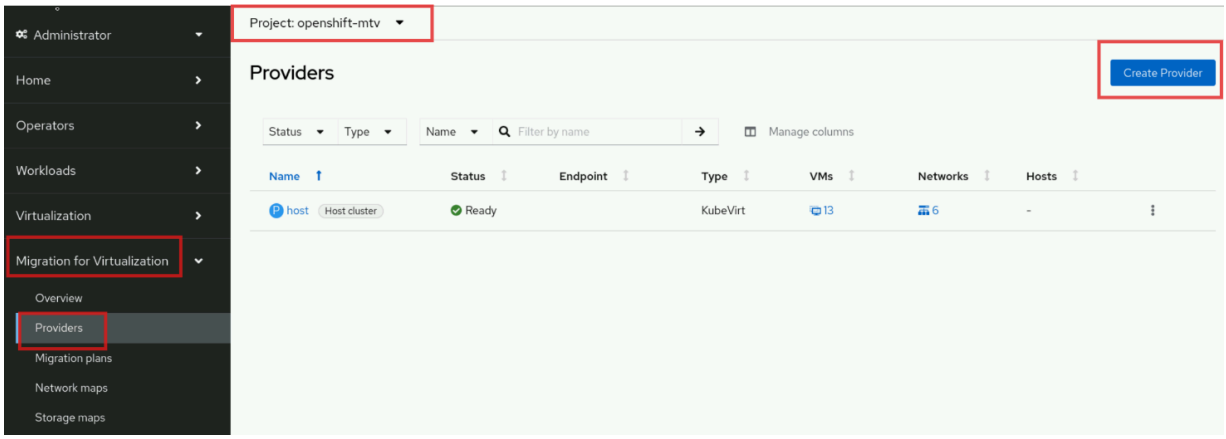
One or more virtual machines images are available for browsing from your workbench system using command: **elinks --dump http://rhgls.domain3.example.com/materials/migration_images**

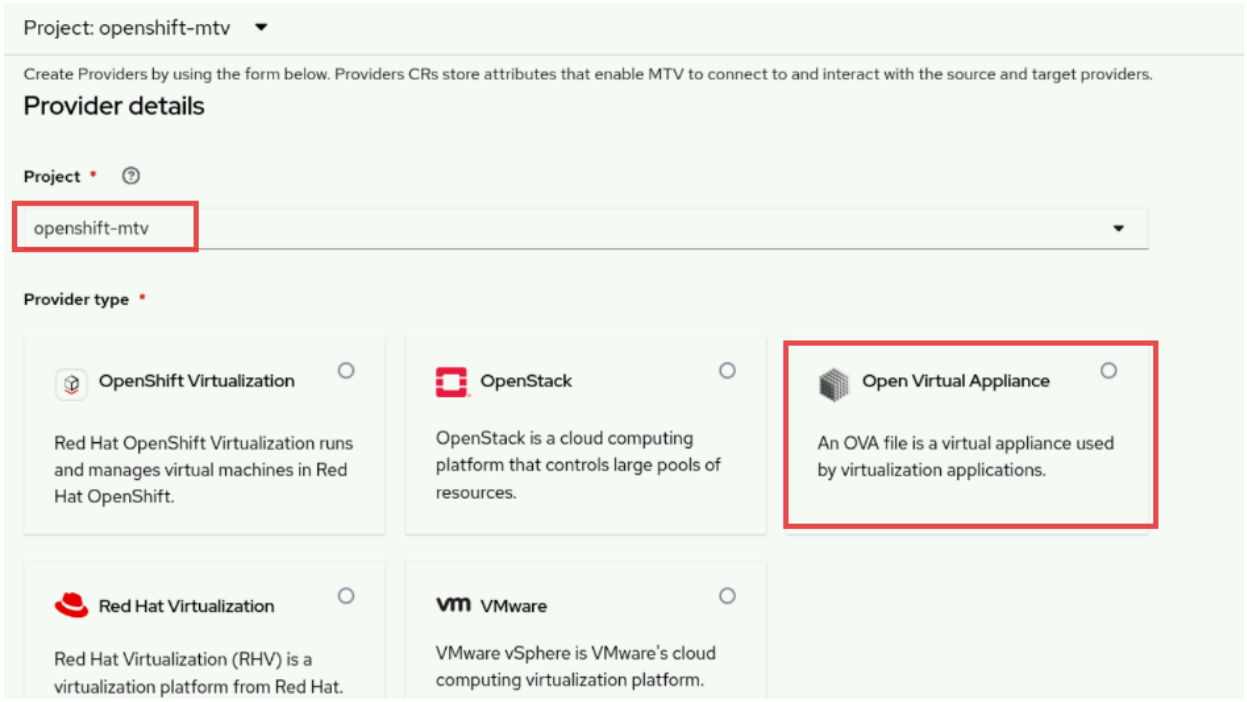
This directory is also exported via NFS at:
rhgls.domain3.example.com:/var/www/html/materials/migration_images
Important the virtual machine called rhel9-web according to the following requirements:

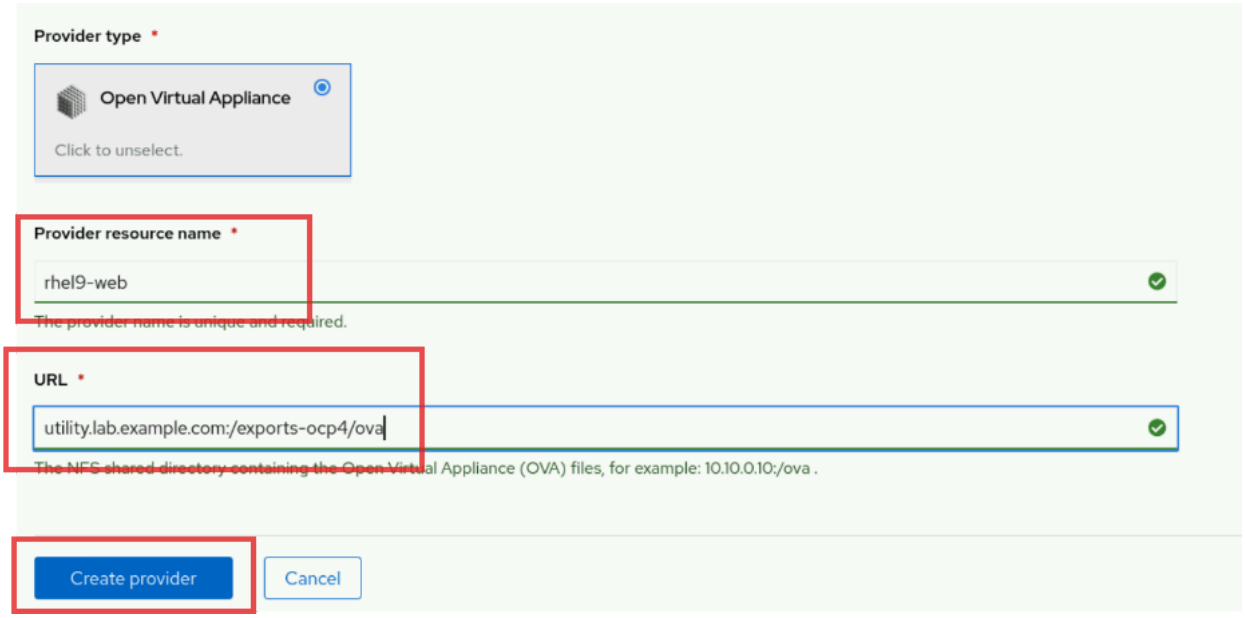
- The important virtual machine exists in the **web-import** project
- The network of the source virtual machine has been mapped to the Pod network of the import virtual machine
- The imported virtual machine uses the **ocs-external-storagecluster-ceph-rbd** stores class
- The imported virtual machine is running

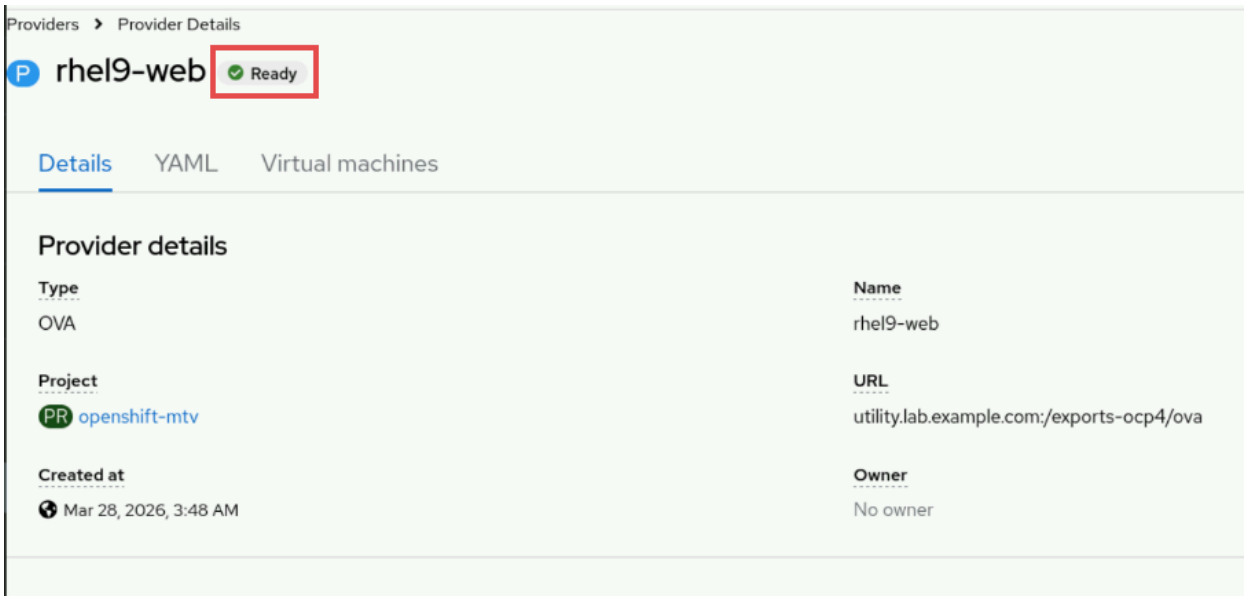
ANS:
NOTE: you have to execute below lab cmd inorder to perform the steps in lab
In 4.16 - lab start -t do316 advanced-import
In 4.18 lab start migrate-mtv

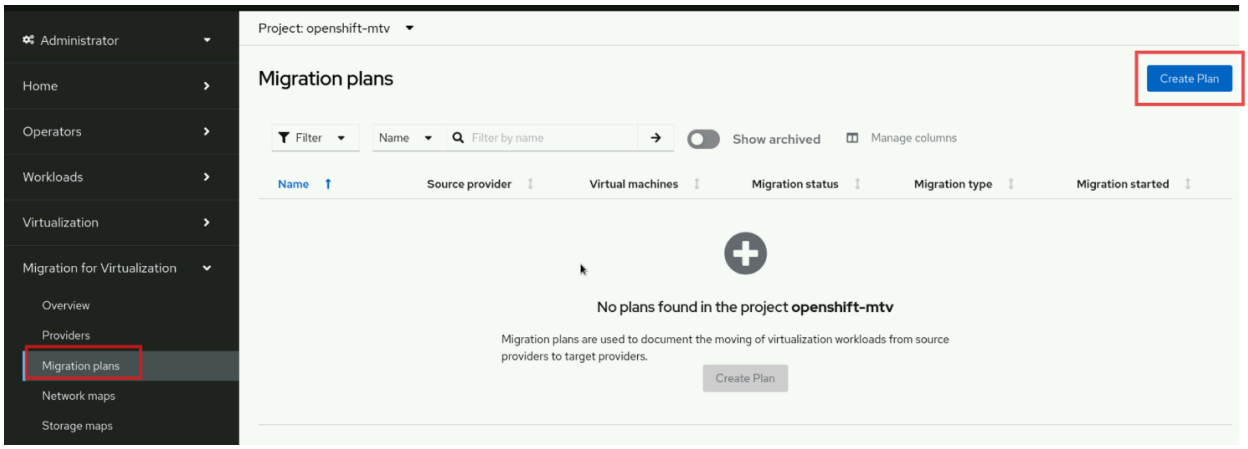
- First install the operator, and then if web-import project not created, then create it. In lab as well as in exam

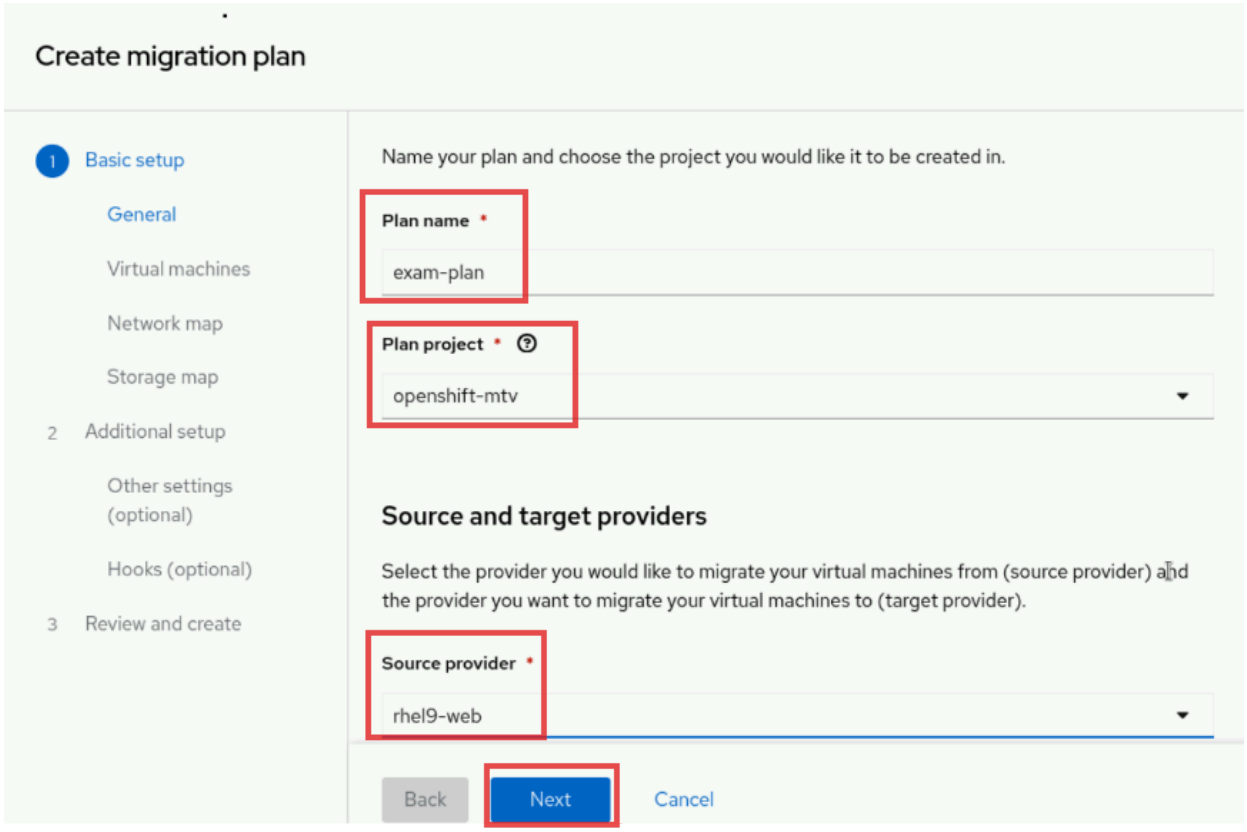


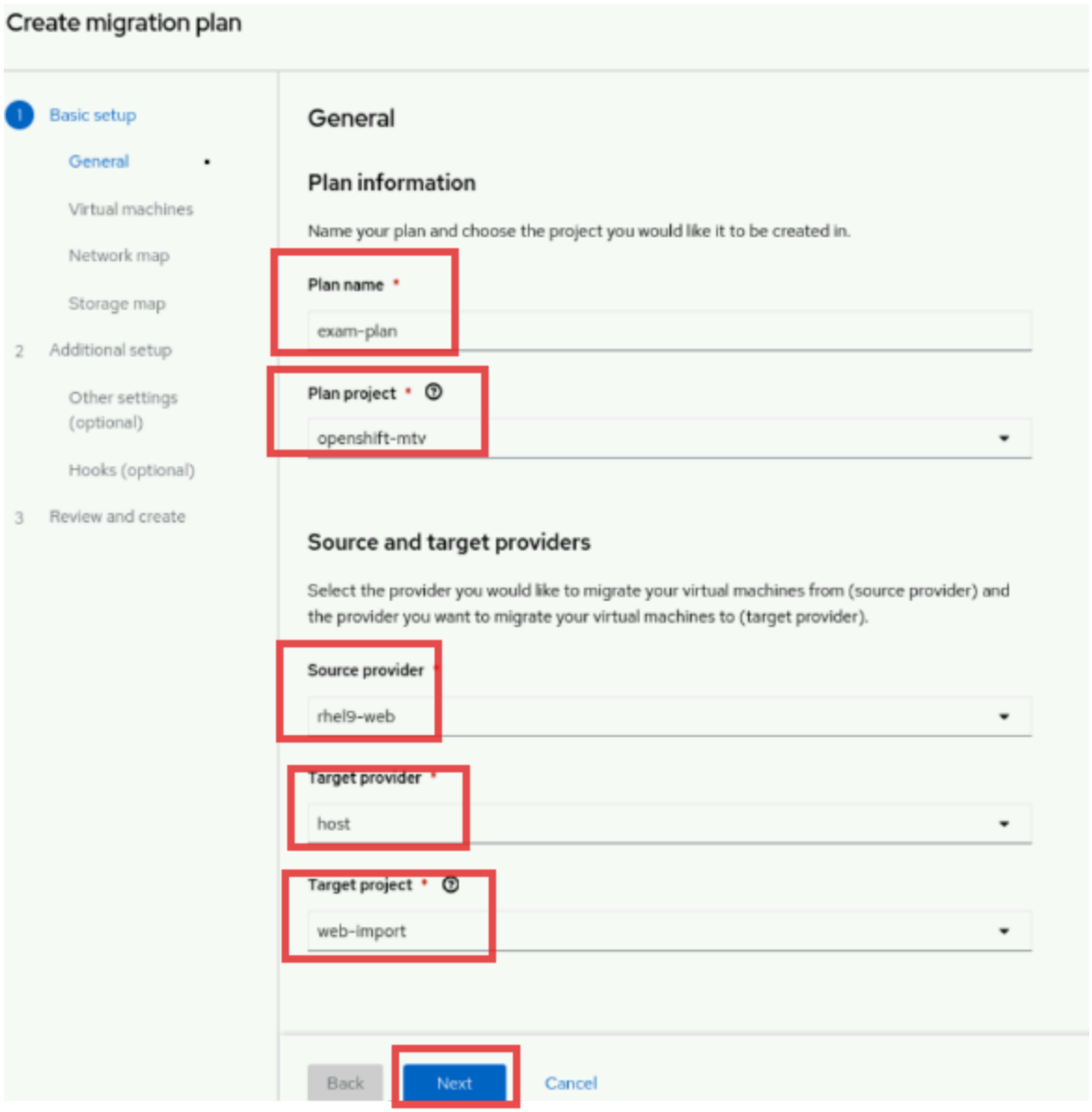

Project: openshift-mtv
Create Providers by using the form below. Providers CRs store attributes that enable MTV to connect to and interact with the source and target providers.
Provider details
Project *
openshift-mtv
Provider type *
OpenShift Virtualization
Red Hat OpenShift Virtualization runs and manages virtual machines in Red Hat OpenShift.
OpenStack
OpenStack is a cloud computing platform that controls large pools of resources.
Open Virtual Appliance
An OVA file is a virtual appliance used by virtualization applications.
Red Hat Virtualization
Red Hat Virtualization (RHV) is a virtualization platform from Red Hat.
VMware
VMware vSphere is VMware's cloud computing virtualization platform.

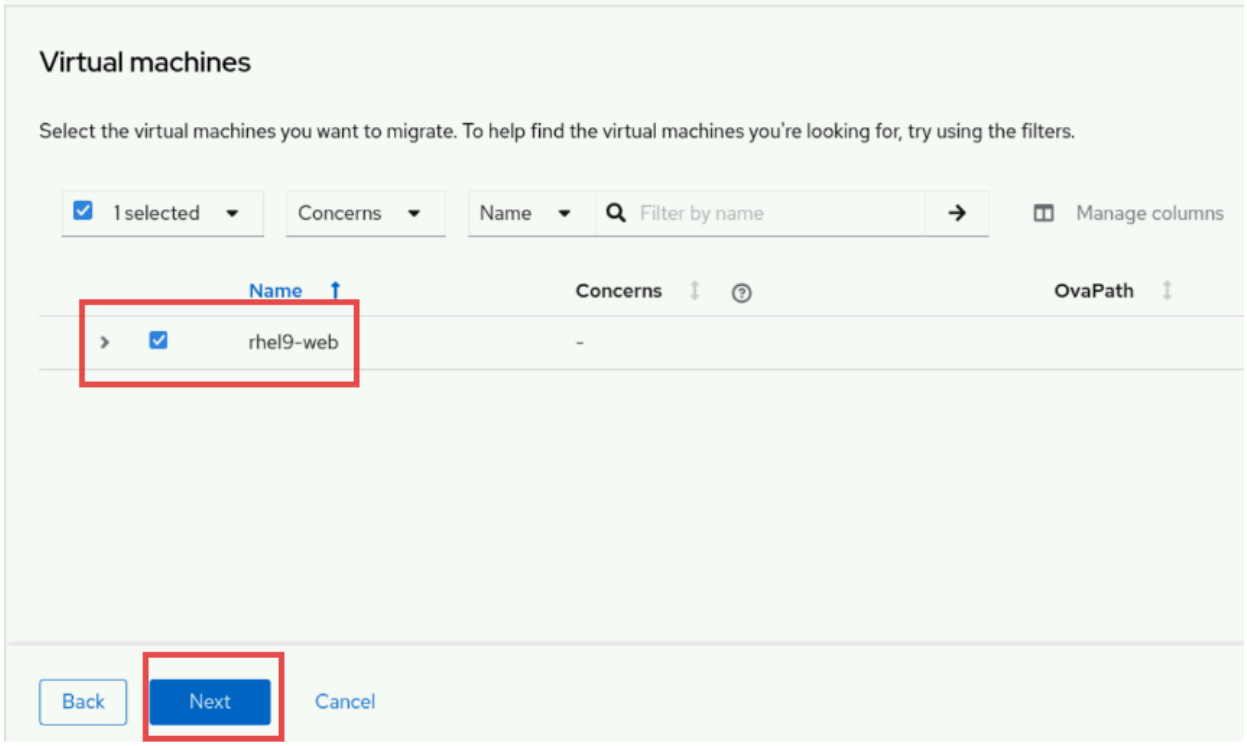

Provider type *
Open Virtual Appliance
Click to unselect.
Provider resource name *
rhel9-web
The provider name is unique and required.
URL *
utility.lab.example.com:/exports-ocp4/ova
The NFS shared directory containing the Open Virtual Appliance (OVA) files, for example: 10.10.0.10:/ova .
Create provider
Cancel

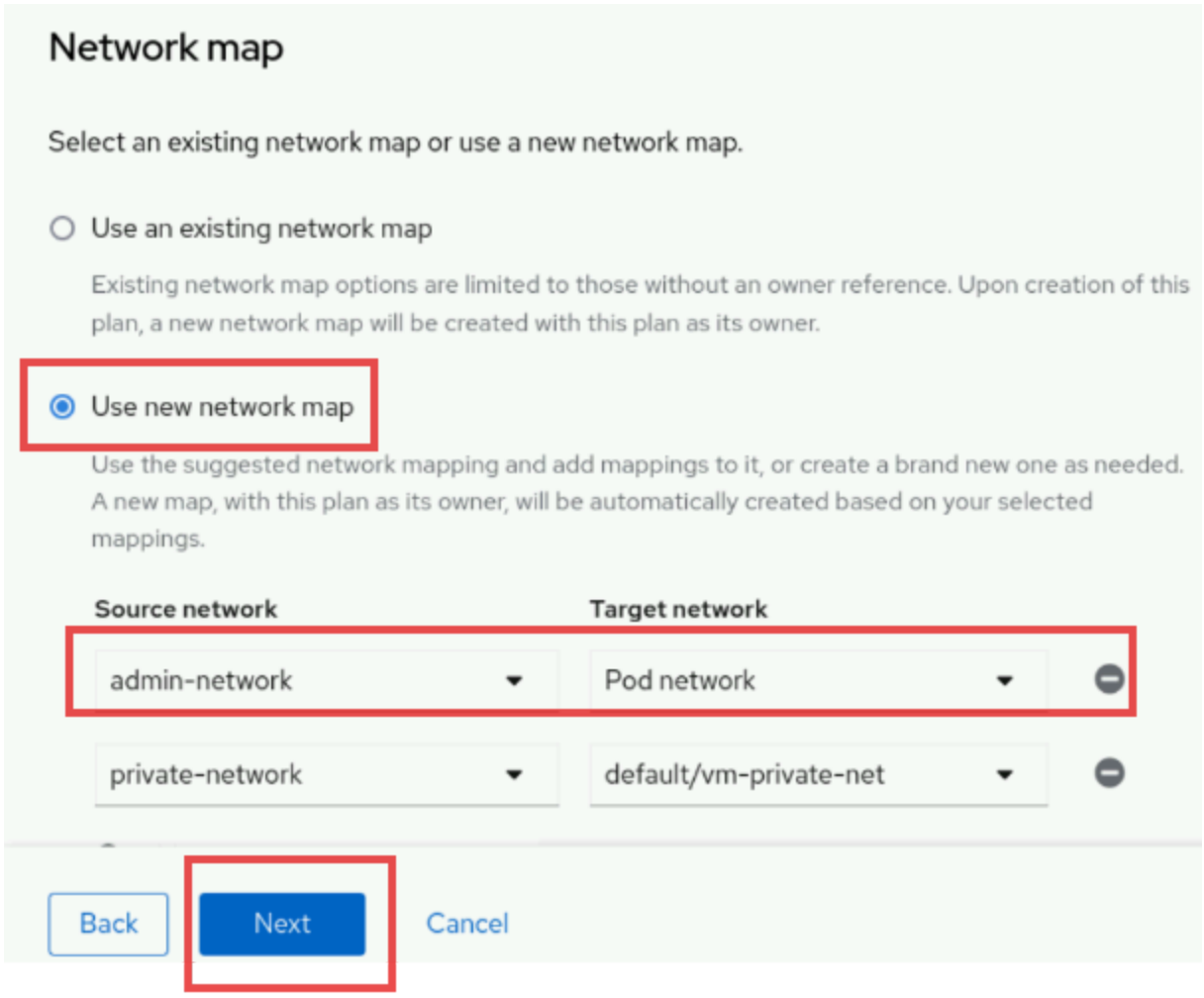

Providers
Provider Details
rhel9-web
Ready
Details
YAML
Virtual machines
Provider details
Type
OVA
Name
rhel9-web
Project
openshift-mtv
URL
utility.lab.example.com:/exports-ocp4/ova
Created at
Mar 28, 2026, 3:48 AM
Owner
No owner

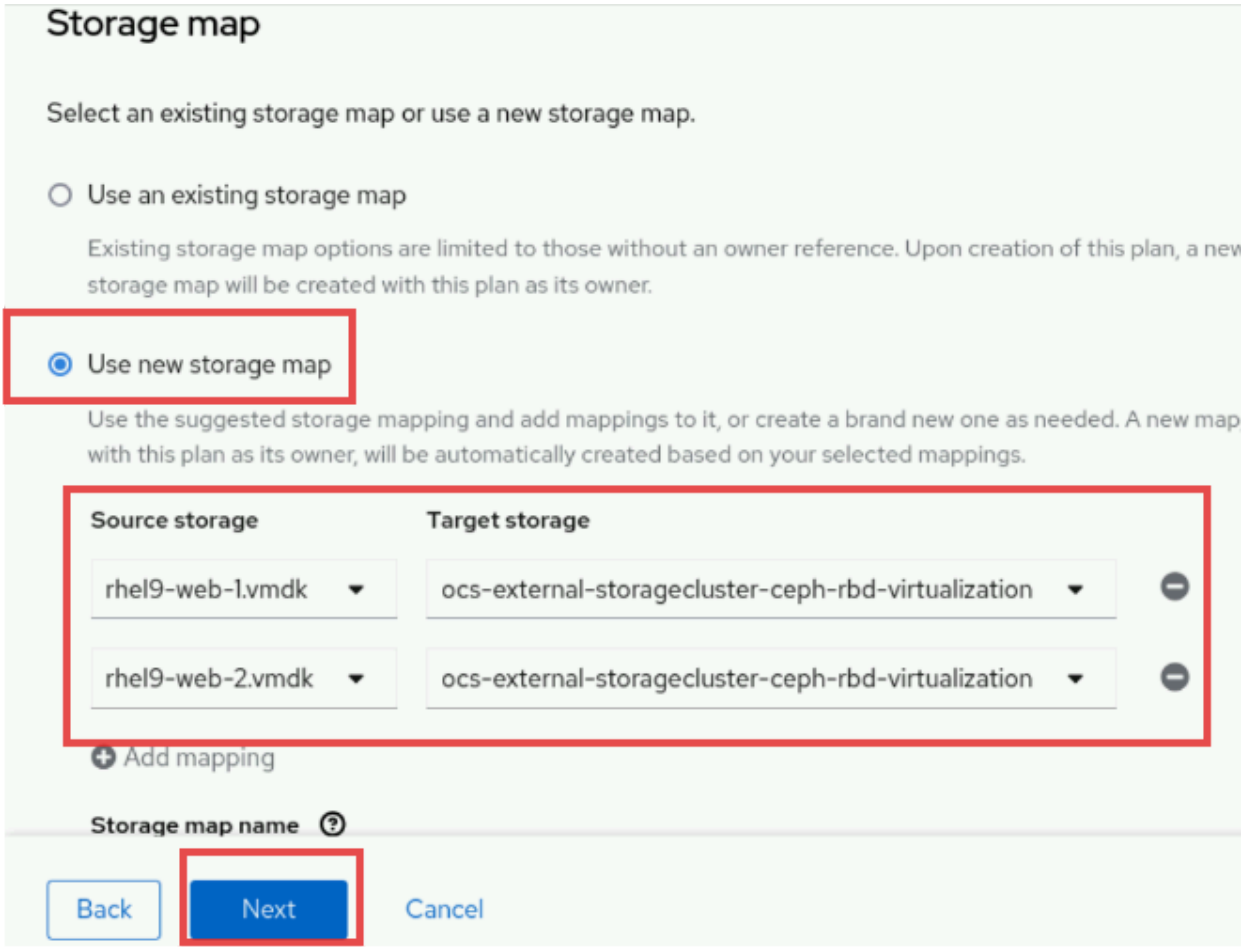

Administrator
Home
Operators
Workloads
Virtualization
Migration for Virtualization
Overview
Providers
Migration plans
Network maps
Storage maps
Project: openshift-mtv
Migration plans
Create Plan
Filter
Name
Filter by name
Show archived
Manage columns
Name
Source provider
Virtual machines
Migration status
Migration type
Migration started
No plans found in the project openshift-mtv
Migration plans are used to document the moving of virtualization workloads from source providers to target providers.
Create Plan

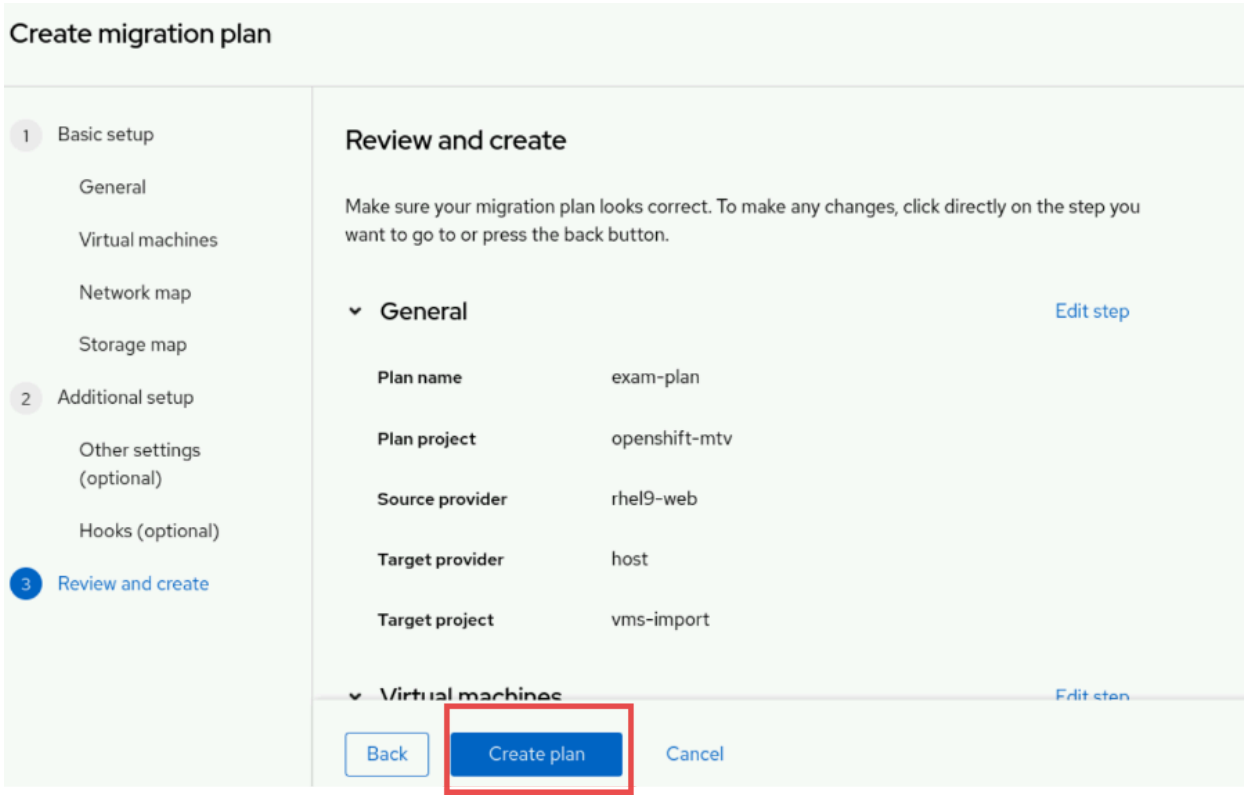

Create migration plan
1 Basic setup
General
Virtual machines
Network map
Storage map
2 Additional setup
Other settings (optional)
Hooks (optional)
3 Review and create
Name your plan and choose the project you would like it to be created in.
Plan name *
exam-plan
Plan project *
openshift-mtv
Source and target providers
Select the provider you would like to migrate your virtual machines from (source provider) and the provider you want to migrate your virtual machines to (target provider).
Source provider *
rhel9-web
Back
Next
Cancel

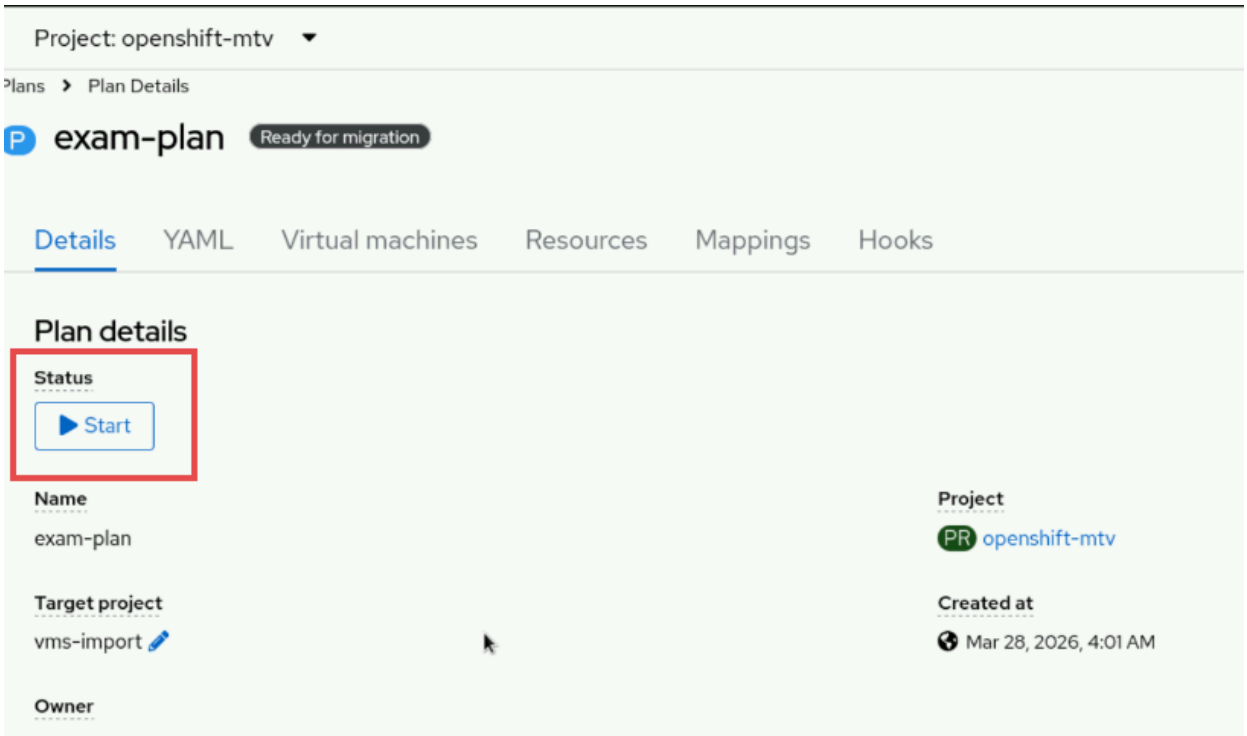

Create migration plan
1 Basic setup
General
Virtual machines
Network map
Storage map
2 Additional setup
Other settings (optional)
Hooks (optional)
3 Review and create
General
Plan information
Name your plan and choose the project you would like it to be created in.
Plan name *
exam-plan
Plan project *
openshift-mtv
Source and target providers
Select the provider you would like to migrate your virtual machines from (source provider) and the provider you want to migrate your virtual machines to (target provider).
Source provider *
rhel9-web
Target provider *
host
Target project *
web-import
Back
Next
Cancel

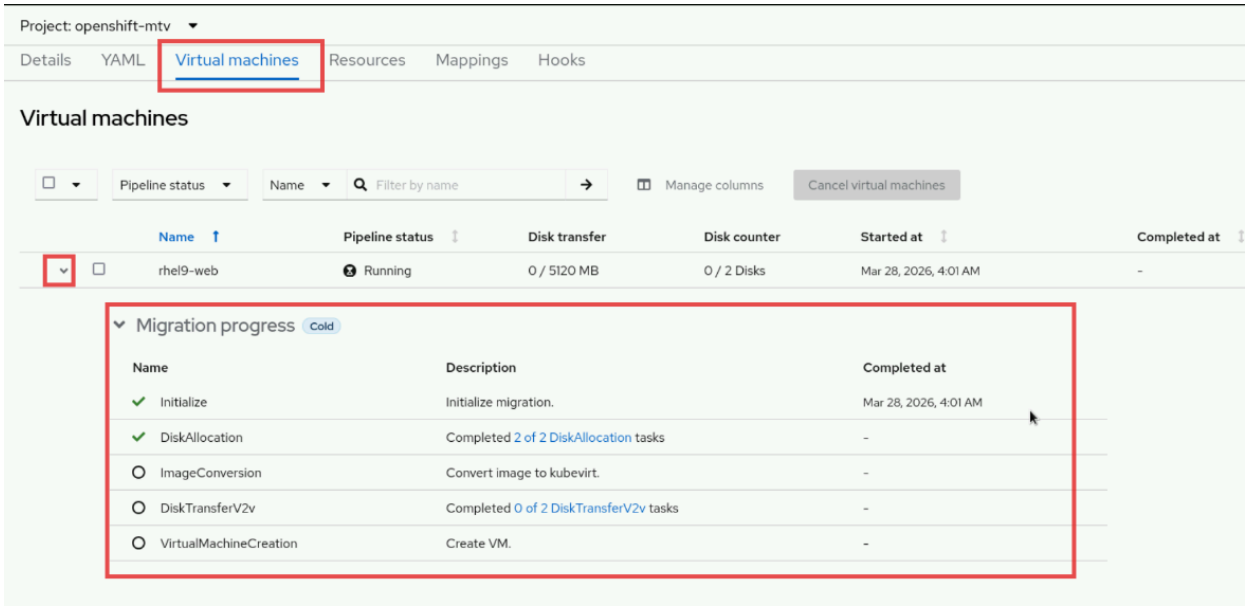

Virtual machines
Select the virtual machines you want to migrate. To help find the virtual machines you're looking for, try using the filters.
1 selected
Concerns
Name
Filter by name
Manage columns
Name
Concerns
OvaPath
rhel9-web
-
Back
Next
Cancel


Network map
Select an existing network map or use a new network map.
Use an existing network map
Existing network map options are limited to those without an owner reference. Upon creation of this plan, a new network map will be created with this plan as its owner.
Use new network map
Use the suggested network mapping and add mappings to it, or create a brand new one as needed. A new map, with this plan as its owner, will be automatically created based on your selected mappings.
Source network
Target network
admin-network
Pod network
private-network
default/vm-private-net
Back
Next
Cancel


Storage map
Select an existing storage map or use a new storage map.
Use an existing storage map
Existing storage map options are limited to those without an owner reference. Upon creation of this plan, a new
storage map will be created with this plan as its owner.
Use new storage map
Use the suggested storage mapping and add mappings to it, or create a brand new one as needed. A new map
with this plan as its owner, will be automatically created based on your selected mappings.
Source storage
Target storage
rhel9-web-1.vmdk
ocs-external-storagecluster-ceph-rbd-virtualization
rhel9-web-2.vmdk
ocs-external-storagecluster-ceph-rbd-virtualization
Add mapping
Storage map name
Back
Next
Cancel


Create migration plan
1 Basic setup
General
Virtual machines
Network map
Storage map
2 Additional setup
Other settings (optional)
Hooks (optional)
3 Review and create
Review and create
Make sure your migration plan looks correct. To make any changes, click directly on the step you want to go to or press the back button.
General
Edit step
Plan name
exam-plan
Plan project
openshift-mtv
Source provider
rhel9-web
Target provider
host
Target project
vms-import
Back
Create plan
Cancel


Project: openshift-mtv
Plans > Plan Details
exam-plan
Ready for migration
Details
YAML
Virtual machines
Resources
Mappings
Hooks
Plan details
Status
Start
Name
exam-plan
Project
openshift-mtv
Target project
vms-import
Created at
Mar 28, 2026, 4:01 AM
Owner



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THE END

Best of Luck